

INVARIANT ARITHMETIC OF CLOSED GEODESICS

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ABSTRACT. We study rational valued counts of geodesic strings (reparametrization equivalence classes of closed geodesics) for complete Riemann-Finsler manifolds. There are two main conceptual results. The count (when finite) is a topological invariant on orientable surfaces and on aspherical manifolds with free abelian fundamental group, tori in particular; this combines group theory with loop space energy flow analysis and Fuller theory. The count satisfies a product type formula; this uses index theory and dissipative perturbation ideas. Combined with Gromov hyperbolicity, we get the following new sample phenomenon. Let g_- be a Riemannian metric of negative curvature on a closed orientable surface and β_0 an indivisible free homotopy class. Then for every k , each C^1 -neighborhood of g_- has a C^2 -open nonempty subset whose members have prime geodesic strings in all the classes $2\beta_0, 4\beta_0, \dots, 2^k\beta_0$. Such forcing is invisible to classical bifurcation and min-max methods: it hinges on invariant arithmetic. Similar forcing happens on T^2 but with KAM confinement near a Clairaut metric replacing Gromov hyperbolicity. We also obtain arithmetic constraints on counts of closed geodesics in mapping tori and flat bundles, and constraints on the existence of negative sectional curvature metrics. Finally, we conjecture a complete topological invariance phenomenon of these counts; as verified for surfaces and tori, such invariance would give analogous arithmetic laws for all taut regular metrics on manifolds having the tautness property studied here.

1. INTRODUCTION

Let g be a metric on a manifold X , throughout assumed orientable. A *geodesic string* will be a short name for the equivalence class of a closed, constant speed g -geodesic up to the reparametrization S^1 action. We say it is *prime* if it does not have a multiply covered representative. More generally, we say that a smooth loop is *prime* if it is not multiply covered.

We will study arithmetic properties of geodesic string counts, which are outside the scope of (S^1 -equivariant) Morse homology of the loop space. There are two principal theoretical results: first the topological invariance of the count via Euler/Wall characteristic correspondence of Theorem 1.2. Second, a product formula, Section 7, for this count and its generalizations based on Fuller index theory. The Fuller index is a completely general construction in dynamical systems; the product formula is a more subtle phenomenon, and needs the Riemann-Finsler geometric context. Fuller's theory and the basic definitions are outlined in the Appendix, so no prior knowledge is necessary.

For a nontrivial free homotopy class β of loops in X , we say that a metric g on X is β -*regular* if all of its class β closed geodesics are nondegenerate in the usual sense, or equivalently the closed orbits of the associated geodesic flow are dynamically nondegenerate; g is *regular* if it is β -regular for every nontrivial β .

For a β -regular g with finitely many class β geodesic strings, we define:

$$(1.1) \quad F(g, \beta) = \sum_o \frac{(-1)^{\text{morse}(o)}}{\text{mult}(o)} \in \mathbb{Q},$$

where:

- The sum is over class β g -geodesic strings o .

Key words and phrases. Nonpositive curvature, Fuller index, geodesic counts, S^1 -equivariant homology of the loop space, KAM theory.

Supported by CONAHCYT research grant CF-2023-I.

- $\text{morse}(o)$ denotes the Morse index of a representative of o : the number of negative eigenvalues of the Hessian of the energy at that closed geodesic, defined whether or not the critical point is degenerate, and independent of the representative.
- $\text{mult}(o)$ is the geometric multiplicity, equivalently the order of the corresponding isotropy subgroup of S^1 , where S^1 is acting on the loop space by reparametrization.

We write $\pi_1^{\text{inc}}(X)$ for the set of nontrivial free homotopy classes when X is compact, and in general for the end incompressible ones (Definition 2.1). For $k \geq 1$, $\beta \in \pi_1^{\text{inc}}(X)$, $k\beta$ denotes the class of the k -fold iterates of the loops in β . $\beta \in \pi_1^{\text{inc}}(X)$ is **indivisible** if it is not $k\beta'$ for any $k \geq 2$, that is, if it is not representable by a multiply covered loop.

Let $L_\beta X$ denote the corresponding component of the free loop space. A metric g is β -**taut** if the space of class β closed g -geodesics is compact, and **taut** if it is β -taut for every $\beta \in \pi_1^{\text{inc}}(X)$; see Section 1.1. For β -taut metrics the count (1.1) extends by Theorem 2.8.

The following theorem expresses the basic topological invariance of the F -count. Note that X need not be compact.

Theorem 1.2 (Proofs in Sections 6 and 6.1). (1) *Let X be an orientable manifold and $\beta \in \pi_1^{\text{inc}}(X)$ indivisible. Suppose X admits a β -taut metric. Then the Borel construction*

$$(L_\beta X)_{hS^1} := L_\beta X \times_{S^1} ES^1$$

has homology of finite total rank, and for every β -taut metric g :

$$F(g, \beta) = \chi((L_\beta X)_{hS^1}),$$

its Euler characteristic.

- (2) *Let X be an orientable aspherical manifold with free abelian fundamental group, for instance T^n , or an orientable surface. Let $\beta = e\beta_0 \in \pi_1^{\text{inc}}(X)$, with β_0 indivisible and $e \geq 1$. Then for every β -taut metric g :*

$$F(g, \beta) = \chi_{\text{Wall}}((L_\beta X)_{hS^1}) = \begin{cases} 0, & \pi_1(X) \cong \mathbb{Z}^n, \ n \geq 2, \\ 1/e, & \text{otherwise,} \end{cases}$$

the Wall characteristic. The latter is described fully in Conjecture 1 below, generalizing this theorem.

Part (1) is proved by S^1 -equivariant Morse theory and Fuller theory. For divisible classes, the homology theoretic interpretation of the F -count would require more than ordinary S^1 -equivariant Morse homology for the g -energy functional. One would want a g -invariant up to quasi-isomorphism “orbifold Morse homology chain complex” for the quotient stack $[LX/S^1]$, together with (assuming certain finiteness) an algebraically derived $\mathbb{Q}$ -valued characteristic¹ coinciding with the F -count. This is likely related to Wall-Euler characteristic theory in group cohomology theory, appearing above and in the more basic Conjecture 1.

Existence of this orbifold Morse homology is at the moment unknown; there is not even a finite dimensional candidate. Closest in spirit is the equivariant Morse complex of [10], for finite group actions: a complex of $\mathbb{Z}[G]$ -modules, which remembers the isotropy. For the manifolds of part (2), however, a covering transfer substitutes for the missing orbifold theory. Finite covers unwrap the divisible class $e\beta_0$ into indivisible classes, where part (1) applies, and the multiplicity weights of F match the covering multiplicities exactly (Lemma 6.6). For free abelian π_1 one e -fold cyclic cover does it. On surfaces the divisibility is reduced one prime at a time, through finite quotients in which a representative of β_0 has prescribed prime order, supplied by residual freeness of surface groups [11] (Lemma 6.10); the fundamental group of a noncompact surface is free. This gives $F(g, e\beta_0) = 0$ for

¹I am told that in this context a better term is von Neumann dimension.

free abelian π_1 of rank ≥ 2 , and $F(g, e\beta_0) = 1/e$ on surfaces other than T^2 and for $\pi_1 \cong \mathbb{Z}$, for every $e\beta_0$ -taut metric.

Part (2) of Theorem 1.2, combined with the arithmetic of Section 9, gives the following partial consequence:

Theorem 1.3 (Arithmetic law on surfaces; proof in Section 9). *Let X be a closed orientable surface, β_0 an indivisible free homotopy class, $e \geq 1$, and g an $e\beta_0$ -taut and $e\beta_0$ -regular metric on X . Then:*

- (1) *For e odd, if there is a prime geodesic string in class $e\beta_0$, there is another — except for $e = 1$ on X of genus ≥ 2 , where the number of prime geodesic strings in class β_0 is instead odd.*
- (2) *For $e = 2n$, the class $2n\beta_0$ contains at least as many prime geodesic strings as the class $n\beta_0$ contains inverse-hyperbolic ones (the linearized return map has negative eigenvalues) — a period doubling phenomenon.*

Note that e odd is essential in part (1): for e even the law does not force the pairing, and numerically it fails. In Theorem 1.13 below the class $2\beta_0$ contains a prime string, numerically a single one (Remark 12.2). Tautness is supplied by two mechanisms, with $e\beta_0$ -regularity generic. On T^2 , every metric $C^\aleph$ - $\hbar(g_0)$ -close to a flat or to a Clairaut metric g_0 is taut, that is $\|g - g_0\|_{C^\aleph} < \hbar(g_0)$ with the norm convention of Section 11, by KAM confinement (Theorem 1.6). Here the constant $\aleph \in \mathbb{N}$ is absolute, and $\hbar(g_0) > 0$ depends only on g_0 ; both are instantiated in Section 11, and are effective, that is, in principle computable. Clairaut metrics can be far from flat (Theorem 1.13). Near negative curvature, Gromov hyperbolicity supplies tautness (Proposition 2.7). As a concrete effect we have:

Corollary 1.4. *Let g_- be a Riemannian metric of negative curvature on a closed orientable surface and β_0 an indivisible free homotopy class. Then for every k , each C^1 -neighborhood of g_- has a C^2 -open nonempty subset whose members have prime geodesic strings in all the classes $2\beta_0, 4\beta_0, \dots, 2^k\beta_0$.*

Note that C^1 -closeness to g_- permits pointwise arbitrarily large curvature (Example 3).

Proof outline; details in Section 12. The second part of Theorem 1.3 fires as follows. Theorem 12.4, localized in Lemma 12.6, produces a class β_0 inverse-hyperbolic prime g -geodesic string by Hill type engineering, also used in Theorem 1.13. Here g is Riemannian and arbitrarily C^1 -close to g_- , hence $2\beta_0$ -taut by Proposition 2.7, and $2\beta_0$ -regular. Part (2) of Theorem 1.3 then forces a prime g -geodesic string in the divisible class $2\beta_0$. The existence is in fact forced on a C^2 -open set of Riemannian metrics near g_- , modulo generic regularity: Corollary 12.5. Theorem 12.7 and Remark 12.8 iterate the law through the powers of 2, giving the corollary. $\square$

Near negative curvature, part (1) of Theorem 1.3, including its $e = 1$ variant, is not surface-specific: it holds for $e\beta_0$ -regular Riemannian metrics C^1 - ϵ -close to negative curvature on closed orientable manifolds of any dimension, by Proposition 2.7, Example 4, Lemma 1.8 and Theorem 1.9.

For $e = 1$, the pairing of part (1) of Theorem 1.3 holds on T^2 for every Riemannian metric, with no tautness or regularity assumption: an indivisible class contains a simple closed geodesic, and Lusternik–Schnirelmann min-max of the length over a sweepout family of the class, run through the curve shortening flow [30], produces either a second simple closed geodesic of greater length, or infinitely many of the same length. I thank Marco Mazzucchelli for telling me about this.

Question 1. Let g be an $e\beta_0$ -taut metric on T^2 , for an indivisible class β_0 and $e \geq 3$ odd. If the class $e\beta_0$ contains a prime closed geodesic string, must it contain at least two?

“Yes” does not follow formally from the vanishing of F : a single degenerate prime geodesic string, of vanishing local index (Appendix A), is consistent with $F(g, e\beta_0) = 0$. It might follow from some

min-max theory of the S^1 -equivariant pair $(L_{e\beta_0}T^2, I)$, with I the locus of iterated strings — the locus of nontrivial isotropy — in the spirit of the Lusternik–Schnirelmann argument.

1.1. Tautness revisited. The space of closed geodesics in the definition of tautness carries the C^1 topology of the loop space, as in the formal Definition 2.2; on this space it agrees with the compact open topology. A homotopy $\{g_t\}_{t \in [0,1]}$ between β -taut metrics is said to be β -taut if its endpoint geodesics lie in a compact open subset of the space of pairs (o, t) with o a class β g_t -geodesic: it has no sky catastrophe over the class β , Definition 2.4. Compactness of the whole space of pairs suffices. It is **taut** if it is β -taut for every $\beta \in \pi_1^{inc}(X)$; metrics joined by a β -taut homotopy are β -taut homotopic. Examples of metrics β -taut homotopic to a flat metric, arbitrarily “far” from flat, are supplied by Proposition 1.5 below. If X admits a negative curvature metric and is closed, then there are C^1 open sets of taut Riemannian metrics on X , containing metrics of arbitrarily large curvature, all taut homotopic to a negative curvature metric. See Example 3.

Proposition 1.5 (Clairaut metrics on T^n ; proof in Section 12). *Write $T^n = T^{n-1} \times S^1$, with coordinates (x, y) , and let*

$$g = dy^2 + \sum_{i,j} a_{ij}(y) dx_i dx_j,$$

for $a(y)$ symmetric positive definite, so g is invariant under the T^{n-1} action. Let $\beta = (m, 0)$, $m \neq 0$, be a fiber class. Then g is β -taut, β -taut homotopic to a flat metric, and $F(g, \beta) = 0$, with F as in Theorem 2.8.

For $n \geq 3$ there is no KAM confinement (Remark 11.5), but tautness abounds: the proposition gives far from flat metrics, β -taut for every fiber class β , in every dimension. On T^n , $n \geq 2$, $F(g, \beta) = 0$ for every β -taut metric g and every class β , by Theorem 1.2 (2), so for β indivisible and e odd Theorem 1.9 gives the odd class law of that theorem, $s_e = 0$ in its notation, for every $e\beta$ -taut and $e\beta$ -regular metric on T^n . The Clairaut metrics themselves are not regular. For $n \geq 3$ their parallels — the geodesics along which y is constant — come in T^{n-1} -orbits, $(n-2)$ -dimensional families of geodesic strings; for $n = 2$ the closed geodesics other than the parallels come in circles, by the T^1 -symmetry. Small perturbations restore regularity, and for $n = 2$ they stay taut: Proposition 11.2. Theorem 1.13 exploits this.

In the other direction, β -tautness fails easily.

Example 1. Let g agree with a flat metric outside a small embedded disk in T^2 , inside which it is rotationally symmetric with a “mushroom” profile: a bulb over a neck, the circumference having a strict local minimum at a waist circle γ_0 , which is then a closed geodesic. Let $S \subset T^2$ be the complement of the open bulb: a compact surface with geodesic boundary γ_0 , with $\pi_1(S)$ free on x, y and $[x, y]$ the boundary class, killed by the inclusion into T^2 . For $k \in \mathbb{N}$ the elements $x[x, y]^k$ represent pairwise distinct nontrivial conjugacy classes of $\pi_1(S)$ (their cyclically reduced lengths differ), all mapping to the class β of x in T^2 . As ∂S is geodesic, minimizing length in each of these classes produces closed g -geodesics o_k of class β in T^2 ; and since for each L only finitely many free homotopy classes of the compact S contain a loop of length $\leq L$, $\ell_g(o_k) \rightarrow \infty$. So g is not β -taut, and the argument applies to every nontrivial class. The disk may be taken arbitrarily small, so the failure is not curable by localizing the perturbation: the C^{k_0} -closeness in Proposition 11.2 ahead is essential.

Using geodesic confinement based on KAM theory, we show in Section 11 the following:

Theorem 1.6. *Let g_0 be a Riemannian metric on T^2 invariant under a free circle action: a Clairaut metric, the flat metrics included. Any Finsler metric g on T^2 , C^∞ - $\hbar(g_0)$ -close to g_0 , is taut homotopic to g_0 , and hence to a flat metric.*

The arithmetic behind Theorem 1.3 is general, given a cancellation property.

Definition 1.7. An indivisible class $\beta \in \pi_1^{inc}(X)$ of a manifold X satisfies the *cancellation property* if for all $k, e \geq 1$ and every class α , $k\alpha = e\beta$ implies $k|e$ and $\alpha = (e/k)\beta$.

Lemma 1.8 (Proof in Section 9). *Every indivisible class of X satisfies the cancellation property when $\pi_1(X)$ is free abelian, or torsion-free hyperbolic.*

By this lemma the following theorem applies to T^n or a closed orientable manifold admitting a negative curvature metric.

Theorem 1.9 (Arithmetic law, odd classes; proof in Section 9). *Let X be a closed orientable manifold, β an indivisible class, and $e \geq 1$ odd. Let g be an $e\beta$ -taut and $e\beta$ -regular metric on X , and suppose that β satisfies the cancellation property. Let P_d be the set of prime geodesic strings of class $d\beta$, and $s_d := \sum_{o \in P_d} i(o)$ its signed count, with i the local Fuller index (A.2) — defined, as regularity of the divisor classes is automatic (Section 9). Then*

$$(1.10) \quad e F(g, e\beta) = \sum_{d|e} d s_d,$$

and hence, by Möbius inversion,

$$(1.11) \quad s_e = \frac{1}{e} \sum_{d|e} \mu(e/d) d F(g, d\beta),$$

with μ the Möbius function. In particular,

- (1) *If $F(g, d\beta) = 0$ for every $d|e$ then $s_e = 0$, and if the class $e\beta$ contains a prime geodesic string, it contains another.*
- (2) *If $e > 1$ and $F(g, d\beta) = 1/d$ for every $d|e$ then $s_e = 0$, and if the class $e\beta$ contains a prime geodesic string, it contains another. For $e = 1$, (1.10) reads $s_1 = F(g, \beta) = 1$: the class β contains an odd number of prime geodesic strings.*

The arithmetic of this theorem, and of Section 9, has been formalized in Lean 4, over Mathlib: Möbius inversion and both alternatives, the divisor system (9.8), Proposition 9.5 — with the vanishing of F on the divisors as a hypothesis, and its converse — and Corollary 9.10. The geometry enters as named hypotheses: (1.10) at the divisor levels, the covering identity (9.7), and the sign law of Lemma 9.4.²

Remark 1.12. The second alternative is the negative curvature pattern: there $F(g, d\beta) = 1/d$ by Example 4. Unconditionally, Proposition 2.7 provides it on the C^1 -balls about negative curvature metrics. Combined with Conjecture 1 of Section 1.2, it would apply to every $e\beta$ -taut and $e\beta$ -regular metric on a closed orientable manifold admitting a negative curvature metric — the cancellation property holds there unconditionally, as $\pi_1(X)$ is torsion-free hyperbolic (Lemma 1.8). For closed orientable surfaces of genus ≥ 2 this is unconditional, by Theorem 1.2 (2): Corollary 9.10.

In the notation of Theorem 1.9, with $\beta = \beta_0$, write b_e for the number of inverse-hyperbolic strings in P_e . On T^2 , Theorem 1.3 rests on the even class identity

$$s_e = -b_{e/2}, \quad b_{e/2} := 0 \text{ for } e \text{ odd,}$$

of Proposition 9.5, valid for every $e\beta_0$ -taut and $e\beta_0$ -regular metric: for e odd a vanishing sum of ± 1 's has an even number of terms, and for $e = 2n$, $\#P_{2n} \geq |s_{2n}| = b_n$. The inverse-hyperbolic type entering the identity is realized far from the flat metrics, and the identity fires there forcing prime geodesics:

²Lean sources: <https://yashamon.github.io/web2/papers/surfacegroupformalization/>.

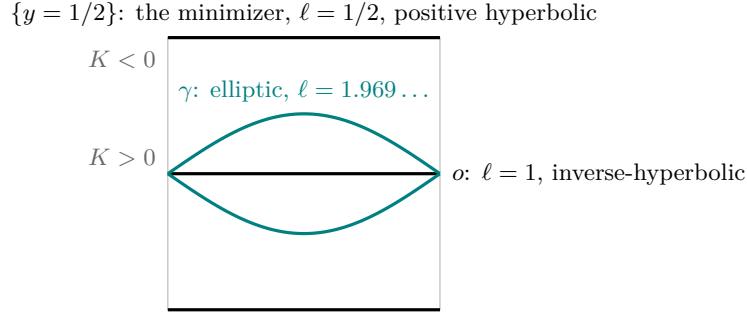


FIGURE 1. Schematic of Theorem 1.13 on the fundamental domain $[0, 1] \times [-\frac{1}{2}, \frac{1}{2}]$, sides identified. The two teal strands form a prime geodesic string γ of class $2\beta_0$: it winds twice in x while oscillating once about o , born of o at a period doubling. Lengths are with respect to g_ϵ at $\epsilon = 5/\pi^2$, where numerically γ is the only prime string of its class (Remark 12.2); the picture is topological.

Theorem 1.13 (Proof in Section 12). *On $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ let*

$$g_\epsilon = dy^2 + A_\epsilon(x, y)^2 dx^2, \quad A_\epsilon(x, y) = 1 - \frac{1}{2}(1 + \epsilon \cos 2\pi x) \sin^2 \pi y, \quad 0 \leq \epsilon < 1,$$

let $g_A := g_{\epsilon=0}$, and let β_0 be the class of the circle $\{y = 0\}$. There are $\delta > 0$ and $\epsilon_0 > 0$ such that:

- (1) *Every metric g with $\|g - g_A\|_{C^\infty} < \delta$ is taut, and taut homotopic to g_A , hence to a flat metric. For $0 \leq \epsilon < \epsilon_0$, g_ϵ is such a metric.*
- (2) *For $0 < \epsilon < \epsilon_0$ the circle $\{y = 0\}$ is an inverse-hyperbolic prime g_ϵ -geodesic string, and $\{y = 1/2\}$ a positive hyperbolic one (positive real eigenvalues), the minimizer of its class.*
- (3) *For $0 < \epsilon < \epsilon_0$ there is a C^∞ -neighborhood U of g_ϵ whose $2\beta_0$ -regular members, C^∞ -generic in U , have a prime geodesic string in class $2\beta_0$.*

The metric $g_A = dy^2 + A(y)^2 dx^2$, $A = \frac{3}{4} + \frac{1}{4} \cos 2\pi y$, is a Clairaut metric with $A_{\max}/A_{\min} = 2$, far from flat: the fat parallel $\{y = 0\}$ has curvature π^2 , the period doubling threshold, where the normal Jacobi fields rotate by π over one period, and $\{y = 1/2\}$ is the skinny one. The perturbation is Hill engineering along the fat parallel, C^∞ -small, and tautness is KAM confinement around g_A . Numerically g_A has no other class β_0 strings; if this holds then, for ϵ and U small, the metrics of (3) have exactly two, with $F(g, \beta_0) = 1 - 1 = 0$, and their $2\beta_0$ -regular members have an odd number of prime class $2\beta_0$ strings, born at the period doubling. See Figure 1 and Section 12.2.

Theorem 1.6 settles the following question near the flat and the Clairaut metrics.

Question 2. Is every β -taut metric on T^2 β -taut homotopic to a flat metric?

1.2. The topology connection. What follows is a special version of Conjecture 2 (appearing further ahead) and generalizes Theorem 1.2 (2). A group is **Wall-Euler finite** if it is of finite homological type in the sense of [23, Chapter IX]: of finite virtual cohomological dimension, with a torsion-free subgroup of finite index whose homology is finitely generated in every degree. Wall's rational Euler characteristic χ_{Wall} [60] is then the Euler characteristic of such a subgroup divided by its index, independent of the choice. A space is Wall-Euler finite if it is aspherical with Wall-Euler finite fundamental group.

Conjecture 1 (Aspherical topological formula for F). *Let X be an orientable aspherical manifold, and let $\beta \in \pi_1^{\text{inc}}(X)$. Suppose that X admits a β -taut metric. Then the Borel construction $(L_\beta X)_{hS^1}$ of Theorem 1.2 is Wall-Euler finite, and for every β -taut metric g ,*

$$F(g, \beta) = \chi_{\text{Wall}}((L_\beta X)_{hS^1}).$$

Equivalently, set $G = \pi_1(X)$, let $C_G(\gamma_\beta)$ be the centralizer of γ_β in G and $Q_\beta = C_G(\gamma_\beta)/\langle \gamma_\beta \rangle$, where $\beta = [\gamma_\beta]$ is the conjugacy class of $\gamma_\beta \in G$. Then Q_β is Wall-Euler finite and

$$F(g, \beta) = \chi_{\text{Wall}}(Q_\beta).$$

The two forms agree. As X is aspherical, $L_\beta X$ is a $K(C_G(\gamma_\beta), 1)$. In the fibration $L_\beta X \rightarrow (L_\beta X)_{hS^1} \rightarrow BS^1$ the generator of $\pi_2(BS^1)$ maps to the orbit loop of a base loop, that is to γ_β , an element of infinite order. So $(L_\beta X)_{hS^1}$ is a $K(Q_\beta, 1)$, and the Wall characteristic of an aspherical space is that of its fundamental group. The conjecture implies that $F(g, \beta)$ is independent of the β -taut metric g , for aspherical manifolds. For background on rational Euler characteristics of groups, we refer to Wall's original note [60] and to Brown's treatment of Euler characteristics of groups in [23, Chapter IX]; see also Brown [22] for Euler characteristics of discrete groups and G -spaces. By Theorem 1.2 (2), the conjecture holds for orientable surfaces and for aspherical manifolds with free abelian fundamental group.

The following is a rather important open question in our context:

Question 3. Suppose that X is closed. If X admits a β -taut metric for some nontrivial β , must X be aspherical?

Remark 1.14. A possible route to a counterexample for Question 3 is a mapping torus of a non-aspherical Z by a diffeomorphism with “hyperbolic behavior”. In the other direction, asphericity is not sufficient: by results of Mess [46] and Belegradek [13, Corollary 5.1], in every dimension $n \geq 4$ there are closed aspherical n -manifolds whose fundamental groups contain nontrivial infinitely divisible elements, and hence, by Proposition 1.15 below, classes β for which no β -taut metric exists.

Proposition 1.15. *Let X be a closed manifold and let $\beta \neq 1$ be a free homotopy class which is infinitely divisible in $\pi_1(X)$: for arbitrarily large $n \in \mathbb{N}$, a representative of β in $\pi_1(X)$ is an n -th power. Then X admits no β -taut metric.*

Proof. Take any metric g and let $l > 0$ be the systole of g , that is

$$l = \inf_{\{o \mid o \text{ is noncontractible}\}} \ell_g(o).$$

For n as in the statement, write a representative of β as h_n^n , with h_n necessarily noncontractible since $\beta \neq 1$. A length-minimizing closed geodesic in the free homotopy class of h_n exists as X is closed, and its n -fold cover is a class β closed g -geodesic of length at least $n \cdot l$. So the class β closed geodesics of g have unbounded length and g is not β -taut. $\square$

1.3. Geodesics in mapping tori and negative sectional curvature. A theorem of Preissmann [54] says that there are no negative sectional curvature metrics on compact products. For fibrations, the most obvious analogue of Preissmann's theorem fails even assuming compactness. By Thurston's fibered hyperbolization theorem, a closed surface bundle over S^1 with fiber of negative Euler characteristic is hyperbolic if and only if its monodromy is pseudo-Anosov; see [59, 58]. What follows is a kind of partial, “only if” Finsler analogue in higher dimensions, noting that an isometry of a manifold has “anti-Anosov behavior” (for some reasonable generalization of pseudo-Anosov to higher dimensions).

Recall that $\pi_1^{\text{inc}}(X)$ denotes the set of end incompressible free homotopy classes (Definition 2.1).

Corollary 1.16 (Corollary of Corollary 2.26). *Let X_ϕ be the mapping torus of an isometry $\phi : Z \rightarrow Z$ of a complete taut Riemannian metric on Z , and let $\beta_Z \in \pi_1^{\text{inc}}(Z)$ be indivisible. Suppose that the image β of β_Z under fiber inclusion lies in $\pi_1^{\text{inc}}(X_\phi)$. Then:*

$$(1) \quad \chi((L_\beta X_\phi)_{hS^1}) = 0.$$

- (2) X_ϕ does not admit a forward complete, β -regular Riemann-Finsler metric with an odd number of class β geodesic strings.
- (3) X_ϕ does not admit a complete negative sectional curvature Riemannian metric.

Example 2. The corollary applies to the mapping torus of an isometry of a surface of finite genus at least one, with β_Z the class of a nonseparating simple closed curve (such a curve meets the compact core carrying the genus, as every simple closed curve in a planar surface separates, so the images $\phi^k(c)$ of a representative c , of fixed length, meet one compact set and fall into finitely many classes; the fiber class β then lies in $\pi_1^{inc}(X_\phi)$ by the paragraph before Example 6; we also need a suitable taut metric e.g. a nonpositive curvature metric, with an isometry). The conclusion that X_ϕ does not admit a complete negative sectional curvature metric follows from Thurston's classification³. However the arithmetic constraints on geodesic counts seem interesting even for a surface map.

If we further assume that Z is aspherical, the first part of Corollary 1.16 can be proved via group cohomology theory, for which we already provided some references. But the main point is that in our setup the proof is elementary, once the general product formula of Theorem 7.1 for the F -count is established. The proof of the latter does involve a novel type of perturbation, but Fuller index theory allows us to do this abstractly, perturbing a vector field rather than the metric/energy functional.

Organization of the paper. Section 2 sets up taut metrics and taut deformations, states the supporting theorems, and contains the applications to sky catastrophes. Section 3 proves the compactness results underlying tautness, in particular Theorem 2.5. Section 4 collects Reeb-flow preliminaries, and Section 5 constructs the invariant F and proves Theorem 2.8. Section 6 proves Theorem 1.2: the Euler characteristic formula for indivisible classes, and then the divisible classes on tori and surfaces by covering transfer (Section 6.1). Sections 7 and 8 prove the product and fibration formulas. Section 9 proves Theorem 1.9, the arithmetic law of Proposition 9.5 and Theorem 1.3. Section 10 proves the C^1 stability Proposition 2.7 via Gromov hyperbolicity, and Section 11 proves the near-Clairaut rigidity Proposition 11.2 by KAM confinement. Section 12 proves the Clairaut Proposition 1.5, Theorem 1.13 and Corollary 1.4. Appendix A develops the Fuller index and its continuation theory in the noncompact setting, Appendix B performs a local fibered computation of F used in the product formula, and Appendix C recalls the sky catastrophes behind Definition 2.4 and the facts about sky catastrophes used in Section 2.2.

2. SETUP AND SUPPORTING THEOREMS

Proofs of many results here are postponed till subsequent sections.

Terminology 1. From now on, all our metrics are Riemann-Finsler (a.k.a. Finsler) metrics unless specified to be Riemannian, and usually denoted by just g . Completeness always means forward completeness, and it is an assumption for all our metrics. Curvature always means sectional curvature in the Riemannian case and flag curvature in the Finsler case. Thus we will usually just say complete metric g , for a forward complete Riemann-Finsler metric. A reader may certainly choose to interpret all metrics as Riemannian metrics, completeness as standard completeness, and curvature as sectional curvature.

In what follows, $\pi_k(X)$ denotes the usual based homotopy group, with basepoints suppressed when no confusion can arise.

Definition 2.1. Let X be a smooth manifold. Fix an exhaustion by nested compact sets $\bigcup_{i \in \mathbb{N}} K_i = X$, $K_i^\circ \supset K_{i-1}$ for all $i \geq 1$. A free homotopy class β of loops in X is called **end compressible** if, for every i , it has a representative contained in $X - K_i$. We say that β is **end incompressible** (or incompressible to the ends) if it is not end compressible.

³Suitably adjusted, one may also need finite geometry (volume) assumptions.

Let $\pi_1^{inc}(X)$ denote the set of such end incompressible free homotopy classes. When X is compact, we set $\pi_1^{inc}(X)$ to be the set of nontrivial free homotopy classes of loops in X .

It is easily seen that the above is well defined (independent of the choice of an exhaustion) and moreover any homeomorphism $X_1 \rightarrow X_2$ of a pair of manifolds induces a set isomorphism $\pi_1^{inc}(X_1) \rightarrow \pi_1^{inc}(X_2)$.

Denote by $L_\beta X$ the class $\beta \in \pi_1^{inc}(X)$ component of the free loop space of X , with the C^1 topology. Let g be a complete metric on X , and let $S(g, \beta) \subset L_\beta X$ denote the subspace of all constant speed parametrized, smooth, closed g -geodesics in class β .

Definition 2.2. We say that a metric g on X is β -**taut** if it is complete and $S(g, \beta)$ is compact. We will say that g is **taut** if it is β -taut for each $\beta \in \pi_1^{inc}(X)$.

Lemma 2.3. *A complete Riemannian metric g with nonpositive curvature satisfies:*

- *All of its closed geodesics are minimizing in their free homotopy class.*
- *It is taut.*

Proof. The first part is a standard consequence of the Cartan-Hadamard theorem. The second part follows by the first part and Corollary 3.3. $\square$

Definition 2.4. Let $\beta \in \pi_1^{inc}(X)$, and let g_0, g_1 be a pair of β -taut metrics on X . Let $\{g_t\}$, $t \in [0, 1]$, be a family of complete metrics on X with these endpoints, continuous in the topology of C^1 convergence on compact sets, and set

$$S(\{g_t\}, \beta) := \{(o, t) \in L_\beta X \times [0, 1] \mid o \in S(g_t, \beta)\},$$

with endpoint slices $S(g_0, \beta) \times \{0\}$ and $S(g_1, \beta) \times \{1\}$. The family is a β -**taut deformation** between g_0, g_1 if its endpoint slices are contained in a compact open subset of $S(\{g_t\}, \beta)$. This holds in particular when $S(\{g_t\}, \beta)$ is compact. We also say in this case that g_i are β -taut homotopic.⁴ We say that $\{g_t\}$ is a **taut deformation** if it is β -taut for each $\beta \in \pi_1^{inc}(X)$.

In the language of Appendix C, Definition C.2, $\{g_t\}$ is a β -taut deformation exactly when the associated family of Reeb vector fields has no sky catastrophe over the class β : tautness of a deformation is the absence of a sky catastrophe. A useful criterion for β -tautness is the following.

Theorem 2.5 (Proof in Section 3.1). *Let $\beta \in \pi_1^{inc}(X)$. Let $\{g_t\}_{t \in [0, 1]}$ be a family of complete metrics on X , continuous as in Definition 2.4. Suppose that:*

$$\sup_t \left| \sup_{o \in S(g_t, \beta)} \ell_{g_t}(o) - \inf_{o \in S(g_t, \beta)} \ell_{g_t}(o) \right| < \infty,$$

where ℓ_{g_t} is the length functional with respect to g_t , then $S(\{g_t\}, \beta)$ is compact; in particular the endpoints are β -taut and $\{g_t\}$ is a β -taut deformation.

For example, the hypothesis is trivially satisfied if g_t have the property that all their class β closed geodesics are minimizing in their homotopy class.

Corollary 2.6. *If $\{g_t\}$, $t \in [0, 1]$, is a family of complete Riemannian metrics of nonpositive curvature, continuous as in Definition 2.4, then it is a taut deformation.*

Proof. This follows by Theorem 2.5 and by Lemma 2.3. $\square$

⁴Let us point out that this does not necessarily hold if g_i are path connected by a path of β -taut metrics.

The following supplies more examples of tautness, and the tautness input of Corollary 1.4: strictly negative curvature gives tautness an open C^1 neighborhood in the space of Riemannian metrics.

Proposition 2.7 (C^1 stability of tautness in negative curvature; proof in Section 10). *Let g_- be a metric of negative sectional curvature on a closed orientable manifold X . There is $\epsilon = \epsilon(g_-) > 0$ such that every Riemannian metric g' with $\|g' - g_-\|_{C^1} < \epsilon$ is taut. Any two metrics in this ball are joined by a taut deformation. In particular $F(g', \beta) = F(g_-, \beta)$ for every nontrivial β , with F as in Theorem 2.8.*

C^1 -smallness permits arbitrarily large curvature.

Example 3. On a closed hyperbolic surface (Σ, g_-) take $g' = e^{2u}g_-$, with $u = \delta w \phi(d_{g_-}(\cdot, x_0)/w)$ for a fixed even bump function ϕ with a nondegenerate maximum at 0. Then $\|u\|_{C^1} \leq C\delta$, while $K(g') = e^{-2u}(-1 - \Delta u)$ is of order δ/w at x_0 . For w small this produces metrics in the ball of Proposition 2.7 with curvature maxima as large as desired — so C^2 -far from every metric of nonpositive curvature — and any number of such bumps may be placed. The same bumps on a negatively curved (X, g_-) of any dimension give sectional curvature of order δ/w in every plane at x_0 , by the conformal curvature formula at a critical point of u . These metrics are taut with $F(g', \beta) = 1/n$, for β represented by a multiplicity n closed g_- -geodesic, by Proposition 2.7 and Example 4. In particular the $1/d$ ray values of Remark 1.12 hold for them unconditionally, and, for β indivisible, Theorem 1.9 applies to their $e\beta$ -regular members, e odd; on Σ so does Theorem 1.3. The mushroom of Example 1, planted on Σ , shows that smallness of the support does not suffice.

2.1. The geodesic string counting invariant F . Let $\mathcal{G}(X)$ be the set of complete metrics on X . The following is proved in Section 5.

Theorem 2.8. *For each orientable manifold X there is a functional:*

$$F : \mathcal{G}(X) \times \pi_1^{inc}(X) \rightarrow \mathbb{Q} \sqcup \{\infty\},$$

extending the functional (1.1).

- (1) $F(g, \beta) \neq \infty$ if and only if g is β -taut.
- (2) If g is β -taut and β -regular, then it has finitely many class β geodesic strings, and $F(g, \beta)$ is given by (1.1).
- (3) If g_0, g_1 are joined by a β -taut deformation, then $F(g_0, \beta) = F(g_1, \beta)$.
- (4) If $\phi : X' \rightarrow X$ is a diffeomorphism, then $F(\phi^*g, \beta') = F(g, \phi_*\beta')$ for every $\beta' \in \pi_1^{inc}(X')$.

For a β -taut g , $F(g, \beta)$ is defined in terms of the Fuller index.

Corollary 2.9. *Let $\beta \in \pi_1^{inc}(X)$, and let g_0, g_1 be β -taut metrics on X with*

$$F(g_0, \beta) \neq F(g_1, \beta).$$

Then there is no β -taut deformation between g_0 and g_1 : every family $\{g_t\}$ of complete metrics joining them, continuous as in Definition 2.4, has a sky catastrophe over the class β (Appendix C).

Proof. This is the contrapositive of Theorem 2.8(3). □

I was unable to find such a pair g_0, g_1 , in fact there is strong evidence to believe the following:

Conjecture 2. $F(g, \beta)$ does not depend on the choice of a smooth structure and of a β -taut metric g . In other words we have the following. Let X be an orientable topological manifold, define:

$$\mathcal{S}(X) : \pi_1^{\text{inc}}(X) \rightarrow \mathbb{Q} \sqcup \{\infty\}$$

by:

$$\mathcal{S}(X)(\beta) = \begin{cases} \infty, & \text{if } X \text{ does not admit a smooth structure and a } \beta\text{-taut Finsler metric.} \\ F(g, \beta), & \text{if } g \text{ is a } \beta\text{-taut Finsler metric on } X. \end{cases},$$

then $\mathcal{S}$ is well-defined and hence determines a topological invariant of topological manifolds. So if $f : X_1 \rightarrow X_2$ is a homeomorphism then $\mathcal{S}(X_1)(\beta) = \mathcal{S}(X_2)(f_*(\beta))$.

Theorem 1.2 (1) proves the conjecture on indivisible classes. Indeed, the Borel construction $(L_\beta X)_{hS^1}$ depends only on the pair (X, β) up to homotopy equivalence, and a homeomorphism $f : X_1 \rightarrow X_2$ induces one for (X_1, β) and $(X_2, f_*\beta)$. So for indivisible β , $\mathcal{S}(X)(\beta)$ is well defined: it is $\chi((L_\beta X)_{hS^1})$ if some smooth structure on X carries a β -taut metric, and ∞ otherwise. Theorem 1.2 (2) proves the conjecture on surfaces, and on aspherical manifolds with free abelian fundamental group: the value 0 or $1/e$ depends only on X and on the divisibility e of β (Theorems 6.8 and 6.12).

Remark 2.10. The smooth manifold version of this conjecture, i.e. that $\mathcal{S}(X)$ is a smooth manifold invariant, is implied by the Reeb sky catastrophe conjecture described in the next subsection. However, the above seems to be much more basic as will be apparent from the proof of Theorem 1.2.

2.2. Applications to sky catastrophes.

Definition 2.11 (Preliminary). A **sky catastrophe** in class β for a smooth family $\{V_t\}$, $t \in [0, 1]$, of nonvanishing vector fields on a closed manifold M is a continuous family of closed orbit strings in class β , $\tau \mapsto o_{t_\tau}$, $\tau \in [0, \infty)$, where o_{t_τ} is an orbit string of V_{t_τ} (orbit strings are defined in Appendix A), with $t_0 \in \{0, 1\}$, such that the period of o_{t_τ} is unbounded from above.

Sky catastrophes first appeared in the work of Fuller [28]. One general definition appears in [57], which we review in the Appendix C, the main point of this definition is that there are no regularity assumptions on $\{V_t\}$. The preliminary definition becomes equivalent to the more general definition given certain regularity conditions on the family $\{V_t\}$; the anchoring $t_0 \in \{0, 1\}$ is what the general definition keeps.

The following corollary of Theorem 2.5 may be of independent interest.

Corollary 2.12. *Sky catastrophes of vector fields on closed manifolds are not geodesible by metrics all of whose geodesics are minimizing. That is, if a family $\{V_t\}$ has a sky catastrophe, there is no family $\{g_t\}$ of metrics (all of whose geodesics are minimizing in their homotopy class) such that the orbits of V_t are unit speed parametrized g_t -geodesics.*

Fuller at the end of [29] has asked for any metric conditions on vector fields to rule out sky catastrophes. By the above, nonpositivity of curvature is one such condition. So this is a partial answer to his question.

The main conjecture in [57] is that geodesible, and more generally Reeb sky catastrophes ($\{V_t\}$ are Reeb) are unstable, in the sense that there is a C^0 nearby family $\{V'_t\}$, with the same endpoints, which does not have a sky catastrophe in the fixed class β . Let's call this **Reeb sky catastrophe conjecture**. The reason to conjecture such a thing is that [57, Theorem 1.10] says that if $\{o_{t_\tau}\}_\tau$ is a sky catastrophe for a family of Reeb vector fields $\{V_t\}$ then the length of the corresponding projection $\tau \mapsto t_\tau$ is infinite. (Assuming some minor regularity on $\{o_{t_\tau}\}$.) The latter suggests that the structure of such a sky catastrophe is very fragile.

Remark 2.13. The Reeb sky catastrophe conjecture implies the geodesible Seifert conjecture (apparently still an open problem), by the main result of [57, Theorem 1.5]. Hence, this is a subtle question.

Corollary 2.9 is a version of [57, Theorem 3.2]; the tautness hypothesis permits the short self-contained proof of Section 5. So if a pair g_0, g_1 as in the corollary exists, the Reeb sky catastrophe conjecture is disproved, and in a rather extreme fashion.

2.3. Basic results on the invariant F .

Definition 2.14. Let β be a free homotopy class of loops in X . Choose a representative of β and a base point x_0 on its image. This choice determines an element

$$\beta_{x_0} \in \pi_1(X, x_0)$$

well-defined up to inner automorphism.

- An element $\gamma \in \pi_1(X, x_0)$ is **indivisible** if whenever $\gamma = \alpha^k$ for some $\alpha \in \pi_1(X, x_0)$ and $k > 0$, then $k = 1$.
- An element $\gamma \in \pi_1(X, x_0)$ is a **k -power** if $\gamma = \alpha^k$, $k > 1$, for some indivisible α .
- An element $\gamma \in \pi_1(X, x_0)$ is **atomic** if $\gamma = \alpha^k$, for some $k \geq 1$, with α indivisible.

We say that the free homotopy class β is indivisible, respectively a k -power, respectively atomic, if β_{x_0} has the corresponding property for one, equivalently any, such choice of x_0 and representative. This is well-defined because the corresponding notions are invariant under inner automorphism. Equivalently, β is a k -power precisely when it admits a smooth representative with multiplicity k whose underlying prime representative represents an indivisible class.

Note that a class $\beta \in \pi_1^{inc}(X)$ is indivisible iff any representative of this class is prime.

Example 4. Let g be a Riemannian metric with negative sectional curvature on a closed orientable manifold X and $\beta \in \pi_1^{inc}(X)$ a class represented by a multiplicity n closed geodesic, then

$$(2.15) \quad F(g, \beta) = \frac{1}{n}.$$

In particular, if β is indivisible then $F(g, \beta) = 1$. More generally, (2.15) holds whenever g has a unique and nondegenerate geodesic string in class β , of even Morse index, where nondegenerate is as in the Introduction.

If $\beta \in \pi_1^{inc}(X)$ is indivisible, then it is easy to see that the reparametrization S^1 action on $L_\beta X$ is free (Section 6), so that $(L_\beta X)_{hS^1} \simeq L_\beta X / S^1$. Moreover, part (1) of Theorem 1.2 computes F in this situation; it is proved in Section 6. Taut metrics distinct from nonpositive curvature metrics can be constructed explicitly in every dimension, recall Proposition 1.5, Example 3 and Theorem 1.13.

We can use the above, and the product formula of Theorem 7.1 to get:

Theorem 2.16 (Proof in Section 8). *Every rational number has the form $F(g, \beta)$ for some β -taut Riemannian g on some compact manifold X and for some β .*

2.4. Applications to existence of negative curvature metrics. We are going to give a certain generalization of Preissmann's theorem to fibrations, with possibly noncompact fibers, also replacing the negative sectional curvature condition by a significantly weaker condition. Though the definitions may at first seem hard to understand, we just want enough generality to incorporate interesting “flat bundle” examples (with anti-Anosov or periodic holonomy behavior). The latter are of interest in 3-manifold theory, for instance vis-à-vis Thurston's classification [58].

Definition 2.17. For a fiber bundle $Z \hookrightarrow X \rightarrow Y$, we say that $\beta \in \pi_1^{\text{inc}}(X)$ is a **fiber class** if

$$\beta = (i_Z)_*(\alpha) \quad \text{for some } \alpha \in \pi_1^{\text{inc}}(Z),$$

where i_Z is the inclusion of a fiber and $(i_Z)_*$ the induced map on free homotopy classes.

Lemma 2.18. *Let*

$$Z \hookrightarrow X \xrightarrow{p} Y$$

be a smooth fiber bundle with connected fiber and connected base. Assume

$$\pi_2(Y) = 0.$$

Fix $y_0 \in Y$, and let

$$H = \pi_1(Y, y_0)$$

act on $\pi_1^{\text{inc}}(Z_{y_0})$ by holonomy. Let $\beta \in \pi_1^{\text{inc}}(X)$ be a fiber class, and define

$$\mathcal{R}_{y_0}(\beta) = \{ \alpha \in \pi_1^{\text{inc}}(Z_{y_0}) \mid (i_{y_0})_*(\alpha) = \beta \}.$$

Then $\mathcal{R}_{y_0}(\beta)$ is a single H -orbit. In particular, if

$$\beta_Z, \beta'_Z \in \mathcal{R}_{y_0}(\beta),$$

then

$$H \cdot \beta_Z = H \cdot \beta'_Z.$$

Thus the holonomy orbit associated to the fiber class β is independent of the choice of fiber representative β_Z .

Proof. Choose basepoints $z_0 \in Z_{y_0}$ and $x_0 = i_{y_0}(z_0)$. Since $\pi_2(Y) = 0$, the homotopy long exact sequence gives an exact sequence of based groups

$$1 \rightarrow \pi_1(Z_{y_0}, z_0) \xrightarrow{(i_{y_0})_*} \pi_1(X, x_0) \xrightarrow{p_*} \pi_1(Y, y_0) \rightarrow 1.$$

Thus we identify $\pi_1(Z_{y_0}, z_0)$ with a normal subgroup of $\pi_1(X, x_0)$, and the quotient is $\pi_1(Y, y_0)$.

Let $\alpha, \alpha' \in \mathcal{R}_{y_0}(\beta)$. Choose based representatives $a, a' \in \pi_1(Z_{y_0}, z_0)$ of these fiber free homotopy classes. Since they determine the same total free homotopy class β in X , the corresponding elements of $\pi_1(X, x_0)$ are conjugate. Thus, after conjugating inside $\pi_1(Z_{y_0}, z_0)$ if necessary, there exists $x \in \pi_1(X, x_0)$ such that

$$a' = xax^{-1}.$$

Let

$$h = p_*(x) \in H.$$

Conjugation by x on the normal subgroup $\pi_1(Z_{y_0}, z_0)$ induces the holonomy action of h , up to an inner automorphism of $\pi_1(Z_{y_0}, z_0)$. Passing to free homotopy classes in the fiber, we get

$$\alpha' = h \cdot \alpha.$$

Thus all fiber representatives of the total free homotopy class β belong to a single holonomy orbit.

Conversely, if $\alpha' = h \cdot \alpha$, choose $x \in \pi_1(X, x_0)$ with $p_*(x) = h$. Then xax^{-1} represents the same total free homotopy class in X as a , and its fiber free homotopy class is $h \cdot \alpha$. Hence the holonomy orbit is exactly the set of fiber representatives of β . $\square$

We assume from now on that Z, Y are connected. For a fiber class $\beta \in \pi_1^{\text{inc}}(X)$, set

$$\mathcal{R}(\beta) = \{ \alpha \in \pi_1^{\text{inc}}(Z) \mid (i_Z)_*(\alpha) = \beta \}.$$

Assume that $\pi_2(Y) = 0$, and put

$$H = \pi_1(Y, y_0).$$

By Lemma 2.18, the holonomy orbit

$$S = H \cdot \beta_Z$$

is independent of the choice of the fiber representative $\beta_Z \in \mathcal{R}(\beta)$. If another fiber inclusion is chosen, the corresponding holonomy actions are abstractly conjugate.

Definition 2.19. Let

$$Z \hookrightarrow X \xrightarrow{p} Y$$

be a smooth fiber bundle, with Z, Y connected and $\pi_2(Y) = 0$, and let $\beta \in \pi_1^{\text{inc}}(X)$ be a fiber class. We say that p is **β -holonomy-finite** if for one, equivalently any, $\beta_Z \in \mathcal{R}(\beta)$, the orbit

$$S = \pi_1(Y, y_0) \cdot \beta_Z$$

is finite.

Definition 2.20. Let

$$Z \hookrightarrow X \xrightarrow{p} Y$$

be a smooth fiber bundle, and let g be a Riemannian metric on X . Let

$$T^{\text{vert}}X = \ker(dp)$$

and let

$$\pi^{\text{vert}} : TX \rightarrow T^{\text{vert}}X$$

be the g -orthogonal projection. We say that p is **g -parallel** if

$$\nabla \pi^{\text{vert}} = 0.$$

Equivalently,

$$\nabla_A(\pi^{\text{vert}}B) = \pi^{\text{vert}}(\nabla_A B)$$

for all vector fields A, B on X .

Definition 2.21. Let $\beta \in \pi_1^{\text{inc}}(X)$ be a fiber class, for a smooth fibration

$$p : Z \hookrightarrow (X, g) \rightarrow (Y, g_Y).$$

We say that this is a **β -periodic fibration** if:

- (1) g is a complete Riemannian metric.
- (2) Z, Y are connected and $\pi_2(Y) = 0$.
- (3) p is g -parallel and is a Riemannian submersion.
- (4) p is β -holonomy-finite.
- (5) the fiber metrics g_y on Z_y are taut.

The metrics g, g_Y may be kept implicit. In the above definition of a β -periodic fibration and the hypothesis of the following theorem we need the auxiliary metric g on X to be Riemannian, and there is no obvious extension of the theorem to the Riemann-Finsler case. However, the conclusions of the theorem are for Riemann-Finsler metrics. The following is perhaps the main technical result of the paper, proved in Section 8.

Theorem 2.22. *Let $p : Z \hookrightarrow (X, g) \rightarrow (Y, g_Y)$ be a β -periodic fibration, where $\beta \in \pi_1^{\text{inc}}(X)$ is a fiber class. Suppose further that Y is connected, closed, $\chi(Y) \neq \pm 1$ and is such that all smooth closed contractible g_Y -geodesics in Y are constant. Then the following holds:*

- X does not admit a complete Riemannian metric with negative curvature.
- X does not admit a forward complete Riemann-Finsler metric with a unique nondegenerate class β geodesic string.

Note that $\chi(Y) \neq 1$ is of course essential, as the trivial fibration $X \rightarrow \{pt\}$, with X admitting a complete negatively curved metric, will satisfy the hypothesis. The condition that there is a fiber class $\beta \in \pi_1^{inc}(X)$ is also essential; as any vector bundle over a closed manifold admitting a Riemannian metric of negative curvature also admits a metric of negative curvature, Anderson [4].

Corollary 2.23 (Proof in Section 8). *Let Y, Z be connected smooth manifolds, let $\beta_Z \in \pi_1^{inc}(Z)$, and assume that $\chi(Y) \neq \pm 1$. Suppose that Y is closed and admits a Riemannian metric all of whose contractible closed geodesics are constant, for instance a nonpositive curvature metric, and suppose that Z admits a taut Riemannian metric. Let*

$$X = Y \times Z$$

with projection $X \rightarrow Y$, let $i : Z \rightarrow X$ be a fiber inclusion, and set

$$\beta = i_*(\beta_Z).$$

Then:

- *X does not admit a complete Riemannian metric of negative sectional curvature.*
- *X does not admit a forward complete Riemann-Finsler metric with a unique and nondegenerate class β geodesic string.*

Example 5. Take Y any nonpositively curved closed manifold with $\chi(Y) \neq \pm 1$, and Z a surface of genus at least one, possibly noncompact, even of infinite genus.

This theorem is very close to being sharp, for example the conclusion of the corollary is false if $Y = S^1$ and $Z = S^1 \times \mathbb{R}$. As $X = T^2 \times \mathbb{R}$ admits the warped product metric $g_{\mathbb{R}} \times_{e^t} g_{T^2}$ (with respect to the function $f = e^t$ on $\mathbb{R}$) where $g_{T^2}, g_{\mathbb{R}}$ are the flat metrics. This warped product has constant negative sectional curvature -1 . So it is essential that not only $\pi_1(Z)$ be nontrivial but also that there is a class incompressible to the ends. Of course, $\chi(Y) \neq 1$ is also essential, otherwise we may take $Y = pt$. The condition $\chi(Y) \neq -1$ is however not obviously essential.

A further basic set of examples for the theorem is obtained by starting with a Riemannian manifold (Y, g_Y) , and any homomorphism

$$(2.24) \quad \phi : \pi_1(Y, y_0) \rightarrow \text{Isom}(Z, g_Z), \quad (\text{the group of all isometries}),$$

where g_Z is a taut Riemannian metric and $\beta_Z \in \pi_1^{inc}(Z)$. (For example (Z, g_Z) is a non-simply connected complete hyperbolic surface of genus at least one). Suppose further:

- (1) The orbit

$$\text{Orb}(\beta_Z) = \{\phi(\gamma)_*(\beta_Z) \mid \gamma \in \pi_1(Y, y_0)\}$$

is finite.

- (2) Y is closed and connected, with $\pi_2(Y) = 0$ and $\chi(Y) \neq \pm 1$, and Z is connected.

- (3) All contractible smooth closed g_Y geodesics in Y are constant.

We have the obvious induced diagonal action

$$\pi_1(Y, y_0) \rightarrow \text{Diff}(Z \times \tilde{Y}), \quad (\text{the group of all diffeomorphisms}),$$

$$\gamma \mapsto ((z, y) \mapsto (\phi(\gamma)(z), \gamma \cdot y)),$$

for $\tilde{Y}$ the universal cover of Y . Taking the quotient of $Z \times \tilde{Y}$ by this action, we get an associated “flat” bundle $Z \hookrightarrow X_\phi \xrightarrow{p} Y$, with a metric g_ϕ induced from the product metric $\tilde{g} = g_Z \oplus \tilde{g}_Y$, with $\tilde{g}_Y$ the lift of g_Y , on the covering space $q : Z \times \tilde{Y} \rightarrow X_\phi$. The natural projection $p : X_\phi \rightarrow Y$ is then a Riemannian submersion and is g_ϕ -parallel. Let β be the class of X_ϕ represented by the loops of β_Z in a fiber. It is end incompressible. Indeed, the isometries $\phi(\gamma)$ permute the finitely many classes of $\text{Orb}(\beta_Z)$, and so preserve the compact nonempty set of closed g_Z -geodesics in these classes, nonempty by Lemma 3.4

and compact by tautness; hence they lie in a compact group of isometries. Each class of $\text{Orb}(\beta_Z)$ is end incompressible, so some compact $K_Z \subset Z$ meets all of their representatives, and K_Z may be taken invariant under that compact group. A loop of class β lifts to $Z \times \tilde{Y}$, with Z -projection of a class in $\text{Orb}(\beta_Z)$; after a deck translation it meets $K_Z \times D$, D a compact fundamental domain of Y . So every class β loop meets the compact set $q(K_Z \times D)$. The condition on the finiteness of Orb implies that p is β -holonomy finite. It follows that p is a β -periodic fibration. Hence $p : (X_\phi, g_\phi) \rightarrow (Y, g_Y)$ satisfies the hypothesis of Theorem 2.22.

Example 6. Suppose we have $\beta_Z \in \pi_1^{\text{inc}}(Z)$, and let $\phi : Z \rightarrow Z$ be an isometry of a complete taut Riemannian metric g_Z on a connected Z , such that $\text{Orb}(\beta_Z)$ is finite. Then for $\beta = (i_Z)_*(\beta_Z)$, by the construction above, the mapping torus

$$(Z, g_Z) \hookrightarrow (X_\phi, g_\phi) \xrightarrow{\pi} S^1$$

has the structure of a β -periodic fibration, satisfying the hypothesis of Theorem 2.22.

The finiteness of the orbit is automatic in the situation of interest.

Lemma 2.25. *Let (Z, g_Z) be a complete Riemannian manifold, ϕ an isometry of Z , $\beta_Z \in \pi_1^{\text{inc}}(Z)$, and β the class of X_ϕ represented by the loops of β_Z in a fiber. If $\beta \in \pi_1^{\text{inc}}(X_\phi)$, then $\text{Orb}(\beta_Z)$ is finite.*

Proof. Write $X_\phi = Z \times \mathbb{R} / ((z, s) \sim (\phi(z), s + 1))$. Choose a loop c representing β_Z , of length L . The loop $\phi^k(c)$ has length L and represents $\phi_*^k(\beta_Z)$. In X_ϕ the loops $\phi^k(c) \times \{s\}$, s between 0 and k , form a free homotopy from $\phi^k(c) \times \{0\}$ to $\phi^k(c) \times \{k\} = c \times \{0\}$. So every $\phi^k(c) \times \{0\}$ represents β .

Let $K \subset X_\phi$ be compact. Then $K_Z := \{z \mid (z, 0) \in K\}$ is compact in Z , and a loop of length at most L meeting K_Z lies in the closed L -neighborhood $N_L(K_Z)$, compact as Z is complete. A compact domain of a Riemannian manifold, after a harmless thickening of $N_L(K_Z)$, contains only finitely many free homotopy classes with representatives of length at most L : choose a finite good cover with Lebesgue number $\delta > 0$; a loop of length at most L is encoded by an edge loop in the nerve of the cover, of combinatorial length bounded by $N(L, \delta)$, and there are finitely many such combinatorial loops. So if the orbit is infinite, the loops $\phi^k(c) \times \{0\}$ meet K for finitely many k only. Applied to an exhaustion of X_ϕ by compact sets, this makes β end compressible. $\square$

Corollary 2.26. *Let*

$$(Z, g_Z) \hookrightarrow (X_\phi, g_\phi) \rightarrow S^1$$

be the mapping torus of an isometry of a connected complete taut Riemannian manifold (Z, g_Z) . Let $\beta \in \pi_1^{\text{inc}}(X_\phi)$ be the image of some $\beta_Z \in \pi_1^{\text{inc}}(Z)$ under fiber inclusion, as previously.

- (1) X_ϕ does not admit a complete Riemannian metric with negative curvature.
- (2) X_ϕ does not admit a forward complete Riemann-Finsler metric with a unique nondegenerate class β geodesic string.
- (3) If β is indivisible, $\chi((L_\beta X_\phi)_{hS^1}) = 0$ and so X_ϕ does not admit a β -regular forward complete Finsler metric with an odd number of β -class geodesic strings.

3. COMPACTNESS PRELIMINARIES

Lemma 3.1. *Suppose that g is a complete metric on X , $\beta \in \pi_1^{\text{inc}}(X)$ and let $S \subset L_\beta X$ be a subset on which the g -length functional is bounded from above. Then the images in X of elements of S are contained in a fixed compact subset of X .*

Proof. Suppose otherwise. Fix an exhaustion by nested compact sets

$$\bigcup_{i \in \mathbb{N}} K_i = X, \quad K_i^\circ \supset K_{i-1}.$$

Then either there is a sequence $\{o_i\}_{i \in \mathbb{N}}$, $o_i \in S$ such that $o_i \in K_i^c$, for K_i^c the complement of K_i , which contradicts the fact that β is end incompressible. Or there is a sequence $\{o_k\}_{k \in \mathbb{N}}$, $o_k \in S$ such that:

- (1) Each o_k intersects K_{i_0} for some i_0 fixed.
- (2) For each $i \in \mathbb{N}$ there is a $k_i > i$ such that o_{k_i} is not contained in K_i .

Now if $\text{diam}(o_k)$ is bounded in k , then condition 1 implies that o_k are contained in a set of bounded diameter. (Here $\text{diam}(o_k)$ denotes the diameter of image o_k .) Consequently, by Hopf-Rinow theorem [9], o_k are contained in a compact set. But this contradicts condition 2, and the fact that K_i form an exhaustion of X .

Thus, we conclude that $\text{diam}(o_k)$ is unbounded, but this contradicts the hypothesis. $\square$

Lemma 3.2. *Let g be a complete metric on X , and let $\beta \in \pi_1^{\text{inc}}(X)$. Denote by $S_A(g, \beta) \subset S(g, \beta)$ the subset of elements with g -length at most A . Then $S_A(g, \beta) \subset S(g, \beta)$ is compact.*

Proof. By Lemma 3.1, the images of elements of $S_A(g, \beta) \subset S(g, \beta)$ are contained in compact K . Since elements of $S_A(g, \beta) \subset S(g, \beta)$ are parametrized by constant speed, and since we have a length bound $S_A(g, \beta)$ is an equicontinuous collection. The result then follows by the Arzelà–Ascoli theorem and standard arguments. $\square$

Corollary 3.3. *Suppose that g is a complete metric on X , all of whose class $\beta \in \pi_1^{\text{inc}}(X)$ geodesics are minimizing, then g is β -taut.*

Lemma 3.4. *Let g be a complete metric on X , and let $\beta \in \pi_1^{\text{inc}}(X)$. Then there exists a closed g -geodesic minimizing length among all loops in the free homotopy class β .*

Proof. Suppose otherwise and let

$$A = \inf_{\gamma \in L_\beta X} \ell_g(\gamma).$$

By the hypothesis that there is no geodesic with length A , and by Lemma 3.2, there is an $\epsilon > 0$ such that there is no closed geodesic o with $A \leq \ell_g(o) < A + \epsilon$.

By Lemma 3.1 the subset $M_{\beta, A} \subset L_\beta X$ consisting of elements with length less than $A + \epsilon$ is contained in a compact K . Pick $o \in M_{\beta, A}$ and apply the curve shortening process to it. Let o_k , $k \in \mathbb{N}$ be the stages of this process. Then image $o_k \subset K$. In particular $\{o_k\}$ must converge to a geodesic, for instance [47] (we can also argue via Arzelà–Ascoli to prove convergence) but this contradicts the condition on ϵ . $\square$

3.1. Proof of Theorem 2.5. Let $\{g_t\}$, $t \in [0, 1]$ be as in the hypothesis, with

$$(3.5) \quad \sup_t \left| \sup_{o \in S(g_t, \beta)} \ell_{g_t}(o) - \inf_{o \in S(g_t, \beta)} \ell_{g_t}(o) \right| < C,$$

and suppose that

$$\sup_{(o, t) \in S(\{g_t\}, \beta)} \ell_{g_t}(o) = \infty.$$

Then we have a sequence $\{o_k\}$, $k \in \mathbb{N}$, of closed class β g_{t_k} -geodesics in X , satisfying:

- (1) $\lim_{k \rightarrow \infty} t_k = t_\infty \in [0, 1]$.
- (2) $\lim_{k \rightarrow \infty} \ell_{g_{t_k}}(o_k) = \infty$, where $\ell_{g_{t_k}}(o_k)$ is the length with respect to g_{t_k} .

Applying Lemma 3.4, we may choose a length-minimizing closed g_{t_∞} -geodesic o_∞ in the class β . And let L denote its g_{t_∞} -length. Let g_{aux} be a fixed auxiliary metric on X , and let L_{aux} be the g_{aux} length of o_∞ .

Define a pseudo-metric d_{C^0} on the space of metrics on X as follows. Set $K = \text{image } o_\infty$. And set

$$V = \{v \in TX \mid \pi(v) \in K \text{ for } \pi : TX \rightarrow X \text{ the canonical projection, and } |v|_{aux} = 1\},$$

where $|v|_{aux}$ is the norm taken with respect to g_{aux} .

Then define:

$$d_{C^0}(g_1, g_2) = \sup_{v \in V} ||v|_{g_1} - |v|_{g_2}|.$$

By Properties 1 and 2, for every $\epsilon > 0$ we may find a $k > 0$ such that:

$$(3.6) \quad d_{C^0}(g_{t_k}, g_{t_\infty}) < \epsilon$$

and

$$(3.7) \quad \ell_{g_{t_k}}(o_k) > C + L + L_{aux} \cdot \epsilon.$$

By (3.6), we have:

$$\ell_{g_{t_k}}(o_\infty) < \ell_{g_{t_\infty}}(o_\infty) + L_{aux} \cdot \epsilon = L + L_{aux} \cdot \epsilon.$$

Combining with (3.7) we get:

$$\ell_{g_{t_k}}(o_k) > \ell_{g_{t_k}}(o_\infty) + C.$$

Since we may find a closed g_{t_k} -geodesic o' satisfying $\ell_{g_{t_k}}(o') \leq \ell_{g_{t_k}}(o_\infty)$, we get that

$$\left| \sup_{o \in S(g_{t_k}, \beta)} \ell_{g_{t_k}}(o) - \inf_{o \in S(g_{t_k}, \beta)} \ell_{g_{t_k}}(o) \right| > C,$$

and so we are in contradiction.

Thus,

$$\sup_{(o, t) \in S(\{g_t\}, \beta)} \ell_{g_t}(o) < \infty.$$

Then compactness of $S(\{g_t\}, \beta)$ follows by a slight modification of the proof of Lemma 3.2. $\square$

4. PRELIMINARIES ON REEB FLOW

Let (C^{2n+1}, λ) be a contact manifold with λ a contact form, that is a one form such that $\lambda \wedge (d\lambda)^n \neq 0$. Denote by R^λ the Reeb vector field satisfying:

$$d\lambda(R^\lambda, \cdot) = 0, \quad \lambda(R^\lambda) = 1.$$

Recall that a **closed λ -Reeb orbit** (or just Reeb orbit when λ is implicit) is a smooth map

$$o : (S^1 = \mathbb{R}/\mathbb{Z}) \rightarrow C$$

such that

$$\dot{o}(t) = cR^\lambda(o(t)),$$

with $\dot{o}(t)$ denoting the time derivative, for some $c > 0$ called the period. Let $S(R^\lambda, \beta)$ denote the space of all closed λ -Reeb orbits in free homotopy class β , with its compact open topology, and set

$$\mathcal{O}(R^\lambda, \beta) = S(R^\lambda, \beta)/S^1,$$

where $S^1 = \mathbb{R}/\mathbb{Z}$ acts by reparametrization $t \cdot o(\tau) = o(t + \tau)$.

5. DEFINITION OF THE FUNCTIONAL F AND PROOFS OF AUXILIARY RESULTS

Let X be an orientable manifold. Fix once and for all an auxiliary Riemannian metric g_0 , and set

$$C = S_{g_0}^* X.$$

For any Riemann-Finsler metric g , let F_g^* denote the dual norm, and let

$$\rho_g : C \rightarrow S_g^* X, \quad \rho_g(\xi) = \frac{\xi}{F_g^*(\xi)}$$

be the fiberwise radial projection. We define a contact form on the fixed manifold C by

$$\lambda_g = \rho_g^* \lambda_{\text{can}},$$

where λ_{can} is the Liouville one-form on T^*X . We write

$$R^{\lambda_g}$$

for the Reeb vector field of λ_g on this fixed manifold C .

The same convention applies to the Sasaki metric. If g is Riemannian, let g_S^{nat} denote the usual Sasaki metric on $S_g^* X$, and set

$$g_S = \rho_g^* g_S^{\text{nat}}$$

on the fixed manifold C . Similarly, all metric-dependent functions on $S_g^* X$, such as verticality functions, are regarded as functions on C by pullback through ρ_g . This fixed- C convention will be used throughout.

5.1. Lifted geodesic strings. We recall how closed geodesics in X are lifted to closed Reeb orbits in its cosphere bundle. Let C be the fixed cosphere model used in this paper, and let

$$\text{pr} : C \rightarrow X$$

denote the projection. For a Riemann-Finsler metric g , the contact form λ_g on C has Reeb vector field R^{λ_g} . Its closed orbits project to oriented closed constant-speed g -geodesics in X .

Conversely, every oriented closed constant-speed g -geodesic has a unique Reeb lift to C . In the Riemannian case, if

$$o : S^1 \rightarrow X$$

is a nonconstant closed g -geodesic, its lift to the g -unit cotangent bundle is

$$\tilde{o}_g(t) = \frac{g_{o(t)}(\dot{o}(t), \cdot)}{|\dot{o}(t)|_g} \in S_g^* X.$$

Since o is a geodesic, this covector field is parallel along o . Thus, under the Levi-Civita splitting of $T(T^*X)$, the curve $\tilde{o}_g$ is horizontal. Hence, after the constant reparametrization determined by the speed of o , it is a trajectory of the Reeb vector field. Passing to the fixed model C by radial projection gives the corresponding closed R^{λ_g} -orbit in C . The same statement holds in the Riemann-Finsler case using the Legendre transform.

The projection pr induces a homeomorphism between closed R^{λ_g} -orbit strings in C and oriented closed constant-speed g -geodesic strings in X . Hence, for every class $\beta \in \pi_1^{\text{inc}}(X)$, if $\mathcal{O}(g, \beta)$ denotes the set of class β g -geodesic strings, then its Reeb lift

$$\tilde{\mathcal{O}}(g, \beta) = \{\tilde{o} \mid o \in \mathcal{O}(g, \beta)\} \subset \mathcal{O}(R^{\lambda_g})$$

is homeomorphic to $\mathcal{O}(g, \beta)$. In particular, if $\mathcal{O}(g, \beta)$ is compact, then $\tilde{\mathcal{O}}(g, \beta)$ is compact. Moreover, $\tilde{\mathcal{O}}(g, \beta)$ is open in $\mathcal{O}(R^{\lambda_g})$, since the projected free homotopy class in X is locally constant on the loop space of C .

5.2. Definition via the Fuller index. Suppose then that g is β -taut. Define

$$F(g, \beta) = i(\tilde{\mathcal{O}}(g, \beta), R^{\lambda_g}) \in \mathbb{Q},$$

where the right hand side is the Fuller index, see Appendix A. For g not β -taut, set $F(g, \beta) := \infty$.

For a path of metrics $\{g_t\}_{t \in [0,1]}$, the lifted cobordism of orbit string spaces

$$\tilde{\mathcal{O}}(\{g_t\}, \beta) = \{(\tilde{o}, t) \mid o \in \mathcal{O}(g_t, \beta)\}$$

is likewise homeomorphic to the downstairs cobordism $\mathcal{O}(\{g_t\}, \beta) := S(\{g_t\}, \beta)/S^1$.

Proof of Theorem 2.8. Let $\beta \in \pi_1^{\text{inc}}(X)$ be given.

Part (1) holds by construction.

Part (2). Let g be β -taut and β -regular. Nondegenerate geodesic strings are isolated, so the compact space $\mathcal{O}(g, \beta)$ is finite. Then (A.3) computes $i(\tilde{\mathcal{O}}(g, \beta), R^{\lambda_g})$ as the count (1.1).

Part (3). Let $\{g_t\}_{t \in [0,1]}$ be a β -taut deformation, and let $\mathcal{N} \subset S(\{g_t\}, \beta)$ be a compact open subset containing the endpoint slices. Its S^1 -saturation is again compact and open, so we may take $\mathcal{N}$ to be S^1 -invariant; then $\mathcal{N}/S^1$ is a compact open subset of the downstairs cobordism $\mathcal{O}(\{g_t\}, \beta)$, and its lift $\tilde{\mathcal{N}}$ is a compact open subset of the lifted cobordism $\tilde{\mathcal{O}}(\{g_t\}, \beta)$, hence of the orbit cobordism $\mathcal{O}(\{R^{\lambda_{g_t}}\})$, in which the lifted cobordism is open, the projected free homotopy class in X being locally constant. The endpoint slices of $\tilde{\mathcal{N}}$ are exactly $\tilde{\mathcal{O}}(g_i, \beta)$, $i = 0, 1$: they contain them, as $\mathcal{N}$ contains the endpoint slices of $S(\{g_t\}, \beta)$, and nothing else, as $\mathcal{N} \subset S(\{g_t\}, \beta)$. Basic invariance of the Fuller index, namely (A.6), then gives

$$F(g_0, \beta) = i(\tilde{\mathcal{O}}(g_0, \beta), R^{\lambda_{g_0}}) = i(\tilde{\mathcal{O}}(g_1, \beta), R^{\lambda_{g_1}}) = F(g_1, \beta).$$

Part (4). Given $\beta' \in \pi_1^{\text{inc}}(X')$, set $\beta := \phi_* \beta'$. The map $\phi : (X', \phi^* g) \rightarrow (X, g)$ is an isometry. Hence it maps $S(\phi^* g, \beta')$ homeomorphically onto $S(g, \beta)$, so $\phi^* g$ is β' -taut if and only if g is β -taut, and otherwise both sides are ∞ . In the taut case, the cotangent lift of ϕ , composed with the radial projections, conjugates $R^{\lambda_{\phi^* g}}$ to R^{λ_g} and maps $\tilde{\mathcal{O}}(\phi^* g, \beta')$ onto $\tilde{\mathcal{O}}(g, \beta)$. The Fuller index is invariant under conjugacy. $\square$

6. PROOF OF THEOREM 1.2

For simplicity, as the full argument is already lengthy, we prove the theorem under the assumption that g is Riemannian. In the Finsler case we must use a more modern variational setup for the Hilbert manifold of H^1 -loops. In the Finsler case the energy functional is generally only C^1 , rather than C^2 , on the Hilbert loop space. Nevertheless the required Palais–Smale compactness and critical point theory for closed Finsler geodesics are standard; see Caponio–Javaloyes–Masiello [24], Mercuri [45], and Lu [42]. With these replacements, the compactness argument for end-incompressible classes, and the Fuller-index invariance argument go through without substantial change.

The circle S^1 acts on $L_\beta X$ by reparametrization:

$$(s \cdot \gamma)(t) = \gamma(t + s).$$

If some $\gamma \in L_\beta X$ had nontrivial stabilizer, then γ would factor through a nontrivial covering $S^1 \rightarrow S^1$, contradicting indivisibility. Therefore the S^1 -action on $L_\beta X$ is free, and the quotient map is a principal S^1 -bundle. Hence

$$(L_\beta X)_{hS^1} \simeq L_\beta X / S^1,$$

and we will construct a finite total rank model for the latter.

Let

$$\mathcal{L}_\beta X = W^{1,2}(S^1, X)_\beta$$

be the Hilbert manifold of H^1 -loops in the free homotopy class β . Consider the energy functional

$$\mathcal{E}_g : \mathcal{L}_\beta X \rightarrow \mathbb{R}, \quad \mathcal{E}_g(\gamma) = \int_{S^1} |\dot{\gamma}(t)|_g^2 dt.$$

Since $S(g, \beta)$ is compact, $\mathcal{E}_g$ is bounded above on the critical set of $\mathcal{E}_g$ in class β . Choose

$$(6.1) \quad E_0 > \max_{\gamma \in S(g, \beta)} \mathcal{E}_g(\gamma) =: M^2,$$

which exists by the β -tautness assumption. For a metric h and a number r set

$$\mathcal{U}_r^h = \{\gamma \in \mathcal{L}_\beta X \mid \mathcal{E}_h(\gamma) < r\}, \quad \mathcal{U}_{E_0} := \mathcal{U}_{E_0}^g.$$

Lemma 6.2. *The inclusion*

$$\mathcal{U}_{E_0} \hookrightarrow \mathcal{L}_\beta X$$

is an S^1 -equivariant homotopy equivalence. Consequently,

$$\mathcal{U}_{E_0}/S^1 \simeq \mathcal{L}_\beta X/S^1.$$

Moreover, the inclusion of smooth loops into the Hilbert loop space is an S^1 -equivariant homotopy equivalence, and hence

$$\mathcal{L}_\beta X/S^1 \simeq L_\beta X/S^1.$$

Therefore

$$\mathcal{U}_{E_0}/S^1 \simeq L_\beta X/S^1.$$

Proof. The energy functional $\mathcal{E}_g$ is smooth and S^1 -invariant. We equip $\mathcal{L}_\beta X$ with the standard S^1 -invariant Hilbert metric induced by g . With respect to this metric, the negative gradient flow of $\mathcal{E}_g$ is S^1 -equivariant.

We first recall the Palais–Smale compactness needed for the deformation lemma. Let $\{\gamma_j\} \subset \mathcal{L}_\beta X$ be a Palais–Smale sequence for $\mathcal{E}_g$ with

$$\mathcal{E}_g(\gamma_j) \leq A.$$

Then

$$\ell_g(\gamma_j) \leq \mathcal{E}_g(\gamma_j)^{1/2} \leq A^{1/2}.$$

Since $\beta \in \pi_1^{\text{inc}}(X)$, Lemma 3.1 implies that the images $\gamma_j(S^1)$ lie in a fixed compact subset of X . On this compact set, the usual Hilbert-loop compactness argument for the geodesic energy gives a strongly convergent subsequence in $\mathcal{L}_\beta X$. Thus $\mathcal{E}_g$ satisfies the Palais–Smale condition on sublevels.

By the choice of E_0 , the functional $\mathcal{E}_g$ has no critical points in

$$\{\mathcal{E}_g \geq E_0\}.$$

Therefore, for every $A > E_0$, Palais’ deformation lemma gives an S^1 -equivariant homotopy equivalence

$$\{\mathcal{E}_g < E_0\} \hookrightarrow \{\mathcal{E}_g < A\}.$$

Letting $A \rightarrow \infty$, or equivalently applying the negative gradient flow on the exhaustion by sublevels, we obtain an S^1 -equivariant homotopy equivalence

$$\mathcal{U}_{E_0} \simeq_{S^1} \mathcal{L}_\beta X.$$

Passing to quotients gives

$$\mathcal{U}_{E_0}/S^1 \simeq \mathcal{L}_\beta X/S^1.$$

Finally, the inclusion

$$C^\infty(S^1, X)_\beta \hookrightarrow W^{1,2}(S^1, X)_\beta$$

is an S^1 -equivariant homotopy equivalence by standard equivariant smoothing approximation for loops. Since $L_\beta X$ denotes the same free homotopy component in the smooth or compact-open loop space, we get

$$\mathcal{L}_\beta X \simeq_{S^1} L_\beta X.$$

Combining the quotient equivalences gives

$$\mathcal{U}_{E_0}/S^1 \simeq L_\beta X/S^1.$$

□

We now compare the Fuller index with the equivariant Morse model. The only subtlety is that after perturbing g to a metric g' with nondegenerate class β geodesics of bounded length, the full class β geodesic set of g' need not be compact. We therefore work in an isolating neighborhood of the compact lifted orbit block coming from g .

Let g_S denote the Sasaki metric on the fixed contact manifold C induced by g under the convention of Section 5. Define the S^1 -invariant loop set $\widehat{U}_{\sqrt{E_0}}$ to consist of all loops $\gamma : S^1 \rightarrow C$ such that

$$[\text{pr} \circ \gamma] = \beta, \quad \ell_{g_S}(\gamma) < \sqrt{E_0}.$$

Let

$$U_{\sqrt{E_0}} = \widehat{U}_{\sqrt{E_0}}/S^1$$

be its image in the orbit-string quotient. By (A.10) and the choice of E_0 , this is an open neighborhood of

$$N_0 = \widetilde{\mathcal{O}}(g, \beta)$$

with $\overline{U_{\sqrt{E_0}}} \cap \mathcal{O}(R^{\lambda_g}) = N_0$, as required in Lemma A.7; we call such a neighborhood *isolating*. It is also g_S -controlled, in the sense of Definition A.1, with the lower length bound supplied by the nontrivial projected class β and $K = \text{pr}^{-1}(K_X)$ for the compact $K_X \subset X$ provided by Lemma 3.1.

Choose numbers E'' and E' with

$$\max_{\gamma \in N_0} \ell_{g_S}(\gamma) < \sqrt{E''} < \sqrt{E'} < \sqrt{E_0},$$

and let $W = U_{\sqrt{E''}}$, defined like $U_{\sqrt{E_0}}$ with E'' in place of E_0 : an open neighborhood of N_0 with $\overline{W} \subset U_{\sqrt{E_0}}$. Let ϵ be the Fuller constant of Lemma A.7 for the data

$$(R^{\lambda_g}, N_0, U_{\sqrt{E_0}}, W).$$

Fix also numbers a, b with $M^2 < a < E' < b < E_0$, with M^2 as in (6.1); they depend on g only. Choose $\epsilon' > 0$ and a complete metric g' with $\|g' - g\|_{C^2} < \epsilon'$, agreeing with g outside a compact neighborhood of K_X , all of whose closed geodesics of class β and length $< \sqrt{E_0}$ are nondegenerate. Such g' exist by the bumpy metric theorem [1, 5], whose proof perturbs near the geodesics in question and so applies to the closed geodesics of bounded length through a compact set. Take ϵ' so small that, on the fixed manifold C , the Reeb vector field $R^{\lambda_{g_t}}$ of every metric $g_t = (1-t)g + tg'$, $t \in [0, 1]$, is C^0 ϵ -close to R^{λ_g} on K , that the sublevel sets below satisfy $\mathcal{U}_a^g \subset \mathcal{U}_{E'}^{g'} \subset \mathcal{U}_b^g$, and that the g_S -length and the g'_S -length of every loop in C differ by a factor so close to 1 that, with $\widehat{U}$ the set of loops $\gamma : S^1 \rightarrow C$ with

$$[\text{pr} \circ \gamma] = \beta, \quad \ell_{g'_S}(\gamma) < \sqrt{E'},$$

and $U = \widehat{U}/S^1$, we have

$$W \subset U \subset U_{\sqrt{E_0}}.$$

Lemma A.9, for the family $R^{\lambda_{g_t}}$, gives

$$F(g, \beta) = i(N_1, R^{\lambda_{g'}}), \quad N_1 = U_{\sqrt{E_0}} \cap \mathcal{O}(R^{\lambda_{g'}}) = U \cap \mathcal{O}(R^{\lambda_{g'}}),$$

the last equality as $\mathcal{O}(R^{\lambda_{g'}}) \cap U_{\sqrt{E_0}} \subset W \subset U$ by Lemma A.7.

It remains to compute the last Fuller index. Let

$$\mathcal{U}_{E'}^{g'} = \{\gamma \in \mathcal{L}_\beta X \mid \mathcal{E}_{g'}(\gamma) < E'\}.$$

For a constant-speed g' -geodesic, its canonical lift has g'_S -length equal to its g' -length. Hence the critical circles of $\mathcal{E}_{g'}$ in $\mathcal{U}_{E'}^{g'}$ are in one-to-one correspondence with the elements of N_1 . By the choice of g' , this critical set is a finite union of nondegenerate critical circles

$$C_o \simeq S^1.$$

Although the S^1 -action on $\mathcal{L}_\beta X$ is only continuous, each individual rotation is a smooth isometry preserving $\mathcal{E}_{g'}$, so the negative gradient flow of $\mathcal{E}_{g'}$ commutes with the action, and the deformations below are S^1 -equivariant. The Morse–Bott handlebody $\mathcal{MB}$ obtained from the unstable disk bundles of these critical circles is S^1 -invariant, and the standard Hilbert-manifold Morse–Bott deformation theorem gives an S^1 -equivariant homotopy equivalence

$$\mathcal{MB} \simeq_{S^1} \mathcal{U}_{E'}^{g'}.$$

After passing to the quotient by S^1 , the unstable disk bundle over C_o contributes one cell of dimension $\text{morse}(o)$; here freeness of the action (indivisibility of β) is used, as the quotient of the disk bundle by the free S^1 -action is a single cell. Therefore

$$\chi(\mathcal{U}_{E'}^{g'}/S^1) = \chi(\mathcal{MB}/S^1) = \sum_{o \in N_1} (-1)^{\text{morse}(o)}.$$

We now compare the g' -sublevel with the Hilbert sublevel for g . The numbers a, b fixed above satisfy

$$M^2 < a < E' < b < E_0,$$

so $\mathcal{E}_g$ has no critical points on $\{a \leq \mathcal{E}_g \leq b\}$, and $\mathcal{U}_a^g \subset \mathcal{U}_{E'}^{g'} \subset \mathcal{U}_b^g$ by the choice of ϵ' . So by Lemma 6.3 the inclusion

$$\mathcal{U}_{E'}^{g'} \hookrightarrow \mathcal{U}_b^g$$

is an S^1 -equivariant homotopy equivalence. Since all critical points of $\mathcal{E}_g$ have energy at most $M^2 < b$, the inclusion

$$\mathcal{U}_b^g \hookrightarrow \mathcal{U}_{E_0}^g$$

is also an S^1 -equivariant homotopy equivalence. Thus

$$\mathcal{U}_{E'}^{g'} \simeq_{S^1} \mathcal{U}_{E_0}^g.$$

By Lemma 6.2,

$$\mathcal{U}_{E_0}^g/S^1 \simeq L_\beta X/S^1.$$

Consequently

$$\chi(\mathcal{U}_{E'}^{g'}/S^1) = \chi(L_\beta X/S^1).$$

In particular the homology of $(L_\beta X)_{hS^1} \simeq L_\beta X/S^1$ has finite total rank.

On the other hand, the elements of N_1 are nondegenerate by the choice of g' , so (A.3) gives

$$i(N_1, R^{\lambda_{g'}}) = \sum_{o \in N_1} \frac{(-1)^{\text{morse}(o)}}{\text{mult}(o)}.$$

Since β is indivisible, all multiplicities in this class are 1. Hence

$$i(N_1, R^{\lambda_{g'}}) = \chi(\mathcal{U}_{E'}^{g'}/S^1) = \chi(L_\beta X/S^1).$$

Therefore

$$F(g, \beta) = \chi(L_\beta X/S^1) = \chi((L_\beta X)_{hS^1}).$$

□

We state the following elementary lemma separately, since it may be of independent interest.

Lemma 6.3. *With $\mathcal{U}_r^h$ as above, let $a < b$, and suppose that $\mathcal{E}_g$ satisfies the Palais–Smale condition on the band*

$$\mathcal{B} = \{a \leq \mathcal{E}_g \leq b\}$$

and has no critical points there. Let g' be sufficiently C^2 -close to g , so that $\nabla \mathcal{E}_{g'}$ is C^0 -close to $\nabla \mathcal{E}_g$ on $\mathcal{B}$. Suppose

$$\mathcal{U}_a^g \subset \mathcal{C} \subset \mathcal{U}_b^g,$$

where

$$\mathcal{C} = \{\mathcal{E}_{g'} < c\}$$

is an S^1 -invariant sublevel set of $\mathcal{E}_{g'}$. Then the inclusion

$$\mathcal{C} \hookrightarrow \mathcal{U}_b^g$$

is an S^1 -equivariant homotopy equivalence.

Proof. By the Palais–Smale condition and the absence of critical points on $\mathcal{B}$, there exists $\mu > 0$ such that

$$(6.4) \quad \|\nabla \mathcal{E}_g\| \geq \mu$$

on $\mathcal{B}$. For g' sufficiently close to g , we have

$$(6.5) \quad \|\nabla \mathcal{E}_{g'} - \nabla \mathcal{E}_g\| < \mu/2$$

on the same band. Therefore

$$\langle \nabla \mathcal{E}_g, \nabla \mathcal{E}_{g'} \rangle > 0$$

on $\mathcal{B}$.

Since

$$\mathcal{U}_a^g \subset \mathcal{C} \subset \mathcal{U}_b^g,$$

we have

$$\mathcal{U}_b^g - \mathcal{C} \subset \mathcal{U}_b^g - \mathcal{U}_a^g \subset \mathcal{B}.$$

Now along a negative $\mathcal{E}_{g'}$ -gradient trajectory lying in $\mathcal{B}$ we have:

$$\frac{d}{dt} \mathcal{E}_g = -\langle \nabla \mathcal{E}_g, \nabla \mathcal{E}_{g'} \rangle < 0.$$

Note that the topological boundary of $\mathcal{U}_b^g - \mathcal{C}$ is contained in $\{\mathcal{E}_g = b\} \cup \{\mathcal{E}_{g'} = c\}$. Consequently the negative $\mathcal{E}_{g'}$ -gradient flow line of an element in $\mathcal{U}_b^g - \mathcal{C}$ cannot exit $\mathcal{U}_b^g - \mathcal{C}$ before hitting $\mathcal{C}$. Moreover, $\mathcal{C}$ is forward-invariant because it is a sublevel set of $\mathcal{E}_{g'}$.

The estimates above also imply that $\mathcal{E}_{g'}$ has no critical points on $\mathcal{U}_b^g - \mathcal{C}$. The standard deformation argument of Palais [50, Section 10, Proposition (2)], applied to $\mathcal{E}_{g'}$ on the region $\mathcal{U}_b^g - \mathcal{C}$, therefore gives a deformation retraction of $\mathcal{U}_b^g$ onto $\mathcal{C}$. The energies and the Hilbert metrics are S^1 -invariant, so the gradient flow is S^1 -equivariant. Hence the deformation retraction is S^1 -equivariant. $\square$

6.1. A transfer formula, and the divisible classes on tori and surfaces. The following lemma substitutes for the orbifold Morse homology discussed after Theorem 1.2: finite covers do the orbifold's bookkeeping.

Lemma 6.6. *Let $\pi : \hat{X} \rightarrow X$ be a finite normal covering of manifolds, with deck group G of order n . Let g be a complete Riemann–Finsler metric on X , $\hat{g} = \pi^*g$, and $\beta \in \pi_1^{\text{inc}}(X)$ a class whose based representatives lie in $\pi_1(\hat{X}) \leq \pi_1(X)$. Let $\hat{B} \subset \pi_1^{\text{inc}}(\hat{X})$ be the set of classes of the closed lifts of class β loops. Then g is β -taut if and only if $\hat{g}$ is $\hat{\beta}$ -taut for every $\hat{\beta} \in \hat{B}$, and in that case*

$$(6.7) \quad n F(g, \beta) = \sum_{\hat{\beta} \in \hat{B}} F(\hat{g}, \hat{\beta}).$$

Proof. Since the covering is normal and the conjugacy class of β lies in the subgroup, every class β loop lifts closed at every lift of its basepoint. The lifted classes are nontrivial, as they project to β , and end incompressible: π is proper, so a class with representatives outside every compact set of $\widehat{X}$ projects to one with representatives outside every compact set of X . Every class $\widehat{\beta}$ geodesic string upstairs, $\widehat{\beta} \in \widehat{B}$, projects to a class β geodesic string downstairs, as π is a local isometry. So the total space of lifts maps onto $S(g, \beta)$ finitely-to-one and properly, and the two tautness conditions are equivalent.

Let $w = o^c$ be a class β string, o prime, so w has multiplicity c . The deck holonomy of o is an element $\text{hol}(o) \in G$, defined up to conjugacy; let k be its order. Then $\text{hol}(w) = \text{hol}(o)^c = 1$, so k divides c , and the closed lifts of w are the (c/k) -fold covers of the closed lifts of o^k . A closed lift ρ of o^k is prime: $\rho = \sigma^j$ with $j \geq 2$ gives $\pi \circ \sigma = o^{k/j}$, a closed lift of a smaller power of o , contradicting the minimality of k . So every closed lift $\widehat{w}$ of w has multiplicity c/k . Moreover $\pi \circ \widehat{w} = w$ as parametrized loops, so the second variations of the energies correspond under the local isometry: $\text{morse}(\widehat{w}) = \text{morse}(w)$, and nondegeneracy is equivalent. The n based lifts of w are permuted simply transitively by G , and the stabilizer of a lift, as a string, is the cyclic group $\langle \text{hol}(o) \rangle$ of order k : so w has exactly n/k closed lifts as strings. For β -regular g the contribution of the lifts of w to the right hand side of (6.7) is therefore

$$\frac{n}{k} \cdot \frac{(-1)^{\text{morse}(w)}}{c/k} = n \frac{(-1)^{\text{morse}(w)}}{c},$$

which is n times the contribution of w to the left hand side; summing over w gives (6.7).

Let now g be β -taut and not necessarily β -regular. Take the data of the proof of Theorem 1.2 (1) for (g, β) : the isolating neighborhood $U_{\sqrt{E_0}}$ of $N_0 = \widetilde{\mathcal{O}}(g, \beta)$, and a metric g' with $\|g' - g\|_{C^2} < \epsilon'$, equal to g outside a compact set, whose class β closed geodesics of length $< \sqrt{E_0}$ are nondegenerate. Take ϵ' below the Fuller constants of Lemma A.7 for these data and for their lifts, defined below. Let $g_t = (1-t)g + tg'$. Lemma A.9 gives

$$F(g, \beta) = i(N_1, R^{\lambda_{g'}}), \quad N_1 = U_{\sqrt{E_0}} \cap \mathcal{O}(R^{\lambda_{g'}}),$$

as there.

Upstairs, the cosphere model of $\widehat{X}$ covers that of X with deck group G , the metrics $\widehat{g}_t = \pi^* g_t$ have Reeb vector fields the pullbacks of $R^{\lambda_{g_t}}$, and the $\widehat{g}_S$ -length of a loop is the g_S -length of its projection. Let $\widehat{N}_0 = \bigsqcup_{\widehat{\beta} \in \widehat{B}} \widetilde{\mathcal{O}}(\widehat{g}, \widehat{\beta})$, and let $\widehat{U}_{\sqrt{E_0}}$ be the set of orbit strings of class in $\widehat{B}$ and $\widehat{g}_S$ -length $< \sqrt{E_0}$. A string of class in $\widehat{B}$ projects to a class β string of the same length, and the closed lifts of a class β string have its length. So $\widehat{U}_{\sqrt{E_0}}$ is an isolating neighborhood of $\widehat{N}_0$, and Lemma A.9 applied to $\{\widehat{g}_t\}$ gives

$$\sum_{\widehat{\beta} \in \widehat{B}} F(\widehat{g}, \widehat{\beta}) = i(\widehat{N}_1, R^{\lambda_{\widehat{g}'}}), \quad \widehat{N}_1 = \widehat{U}_{\sqrt{E_0}} \cap \mathcal{O}(R^{\lambda_{\widehat{g}'}}),$$

the set of closed lifts of the strings of N_1 . A closed lift and its projection have the same linearized return map, π being a local isometry, so the strings of $\widehat{N}_1$ are nondegenerate. By (A.3) both Fuller indices are the signed counts of the preceding paragraph, over the strings w of N_1 and their closed lifts, and that paragraph gives $i(\widehat{N}_1, R^{\lambda_{\widehat{g}'}}) = n i(N_1, R^{\lambda_{g'}})$. This is (6.7). $\square$

Theorem 6.8. *Let X be an orientable aspherical manifold with $\pi_1(X) \cong \mathbb{Z}^n$, let $e\beta_0 \in \pi_1^{\text{inc}}(X)$ with β_0 indivisible and $e \geq 1$, and let g be an $e\beta_0$ -taut Riemann-Finsler metric. Then*

$$F(g, e\beta_0) = \begin{cases} 0, & n \geq 2, \\ 1/e, & n = 1. \end{cases}$$

In particular $F(\cdot, \beta)$ is constant on the β -taut metrics on X , for every $\beta \in \pi_1^{\text{inc}}(X)$.

Tori are the basic examples.

Proof. Complete β_0 to a basis $(\beta_0, \gamma_2, \dots, \gamma_n)$ of $\pi_1(X) = \mathbb{Z}^n$, and let $\pi : \widehat{X} \rightarrow X$ be the normal e -fold cover with $\pi_1(\widehat{X}) = H = \langle e\beta_0, \gamma_2, \dots, \gamma_n \rangle$ and deck group $\mathbb{Z}/e$. The group is abelian, so conjugacy is trivial and $e\beta_0 \in H$; all closed lifts of class $e\beta_0$ loops have the single class $\widehat{\beta}_0 := e\beta_0 \in H$, which is a basis element of H , so indivisible. Lemma 6.6 gives $eF(g, e\beta_0) = F(\pi^*g, \beta_0)$, and Theorem 1.2 (1) evaluates the right hand side as $\chi((L_{\widehat{\beta}_0} \widehat{X})_{hS^1})$. As $\widehat{X}$ is aspherical, this Borel construction is a $K(H/\langle \widehat{\beta}_0 \rangle, 1)$, as noted after Conjecture 1, and $H/\langle \widehat{\beta}_0 \rangle \cong \mathbb{Z}^{n-1}$: a torus T^{n-1} , of Euler characteristic 0 for $n \geq 2$, and a point for $n = 1$. The last clause follows as every class is $e\beta_0$ for an indivisible β_0 and $e \geq 1$. Note that $\beta_0 \in \pi_1^{\text{inc}}(X)$ when $e\beta_0$ is: the e -th power of a representative of β_0 outside a compact set is one of $e\beta_0$. $\square$

Remark 6.9. The transfer (6.7) is consistent with the negative curvature ray of Example 4. For a closed hyperbolic surface, β_0 represented by a nonseparating curve with based representative a , take $H = \ker(\varphi)$ for $\varphi : \pi_1 \rightarrow \mathbb{Z}/e$ with $\varphi(a) = 1$. The conjugacy class of a^e meets H in a single H -class, as $|H \backslash \pi_1 / \langle a \rangle| = 1$, and the unique class $e\beta_0$ geodesic has a single closed lift, prime upstairs. Then (6.7) reads $e \cdot \frac{1}{e} = 1$, the value of F upstairs in an indivisible class.

On the remaining surfaces the transfer computes F outright. The covers are supplied by the following lemma. For surface groups it is a consequence of known residual properties, see Remark 6.11; we give a direct proof.

Lemma 6.10. *Let G be a free group, or the fundamental group of a closed orientable surface of genus ≥ 2 ; let $a \in G$ be nontrivial, and p a prime. Then there is a finite quotient $q : G \rightarrow Q$ with $q(a)$ of order exactly p .*

The proof below has been formalized in Lean 4, over Mathlib. The formalization is unconditional, apart from residual freeness entering as a named hypothesis.⁵

Remark 6.11. Lemma 6.10 also follows from the literature. G is residually free [11], and free groups are residually p for every prime p [37]; so G is residually p . And in a finite p -group Q , every $x \neq 1$ has order exactly p in some quotient. Let M be a normal subgroup of Q maximal among those not containing x . Then every nontrivial normal subgroup of Q/M contains xM . The center of Q/M is nontrivial, so it has a subgroup of order p . This subgroup is normal, hence contains xM , and xM has order p . The proof below produces the p -group quotient directly, in the form used by the formalization.

Proof. There is a homomorphism $\psi : G \rightarrow F$ to a finitely generated free group with $w := \psi(a) \neq 1$: for G free, the retraction onto the free factor generated by the finitely many basis elements occurring in a ; for a surface group, residual freeness [11, 12]. It suffices to find a finite quotient of F in which the image of w has order exactly p .

Fix a basis $x_1, \dots, x_k$ of F . For a commutative ring R , let A_R be the ring of formal power series in the noncommuting variables $X_1, \dots, X_k$ over R . Each $1 + X_i$ is a unit of A_R , with inverse $\sum_{j \geq 0} (-X_i)^j$. So $\mu(x_i) = 1 + X_i$ defines a homomorphism $\mu : F \rightarrow A_{\mathbb{Z}}^{\times}$, the Magnus homomorphism. Integer powers expand as $(1 + X_i)^b = \sum_{s \geq 0} \binom{b}{s} X_i^s$, also for $b < 0$, with the generalized integer binomial coefficients.

Write the reduced word of w in syllables: $w = x_{j_1}^{b_1} \cdots x_{j_m}^{b_m}$, with $m \geq 1$, $j_t \neq j_{t+1}$, and $b_t \neq 0$. The coefficient in $\mu(w)$ of the monomial $X_{j_1} X_{j_2} \cdots X_{j_m}$ is $b_1 b_2 \cdots b_m \neq 0$. Indeed, a term of $\prod_t (1 + X_{j_t})^{b_t}$ contributing to this monomial selects from the t -th factor a power $X_{j_t}^{s_t}$, the selected runs concatenating to $X_{j_1} \cdots X_{j_m}$. Adjacent letters of the monomial are distinct, while a run with $s_t \geq 2$ produces two equal adjacent letters: so every $s_t \leq 1$. The lengths sum to m , so $s_t = 1$ for every t , and the coefficient is $\prod_t \binom{b_t}{1}$. In particular $\mu(w) \neq 1$.

⁵Lean sources: <https://yashamon.github.io/web2/papers/surfacegroupformalization/>.

The constant term of $\mu(w)$ is 1. Let $c \geq 1$ be the minimal degree in which $\mu(w) - 1$ has a nonzero coefficient, P the degree c homogeneous part of $\mu(w)$, and s the minimal p -adic valuation of the coefficients of P . Let B be the quotient of $A_{\mathbb{Z}/p^{s+1}}$ by the two-sided ideal of series supported in degrees $> c$: a finite ring. Let $q : F \rightarrow B^\times$ be μ followed by reduction of the coefficients and truncation, a homomorphism to a finite group. By minimality of c : $q(w) = 1 + \overline{P}$, with $\overline{P}$ the reduction of P . Since $\overline{P}$ is concentrated in the top degree c of B : $\overline{P}^2 = 0$, so $q(w)^n = 1 + n\overline{P}$ for all n . By the choice of s : $p\overline{P} = 0$, as p^s divides every coefficient of P ; and $\overline{P} \neq 0$, as some coefficient of P is not divisible by p^{s+1} . So $q(w)^p = 1$ while $q(w) \neq 1$: the order of $q(w)$ is exactly p . Compose with ψ . $\square$

Theorem 6.12. *Let Σ be an orientable surface other than S^2 and T^2 , let $e\beta_0 \in \pi_1^{inc}(\Sigma)$ with β_0 indivisible and $e \geq 1$, and let g be an $e\beta_0$ -taut Riemann-Finsler metric on Σ . Then*

$$F(g, e\beta_0) = \frac{1}{e}.$$

In particular, for every $\beta \in \pi_1^{inc}(\Sigma)$, $F(\cdot, \beta)$ is constant on the β -taut metrics on Σ .

The fundamental group G of such a surface is a surface group of genus ≥ 2 when Σ is closed, and free otherwise; in both cases $C_G(a^e) = \langle a \rangle$ for a indivisible and $e \geq 1$. In a free group centralizers of nontrivial elements are cyclic, generated by a root. In a surface group, torsion-free hyperbolic, $c \in C_G(a^e)$ makes $\langle c, a^e \rangle$ abelian, hence elementary, so $c \in E(a^e) = E(a) = \langle a \rangle$ in the notation of the proof of Lemma 1.8.

Proof. Induction on e , over all such surfaces at once. For $e = 1$, Theorem 1.2 (1) gives $F(g, \beta_0) = \chi((L_{\beta_0}\Sigma)_{hS^1})$, a $K(C_G(a)/\langle a \rangle, 1)$ for a a based representative of β_0 , as noted after Conjecture 1; the group is trivial, so the value is 1.

Let $e > 1$, let p be a prime divisor of e , and let a be a based representative of β_0 ; a is indivisible. By Lemma 6.10 there is a finite quotient $q : G \rightarrow Q$, $|Q| = n$, with $q(a)$ of order exactly p . Let $H = \ker q$, and $\pi : \widehat{\Sigma} \rightarrow \Sigma$ the corresponding normal cover, an orientable surface, again neither S^2 nor T^2 . Then $\langle a \rangle \cap H = \langle a^p \rangle$, and $p|e$ gives $a^e \in H$: the hypothesis of Lemma 6.6 holds for the class $e\beta_0$, and

$$n F(g, e\beta_0) = \sum_{\widehat{\beta} \in \widehat{B}} F(\pi^*g, \widehat{\beta}).$$

The set $\widehat{B}$ consists of the H -classes of the conjugates xa^ex^{-1} . Their number is $|H \backslash G / C_G(a^e)|$, and $C_G(a^e) = \langle a \rangle$. The double cosets correspond to the orbits of right translation by $\langle q(a) \rangle \cong \mathbb{Z}/p$ on Q , a free action: so $|\widehat{B}| = n/p$.

Each $\widehat{\beta} \in \widehat{B}$ is (e/p) times an indivisible class. Indeed $\widehat{\beta}$ is the H -class of $b^e = (b^p)^{e/p}$ for $b = xax^{-1}$, with $b^p \in H$; and b^p is indivisible in H : a root of b^p in G is a power b^j with $j|p$, since $C_G(b^p) = \langle b \rangle$, and $b^j \in H$ forces $p|j$, so $j = p$. By Lemma 6.6 the metric π^*g is $\widehat{\beta}$ -taut, with $\widehat{\beta} \in \pi_1^{inc}(\widehat{\Sigma})$, and the inductive hypothesis on $\widehat{\Sigma}$ gives $F(\pi^*g, \widehat{\beta}) = p/e$ for each $\widehat{\beta}$. So $n F(g, e\beta_0) = (n/p)(p/e)$, and $F(g, e\beta_0) = 1/e$. The last clause follows as every $\beta \in \pi_1^{inc}(\Sigma)$ is $e\beta_0$ with $\beta_0 \in \pi_1^{inc}(\Sigma)$ indivisible. $\square$

Proof of Theorem 1.2 (2). Theorems 6.8 and 6.12 give the values. These are the Wall-Euler characteristics [60] of Conjecture 1. Let a be a based representative of β_0 , so that $\gamma_{e\beta_0} = a^e$. For $G = \mathbb{Z}^n$, $C_G(a^e) = G$ and $Q_{e\beta_0} = \mathbb{Z}^n / \langle a^e \rangle \cong \mathbb{Z}^{n-1} \times \mathbb{Z}/e$, as a is indivisible; its Wall characteristic is that of the index e subgroup $\mathbb{Z}^{n-1}$ divided by e : 0 for $n \geq 2$ and $1/e$ for $n = 1$. On a surface other than T^2 , $C_G(a^e) = \langle a \rangle$, as noted before the proof of Theorem 6.12, so $Q_{e\beta_0} \cong \mathbb{Z}/e$, of Wall characteristic $1/e$. These groups are Wall-Euler finite, and $(L_{\beta}X)_{hS^1}$ is a $K(Q_{\beta}, 1)$, as noted after Conjecture 1. $\square$

Lemma 6.6 is thus the defining device of the Wall characteristic — average over a finite cover unwrapping the torsion — implemented for the geodesic flow. For $\gamma = a^e$ with $C_G(\gamma) = \langle a \rangle$, as on closed manifolds of negative curvature, it reduces Conjecture 1 to:

Question 4. Let Γ be the fundamental group of a closed manifold of negative curvature, or any torsion-free hyperbolic group, $a \in \Gamma$ indivisible, and p a prime. Is there a finite quotient of Γ in which the image of a has order exactly p ?

A positive answer gives, by the proof of Theorem 6.12 run verbatim, $F(g, e\beta_0) = 1/e = \chi_{\text{Wall}}(Q_{e\beta_0})$ for every $e\beta_0$ -taut metric on a closed manifold of negative curvature. Surface groups answer the question through residual freeness (Lemma 6.10). In higher dimensions residual freeness fails, and the omnipotence statements available through virtual specialness, e.g. [3], control orders of images only up to an undetermined multiplicative constant — not enough here, as the order must divide e .

7. PRODUCT FORMULA

In this section we prove a kind of product formula for the F -count. It is used in Section 8 and in Proposition 1.5.

Theorem 7.1 (Product formula). *Let*

$$Z \hookrightarrow (X, g) \xrightarrow{p} (Y, g_Y)$$

be a β -periodic fibration, with Y closed, connected and such that all smooth closed contractible g_Y -geodesics in Y are constant. Then g is β -taut. Fix $y_0 \in Y$, set $Z_{y_0} = p^{-1}(y_0)$, and choose

$$\beta_Z \in \pi_1^{\text{inc}}(Z_{y_0})$$

such that

$$(i_{y_0})_*(\beta_Z) = \beta,$$

where $i_{y_0} : Z_{y_0} \hookrightarrow X$ is the inclusion. Denote by g_{y_0} the restriction of g to the fiber of y_0 . Let

$$H = \pi_1(Y, y_0)$$

act on $\pi_1^{\text{inc}}(Z_{y_0})$ by holonomy, and set

$$S_{y_0} = H \cdot \beta_Z.$$

More generally, for $y \in Y$ let $S_y \subset \pi_1^{\text{inc}}(Z_y)$ be the set of classes of the fiber $Z_y = p^{-1}(y)$ mapping to β under inclusion; by Lemma 2.18 it is the holonomy orbit of any of its elements, so this agrees with S_{y_0} at y_0 . Then

$$F(g, \beta) = \#S_{y_0} \cdot \chi(Y) \cdot F(g_{y_0}, \beta_Z).$$

Proof. We prove β -tautness. We first show that every class β g -geodesic in X is vertical. Let

$$o : S^1 \rightarrow X$$

be a closed g -geodesic representing the class β . Since β is a fiber class, the loop

$$p \circ o : S^1 \rightarrow Y$$

is contractible. Since p is a parallel Riemannian submersion, the local de Rham product decomposition for p implies that $p \circ o$ is a g_Y -geodesic, possibly constant. Hence $p \circ o$ is a smooth closed contractible g_Y -geodesic. By hypothesis, every such geodesic is constant. Therefore $p \circ o$ is constant, and so o is contained in a single fiber

$$Z_y = p^{-1}(y).$$

Thus the class β geodesic strings in X are precisely the vertical geodesic strings whose fiber class represents β in X .

It remains to prove compactness. Let o_j be a sequence of class β g -geodesic strings in X . By the previous paragraph, each o_j is contained in a fiber Z_{y_j} . Since Y is compact, after passing to a subsequence we may assume

$$y_j \rightarrow y_\infty.$$

Choose a contractible coordinate ball

$$D \subset Y$$

around y_∞ . For all large j , we have $y_j \in D$. Since p is a parallel Riemannian submersion, the horizontal and vertical distributions are parallel, so integrable with totally geodesic leaves, and the local de Rham theorem makes X locally a Riemannian product; as g is complete, horizontal lifts of paths in D exist, and as D is simply connected the holonomy along them is trivial. This gives a Riemannian product decomposition

$$p^{-1}(D) \cong Z_{y_\infty} \times D.$$

Under this identification, vertical geodesics in the fibers Z_{y_j} are identified with geodesics of the fiber metric on Z_{y_∞} , and their fiber free homotopy classes lie in the finite holonomy orbit

$$S_{y_\infty}.$$

By the fiber-tautness condition in the definition of a β -periodic fibration, the union of the geodesic string sets

$$\bigcup_{\alpha \in S_{y_\infty}} S(g_{y_\infty}, \alpha)$$

is compact. Since S_{y_∞} is finite, this is a finite union of compact sets. Therefore, after passing to a further subsequence, the vertical geodesic strings o_j , transported to the model fiber Z_{y_∞} , converge to a fiber geodesic string. Transporting back through the local product chart gives convergence of the original o_j in the class β geodesic string space of X .

Thus every sequence of class β geodesic strings has a convergent subsequence. Hence the class β geodesic string set is compact, and g is β -taut.

Let

$$C = S^*X$$

be the unit cotangent bundle of (X, g) , with projection

$$\text{pr} : C \rightarrow X.$$

For this fixed metric calculation, this is equivalent to the fixed- C convention of Section 5. Let R^{λ_g} denote the Reeb vector field on C . The F -invariant is computed from the compact lifted orbit set

$$N_0 = \tilde{\mathcal{O}}(g, \beta).$$

The idea is to localize contributions over critical points of a Morse function on Y .

Let

$$\pi^{\text{vert}} : TX \rightarrow T^{\text{vert}}X = \ker(dp)$$

be the g -orthogonal projection. By assumptions on $p : X \rightarrow Y$,

$$\nabla \pi^{\text{vert}} = 0,$$

where π^{vert} is viewed as a $(1, 1)$ -tensor field.

For $\xi \in C$, let

$$v_\xi \in T_{\text{pr}(\xi)}X$$

be the g -dual unit vector to ξ . Define

$$P : C \rightarrow \mathbb{R}$$

by

$$P(\xi) = |\pi^{\text{vert}} v_\xi|_g^2.$$

Thus $P = 1$ on vertical unit covectors and $P = 0$ on horizontal unit covectors.

Let $f : Y \rightarrow \mathbb{R}$ be a Morse function. Define

$$\tilde{f} : C \rightarrow \mathbb{R}$$

by

$$\tilde{f} = f \circ p \circ \text{pr}.$$

Equip $C = S^*X$ with the Sasaki metric g_S induced by g , as in the Proof of Theorem 1.2. We shall use the following two elementary identities.

First,

$$(7.2) \quad dP(R^{\lambda_g}) = 0.$$

Indeed, let $\xi(\tau)$ be a flow line of R^{λ_g} , and set

$$x(\tau) = \text{pr}(\xi(\tau)).$$

Then $x(\tau)$ is a unit-speed g -geodesic. Let

$$v(\tau) = \dot{x}(\tau) = v_{\xi(\tau)}$$

be its velocity vector field. Since $x(\tau)$ is a geodesic,

$$\nabla_{\tau} v = 0.$$

Since $\nabla \pi^{\text{vert}} = 0$, we have

$$\nabla_{\tau}(\pi^{\text{vert}} v) = \pi^{\text{vert}}(\nabla_{\tau} v) = 0.$$

Therefore

$$\begin{aligned} \frac{d}{d\tau} P(\xi(\tau)) &= \frac{d}{d\tau} |\pi^{\text{vert}} v(\tau)|_g^2 \\ &= 2 \langle \nabla_{\tau}(\pi^{\text{vert}} v), \pi^{\text{vert}} v \rangle_g \\ &= 0. \end{aligned}$$

Hence $dP(R^{\lambda_g}) = 0$.

Second,

$$(7.3) \quad dP(\nabla_{g_S} \tilde{f}) = 0.$$

Since $\tilde{f} = f \circ p \circ \text{pr}$ depends only on the base point $\text{pr}(\xi)$:

$$\nabla_{g_S} \tilde{f} \in T^{\text{hor}} C.$$

More precisely, $\nabla_{g_S} \tilde{f}$ is the horizontal lift of

$$\nabla_g(f \circ p)$$

on X . Let $\xi(s)$ be a horizontal curve in C , and set

$$x(s) = \text{pr}(\xi(s)).$$

By definition of the Levi-Civita horizontal distribution on S^*X , the covectors $\xi(s)$ are parallel along $x(s)$. Equivalently, their g -dual vectors

$$v(s) = v_{\xi(s)}$$

are parallel along $x(s)$:

$$\nabla_s v(s) = 0.$$

Using again $\nabla \pi^{\text{vert}} = 0$, we get

$$\nabla_s(\pi^{\text{vert}} v(s)) = \pi^{\text{vert}}(\nabla_s v(s)) = 0.$$

Hence

$$\begin{aligned}\frac{d}{ds}P(\xi(s)) &= \frac{d}{ds}|\pi^{\text{vert}}v(s)|_g^2 \\ &= 2\langle \nabla_s(\pi^{\text{vert}}v(s)), \pi^{\text{vert}}v(s) \rangle_g \\ &= 0.\end{aligned}$$

Taking $\dot{\xi}(0) = \nabla_{g_S}\tilde{f}(\xi)$ gives

$$dP_\xi(\nabla_{g_S}\tilde{f}) = 0.$$

Since ξ was arbitrary,

$$dP(\nabla_{g_S}\tilde{f}) = 0.$$

Now set

$$V_\varepsilon = R^{\lambda_g} - \varepsilon \nabla_{g_S}(P + \tilde{f}).$$

Recall first the discussion of Fuller continuation in the Appendix. The set $N_0 = \tilde{\mathcal{O}}(g, \beta)$ is compact by β -tautness. Choose a controlled isolating neighborhood U of N_0 and an open neighborhood W with $\overline{W} \subset U$, and let ϵ' be the corresponding Fuller constant. Take $\varepsilon > 0$ small enough so that V_ε is C^0 ϵ' -close to R^{λ_g} on the compact region determined by U . Lemma A.9 gives a compact continued set N_ε and

$$F(g, \beta) = i(N_\varepsilon, V_\varepsilon).$$

Note that the above does not require any further properties of V_ε , such as completeness. We set up the Fuller index theory to sidestep such issues.

We now identify the closed orbits in N_ε . We show that P is a Lyapunov function for V_ε . Namely, along a trajectory $\xi(\tau)$ of V_ε , we have

$$\begin{aligned}\frac{d}{d\tau}P(\xi(\tau)) &= dP(R^{\lambda_g}) - \varepsilon dP(\nabla_{g_S}P) - \varepsilon dP(\nabla_{g_S}\tilde{f}) \\ &= -\varepsilon |\nabla_{g_S}P|_{g_S}^2 \leq 0.\end{aligned}$$

Thus P is nonincreasing along the V_ε -flow.

If $\xi(\tau)$ is a closed orbit of V_ε , then $P(\xi(\tau))$ returns to its initial value after one period. Since P is nonincreasing, it must be constant along the orbit. Hence

$$\nabla_{g_S}P = 0$$

along the orbit.

On each unit cotangent sphere, P is the squared norm of the vertical component of the corresponding unit vector. Therefore the critical locus of P consists of the two critical loci

$$P = 1 \quad \text{and} \quad P = 0,$$

corresponding respectively to vertical and horizontal covectors. At $\varepsilon = 0$, the lifted orbit set under consideration is vertical, hence lies in $P = 1$. Taking the controlled neighborhood U (above) so that all its elements are disjoint (as loops) from $P = 0$, we get that every closed orbit in N_ε lies in the vertical critical locus $P = 1$.

It remains to see over which fibers these vertical orbits occur. On $P = 1$, the Reeb vector field restricts to the geodesic Reeb field of the fiber (Z_y, g_y) , while $\tilde{f}$ depends only on the base point y . The perturbation

$$-\varepsilon \nabla_{g_S}\tilde{f}$$

moves the base point in the negative gradient direction of f . Therefore a vertical closed orbit of V_ε can occur only over a critical point of f . Conversely, if $y \in \text{Crit}(f)$, then

$$\nabla_{g_S}\tilde{f} = 0$$

over Z_y , and the vertical part of V_ε is just the fiber geodesic flow, up to the harmless P -term which vanishes on the vertical critical locus. Thus the closed orbits of V_ε in N_ε are precisely the vertical closed geodesic orbits lying in fibers over critical points of f whose fiber class lies in S_y , the projected class being β on U .

We now compute the index. For $y \in Y$, $S_y \subset \pi_1^{\text{inc}}(Z_y)$ is the set of fiber classes mapping to β , as in the statement.

The Fuller index is additive over disjoint compact open subsets. Thus it remains to compute the local index contribution $i(N_{\varepsilon,y}, V_\varepsilon)$ corresponding to the isolated subset of N_ε lying over a critical point y of f . We defer this computation to the Appendix as Lemma B.1; it yields that

$$(7.4) \quad i(N_{\varepsilon,y}, V_\varepsilon) = (-1)^{\text{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha),$$

with $\text{morse}(y)$ the Morse index of f at y .

Next, we show that $F(g_y, \cdot)$ is constant on S_y . Indeed, let

$$\gamma : [0, 1] \rightarrow Y$$

be a loop based at y , and $h_\gamma : Z_y \rightarrow Z_y$ the holonomy map along its horizontal lifts. As p is g -parallel, the horizontal distribution is parallel, hence integrable with totally geodesic leaves, and X is locally a Riemannian product, as in the local de Rham argument above; so h_γ is an isometry of (Z_y, g_y) . Theorem 2.8 (4) gives

$$F(g_y, \alpha) = F(h_\gamma^* g_y, \alpha) = F(g_y, h_{\gamma,*} \alpha).$$

Thus $F(g_y, \alpha)$ is constant on the holonomy orbit S_y . Hence

$$\sum_{\alpha \in S_y} F(g_y, \alpha) = \#S_y \cdot F(g_y, \alpha_y), \quad \alpha_y \in S_y.$$

If $y, y' \in Y$, holonomy along any path from y to y' is an isometry $(Z_y, g_y) \rightarrow (Z_{y'}, g_{y'})$ identifying S_y with $S_{y'}$, so by the same naturality

$$\sum_{\alpha \in S_y} F(g_y, \alpha)$$

is independent of y . So we get:

$$\begin{aligned} F(g, \beta) &= \sum_{y \in \text{Crit}(f)} i(N_{\varepsilon,y}, V_\varepsilon) \\ &= \sum_{y \in \text{Crit}(f)} (-1)^{\text{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha) \\ &= \sum_{y \in \text{Crit}(f)} (-1)^{\text{morse}(y)} \cdot \#S_{y_0} \cdot F(g_{y_0}, \beta_Z) \\ &= \chi(Y) \cdot \#S_{y_0} \cdot F(g_{y_0}, \beta_Z). \end{aligned}$$

□

Corollary 7.5. *Let $X = Z \times Y$, with Z connected, g_Z a taut Riemannian metric on Z , and (Y, g_Y) closed, connected, Riemannian, with $\pi_2(Y) = 0$ and all smooth closed contractible g_Y -geodesics constant. Let $\beta = (i_Z)_*(\beta_Z) \in \pi_1^{\text{inc}}(X)$ be a fiber class, $\beta_Z \in \pi_1^{\text{inc}}(Z)$. Then $g = g_Z \times g_Y$ is β -taut. Furthermore,*

$$F(g, \beta) = \chi(Y) \cdot F(g_Z, \beta_Z).$$

Proof. This is immediate from the theorem. □

8. PROOF OF THEOREM 2.22 AND ITS COROLLARIES

We prove the first part. A complete negatively curved Riemannian manifold has at most one geodesic string in each nontrivial free homotopy class, by the flat strip theorem; moreover the absence of nontrivial periodic Jacobi fields in negative curvature gives nondegeneracy. Existence of a closed geodesic in the end-incompressible class β follows from Lemma 3.4. Hence a complete negatively curved metric would give a unique nondegenerate class β geodesic string. It thus suffices to prove the second part.

We prove the second part. Suppose first that β is atomic, so that β has a representative that is a k -cover of a loop, $k \geq 1$, whose corresponding class α is indivisible. By the assumption that all contractible g_Y geodesics are constant, Morse theory on the contractible component of the free loop space [38] tells us that Y has vanishing higher homotopy groups $\pi_j(Y, y_0)$, $j \geq 2$: the energy has only the constant loops as critical points there, so evaluation at the base point is a homotopy equivalence from that component to Y , with the constant loops as a section; its fiber, the space of null-homotopic based loops, is therefore weakly contractible, and $\pi_j(Y) = \pi_{j-1}$ of that fiber vanishes for $j \geq 2$. Thus by the long exact sequence of a fibration $i_{Z,*} : \pi_1(Z, z_0) \rightarrow \pi_1(X, i_Z(z_0))$ is a group injection, and image $i_{Z,*} = \ker p_*$. Now $\pi_1(Y, y_0)$ is torsion-free since Y is a finite dimensional $K(\pi_1(Y, y_0), 1)$ space, Hatcher [33, Prop. 2.45]. It follows that α is also a fiber class. First $\alpha \in \pi_1^{\text{inc}}(X)$: otherwise β is not, as the k -th power of a representative of α outside a compact set is one of β . Next, for a based representative a of α ,

$$p_*(a^k) = 1 \implies p_*(a) = 1,$$

so a comes from $\pi_1(Z_{y_0})$; and its class α_Z there is end incompressible, as Z_{y_0} is closed in X and a representative of α_Z outside a compact set of Z_{y_0} is a representative of α outside a compact set of X .

Lemma 8.1. *Let*

$$Z \hookrightarrow (X, g) \xrightarrow{p} (Y, g_Y)$$

be a g -parallel Riemannian submersion with complete total space, and assume that Y is connected with $\pi_2(Y) = 0$. Let

$$\beta, \gamma \in \pi_1^{\text{inc}}(X)$$

be fiber classes such that

$$\beta = m\gamma$$

for some $m \geq 1$. If g is β -taut, then g is γ -taut and p is γ -holonomy-finite.

Proof. We show γ -tautness. Every closed γ -geodesic o_γ has m -fold cover $o_\gamma^m \in S(g, \beta)$, so

$$\ell_g(o_\gamma) = \frac{1}{m} \ell_g(o_\gamma^m) \leq A/m, \quad A := \max_{o \in S(g, \beta)} \ell_g(o) < \infty,$$

where finiteness of A is β -tautness. Hence $S(g, \gamma) = S_{A/m}(g, \gamma)$, which is compact by Lemma 3.2.

We show γ -holonomy-finiteness. Choose

$$\gamma_Z \in \mathcal{R}(\gamma)$$

and set

$$S_{y_0} = \pi_1(Y, y_0) \cdot \gamma_Z.$$

Suppose, toward a contradiction, that S_{y_0} is infinite. For each

$$\gamma_i \in S_{y_0}$$

choose a closed g_{y_0} -geodesic

$$o_i \subset Z_{y_0}$$

representing γ_i . Such geodesics exist by Lemma 3.4. Since the classes γ_i are distinct in Z_{y_0} , the loops o_i are pairwise distinct as elements of the fiber loop space.

We first record a separation property for these loops in the ambient loop space LX . Choose a contractible trivializing neighborhood

$$U \subset Y$$

of y_0 . For a loop $o \in Z_{y_0}$, let

$$\mathcal{V}_{U,o} \subset LX$$

be the set of loops $\eta : S^1 \rightarrow X$ such that

$$p(\eta(S^1)) \subset U$$

and such that η , regarded as a loop in $p^{-1}(U)$, is freely homotopic in $p^{-1}(U)$ to o . The set $\mathcal{V}_{U,o}$ is open in LX : the condition that the image lie in $p^{-1}(U)$ is open, and free homotopy classes are open components of the loop space of $p^{-1}(U)$.

Moreover, if o_i and o_j represent distinct free homotopy classes in Z_{y_0} , then

$$\mathcal{V}_{U,o_i} \cap \mathcal{V}_{U,o_j} = \emptyset.$$

Indeed, an element of the intersection would give a free homotopy between o_i and o_j inside $p^{-1}(U)$. Since U is contractible, the inclusion

$$Z_{y_0} \hookrightarrow p^{-1}(U)$$

is a homotopy equivalence, and hence o_i and o_j would already be freely homotopic inside Z_{y_0} , contrary to construction.

On the other hand, $S(g, \gamma)$ is compact. Thus the infinite family $\{o_i\} \subset S(g, \gamma)$ has an accumulation point $o_\infty \in S(g, \gamma)$. Since every o_i lies in the fiber Z_{y_0} , the limit o_∞ also lies in Z_{y_0} . The open neighborhood $\mathcal{V}_{U,o_\infty}$ of o_∞ therefore contains o_i for infinitely many i , forcing these o_i 's to lie in the same fiber free homotopy class as o_∞ . This contradicts the fact that the classes γ_i are pairwise distinct. Hence S_{y_0} is finite and p is γ -holonomy-finite.

□

By Theorem 7.1 g is β -taut. By the lemma above g is α -taut and p is α -holonomy finite, hence an α -periodic fibration. Now, if $\chi(Y) \neq \pm 1$ then by the product formula of Theorem 7.1 $F(g, \alpha) \neq \pm 1$, since $F(g_{y_0}, \alpha_Z)$, for $\alpha_Z \in \mathcal{R}(\alpha)$, is an integer by Theorem 1.2 (1), α_Z being indivisible as α is.

Now, if X admits a complete metric h with a unique and nondegenerate class β h -geodesic string o_β , then it clearly also has a unique class α h -geodesic string o_α . Furthermore, o_α must be nondegenerate as otherwise o_β , which is a k -fold cover of o_α is degenerate. Then we have:

$$F(h, \alpha) = \pm 1.$$

But this is a contradiction, since $F(h, \alpha) = F(g, \alpha)$ by Theorem 1.2.

We now prove the general case. Suppose by contradiction that X admits a complete metric h with a unique and nondegenerate class β h -geodesic string o_β . By the preceding paragraph, the class β is not atomic. Now o_β covers a multiplicity one geodesic string o_α in some class $\alpha \in \pi_1^{inc}(X)$. Moreover, o_α is the unique geodesic string in its class, otherwise o_β would not be unique in its class. We prove that α is indivisible, which will be a contradiction to β not being atomic and will complete the proof.

Suppose otherwise, so that $\alpha_{x_0} = \rho^k$ for $k > 1$ and $\rho \in \pi_1(X, x_0)$ (with the notation of Definition 2.14). Let o_ρ be a class ρ , closed h -geodesic string (where by slight abuse ρ also denotes the class in $\pi_1^{inc}(X)$ corresponding to the based class ρ .) It is immediate that the k cover of o_ρ , o_ρ^k represents α and is a h -geodesic string. By the uniqueness, $o_\alpha = o_\rho^k$. But this contradicts multiplicity one of o_α . So α is indivisible.

□

Proof of Corollary 2.23. Let g_Z, g_Y be complete Riemannian metrics with g_Z taut, and g_Y not admitting nonconstant contractible closed geodesics. Take the product metric $g = g_Z \times g_Y$ on $X = Z \times Y$. The class $\beta = i_*(\beta_Z)$ is end incompressible: Y is compact, so a representative of β outside $K_Z \times Y$ projects to a representative of β_Z outside K_Z . And $\pi_2(Y) = 0$, as shown at the start of the proof of Theorem 2.22. So the projection $X \rightarrow Y$ is a β -periodic fibration, the holonomy being trivial, and the conclusion follows from Theorem 2.22. $\square$

Proof of Corollary 2.26. The first two parts follow from Theorem 2.22, Lemma 2.25 and Example 6. The last part is deduced as follows. The product formula Theorem 7.1 gives us $F(g_\phi, \beta) = 0$, and g_ϕ is β -taut, so Theorem 1.2 (1) gives $\chi((L_\beta X_\phi)_{hS^1}) = 0$, as β is indivisible. For any β -taut and β -regular g , with its finitely many class β geodesic strings o , the same theorem gives

$$0 = \chi((L_\beta X_\phi)_{hS^1}) = F(g, \beta) = \sum_o (-1)^{\text{morse}(o)}.$$

Then the conclusion readily follows. $\square$

Proof of Theorem 2.16. By Corollary 7.5 with $Y = T^2$, 0 is a value of the invariant F . We first prove that every negative rational number is the value of the invariant. Let p, q be positive integers. Let Y be a closed surface of genus $(p+1) > 1$ with a hyperbolic metric g_Y , let Z be the genus 2 closed surface with a hyperbolic metric g_Z and let $\beta_Z \in \pi_1^{\text{inc}}(Z)$ be the class represented by a $2 \cdot q$ -fold covering of a simple closed loop representing a generator of the fundamental group of Z .

Let $X = Y \times Z$ with the product metric $g = g_Y \times g_Z$ and the canonical projection $X \rightarrow Y$. Taking $\beta = (i_Z)_*(\beta_Z)$, by Corollary 7.5,

$$F(g, \beta) = \chi(Y) \cdot F(g_Z, \beta_Z) = (-2p) \cdot \frac{1}{2q} = -\frac{p}{q}.$$

So we proved our first claim.

Let again p, q be positive integers. Let Y be a closed surface of genus 2, with a hyperbolic metric g_Y . And let Z be a connected manifold satisfying $F(g_Z, \beta_Z) = -\frac{p}{2q}$ for some taut Riemannian metric g_Z on Z and for some class $\beta_Z \in \pi_1^{\text{inc}}(Z)$. This exists by the discussion above: a product of hyperbolic metrics has nonpositive curvature, hence is taut (Lemma 2.3). Let $g = g_Y \times g_Z$ be the product metric on $Y \times Z$, and β as above. Analogously to the discussion above we get:

$$F(g, \beta) = \chi(Y) \cdot F(g_Z, \beta_Z) = (-2) \cdot \frac{-p}{2q} = \frac{p}{q}.$$

$\square$

9. ARITHMETIC LAWS

Throughout this section we use the following notation. For g a metric, β an indivisible class and $d \geq 1$ such that g is $d\beta$ -taut and $d\beta$ -regular, P_d denotes the finite set of prime closed g -geodesic strings of class $d\beta$, and

$$s_d := \sum_{o \in P_d} i(o) \in \mathbb{Z}$$

its signed count, where $i(o)$ is the local Fuller index of o in the normalization (A.2) of Appendix A: $i(o) = (-1)^{n-1} \text{sign det}(I - \mathcal{R}_o) \in \{\pm 1\}$, n the dimension of the manifold, with $\mathcal{R}_o$ the linearized return map of o ; on a surface $i(o) = -\text{sign det}(I - \mathcal{R}_o)$. Under this normalization $i(o) = (-1)^{\text{morse}(o)}$, by (A.3), so that a length minimizing geodesic contributes +1. Finally, b_d denotes the number of elements of P_d of inverse-hyperbolic type ($\mathcal{R}_o$ has negative eigenvalues), and $B_x := \sum_{c|x} c b_c$ for x a positive integer, $B_x := 0$ when x is not an integer.

Proof of Lemma 1.8. Let a, b be based representatives of β, α , so that $xb^kx^{-1} = a^e$ for some $x \in \pi_1(X, x_0)$.

In the free abelian case, write $\pi_1(X, x_0) \cong \mathbb{Z}^N \subset \mathbb{Q}^N$ additively: $kb = ea$, so b is a rational multiple of a . As β is indivisible, a is a primitive vector, and a rational multiple of a primitive vector lying in $\mathbb{Z}^N$ is an integer multiple. So $b = ma$, and $mk = e$ with $m > 0$.

In the torsion-free hyperbolic case, every nontrivial element h lies in a unique maximal elementary subgroup $E(h)$, which is infinite cyclic and contains every abelian subgroup containing h [20], and two maximal elementary subgroups sharing a nontrivial element coincide. As β is indivisible, a is indivisible and $E(a) = \langle a \rangle$. The element $a^e = xb^kx^{-1}$ lies in $E(xbx^{-1})$ and in $E(a)$, so $E(xbx^{-1}) = \langle a \rangle$. Thus $xbx^{-1} = a^m$ for some $m \in \mathbb{Z}$, and $mk = e$ forces $m > 0$. $\square$

Lemma 9.1. *Let X be a closed orientable manifold, β an indivisible class, and $e \geq 1$. Let g be an $e\beta$ -taut and $e\beta$ -regular metric on X , and suppose that β satisfies the cancellation property (Definition 1.7). Then*

$$(9.2) \quad e F(g, e\beta) = \sum_{d|e} d \sum_{o \in P_d} i(o^{e/d}),$$

with P_d the set of prime geodesic strings of class $d\beta$ and i the local Fuller index (A.2).

Proof. Tautness and regularity of the divisor classes $d\beta$ are automatic: covers of strings are strings, and an eigenvalue 1 of $\mathcal{R}_o$ is one of $\mathcal{R}_o^{e/d}$. In particular each P_d , $d|e$, is finite. Every class $e\beta$ geodesic string is the k -fold cover o^k of a unique prime geodesic string o , and o^k has multiplicity k . The class α of o satisfies $k\alpha = e\beta$, so by hypothesis $o \in P_d$ with $dk = e$. So the class $e\beta$ strings are in bijection with the pairs (d, o) , where $d|e$ and $o \in P_d$, the pair corresponding to $o^{e/d}$, of multiplicity e/d . By the definition of F and Appendix A, the contribution of $o^{e/d}$ to $F(g, e\beta)$ is its local Fuller index divided by its multiplicity, that is $i(o^{e/d})/(e/d)$. Summing over the pairs and multiplying by e gives (9.2), since $e/(e/d) = d$. $\square$

Lemma 9.3. *Let o be a closed geodesic string of a complete Riemann-Finsler metric on a manifold, with linearized return map $\mathcal{R} = \mathcal{R}_o$, and let $j \geq 1$ be such that o and o^j are nondegenerate. If j is odd then $i(o^j) = i(o)$. If j is even and o^2 is nondegenerate then $i(o^j) = i(o^2)$.*

Proof. $\det(I - \mathcal{R}^j) = \prod_{\lambda} (1 - \lambda^j)$, over the eigenvalues of $\mathcal{R}$ with multiplicity, and the eigenvalue multiset is invariant under $\lambda \mapsto \lambda^{-1}$ and under conjugation. A pair $\{\lambda, \lambda^{-1}\}$, $\lambda > 1$, contributes $(1 - \lambda^j)(1 - \lambda^{-j}) < 0$ for every j . A pair $\{-\lambda, -\lambda^{-1}\}$, $\lambda \geq 1$, contributes a positive factor for j odd and a negative one for j even, when nonzero. A pair $\{e^{\pm i\theta}\}$ contributes $|1 - e^{ij\theta}|^2 > 0$ when nonzero. A quadruple $\{\lambda, \bar{\lambda}, \lambda^{-1}, \bar{\lambda}^{-1}\}$ off the real and unit loci contributes $|1 - \lambda^j|^2 |1 - \bar{\lambda}^{-j}|^2 > 0$. The factors are nonzero exactly at the nondegenerate iterates. So $\text{sign } \det(I - \mathcal{R}^j)$ depends only on the parity of j , where defined, and by (A.2) so does $i(o^j)$. $\square$

Proof of Theorem 1.9. The hypotheses of Lemma 9.1 hold at every divisor level $e'|e$, tautness and regularity being automatic. For $d|e'$ the exponent e'/d is odd, and the iterates $o^{e'/d}$, $o \in P_d$, are nondegenerate by $e'\beta$ -regularity; so Lemma 9.3 gives $\sum_{o \in P_d} i(o^{e'/d}) = s_d$, and (9.2) at level e' reads

$$e' F(g, e'\beta) = \sum_{d|e'} d s_d.$$

This is (1.10), and Möbius inversion over the divisors of e gives (1.11). If $F(g, d\beta) = 0$ for every $d|e$, then (1.11) gives $s_e = 0$. If $F(g, d\beta) = 1/d$ for every $d|e$, then (1.11) gives

$$s_e = \frac{1}{e} \sum_{d|e} \mu(e/d) = 0$$

for $e > 1$. In both cases each prime geodesic string contributes ± 1 to s_e , so $\#P_e$ is even. $\square$

Lemma 9.4. *Let o be a prime closed geodesic of a Riemann-Finsler metric on a closed surface, with linearized return map $\mathcal{R} = \mathcal{R}_o \in \mathrm{Sp}(2, \mathbb{R})$, and let $k \geq 1$ be such that o and o^k are nondegenerate. Then $\mathcal{R}$ is positive hyperbolic (eigenvalues $\lambda^{\pm 1}$, $\lambda > 1$), inverse-hyperbolic (negative eigenvalues $-\lambda^{\pm 1}$, $\lambda \geq 1$) or elliptic (eigenvalues $e^{\pm i\theta}$, $0 < \theta < \pi$), and*

$$\mathrm{sign} \det(I - \mathcal{R}^k) = \begin{cases} -1 & \mathcal{R} \text{ positive hyperbolic,} \\ +1 & \mathcal{R} \text{ elliptic,} \\ (-1)^{k+1} & \mathcal{R} \text{ inverse-hyperbolic.} \end{cases}$$

Proof. The eigenvalues form a pair $\{\mu, \mu^{-1}\}$, real or on the unit circle, and nondegeneracy of o excludes $\mu = 1$; the three cases are the ones listed. If the eigenvalues are $\lambda^{\pm 1}$, $\lambda > 1$, then

$$\det(I - \mathcal{R}^k) = (1 - \lambda^k)(1 - \lambda^{-k}) < 0.$$

If they are $e^{\pm i\theta}$, then

$$\det(I - \mathcal{R}^k) = |1 - e^{ik\theta}|^2 > 0,$$

nonzero by nondegeneracy of o^k . If they are $-\lambda^{\pm 1}$, $\lambda > 1$, then

$$\det(I - \mathcal{R}^k) = (1 - (-\lambda)^k)(1 - (-\lambda)^{-k}),$$

positive for k odd and negative for k even. If they are $-1, -1$, then $\det(I - \mathcal{R}^k) = (1 - (-1)^k)^2$: this vanishes for k even, which nondegeneracy of o^k excludes, and equals 4 for k odd. $\square$

The vanishing of F on T^2 , Theorem 6.8, has a complete arithmetic expression in terms of prime geodesic strings. A Finsler metric that is β -regular for all β is C^∞ -generic by the bumpy metric theorem, in the Riemannian case due to Abraham [1] and Anosov [5] and in the Finsler case to Rademacher–Taimanov [56]; for such g , in each taut class the geodesic strings are isolated points of a compact set, hence finite in number. Fix an indivisible class β_0 ; the notation of the section preamble applies with $\beta = \beta_0$.

Proposition 9.5. *Let g be a Riemann-Finsler metric on T^2 , β_0 an indivisible class, and $e \geq 1$. Suppose g is $e\beta_0$ -taut and $e\beta_0$ -regular. Then*

$$(9.6) \quad s_e = -b_{e/2},$$

where $b_{e/2} := 0$ for e odd.

Tautness and regularity of the divisor classes are automatic, as in the proof of Lemma 9.1. In particular the counts are defined. The content of Proposition 9.5 lies in the divisible classes, which have no simple representatives, and where the minimizers, as well as whole sweepout families, may consist of iterates of class β_0 geodesics; the prime counts are then invisible to such methods, which do not track isotropy.

Proof of Proposition 9.5. $\pi_1(T^2) = \mathbb{Z}^2$ is free abelian, so β_0 satisfies the cancellation property (Lemma 1.8) and Lemma 9.1 applies at every divisor level d of e (tautness and regularity being automatic, as noted above), giving

$$(9.7) \quad dF(g, d\beta_0) = \sum_{c|d} c \sum_{o \in P_c} i(o^{d/c}).$$

Now let u_c be the number of positive hyperbolic minus the number of elliptic elements of P_c , so that $s_c = u_c - b_c$ and, by Lemma 9.4, $\sum_{o \in P_c} i(o^k) = u_c + (-1)^k b_c$ for every k with $ck | e$: o and o^k are then

nondegenerate, their classes $c\beta_0$ and $ck\beta_0$ being divisor classes. Since $(-1)^{d/c} = 1$ exactly when d/c is even, that is when c divides $d/2$, substitution in (9.7), with $k = d/c$, gives

$$dF(g, d\beta_0) = \sum_{c|d} c(u_c - b_c) + 2 \sum_{c|d/2} cb_c = \sum_{c|d} cs_c + 2 \sum_{c|d/2} cb_c,$$

the second sum being empty for d odd. With B_x as in the section preamble, the vanishing $F(g, d\beta_0) = 0$ of Theorem 6.8, on the divisors d of e , reads

$$(9.8) \quad \sum_{c|d} cs_c = -2B_{d/2} \quad \text{for every divisor } d \text{ of } e.$$

Now for any d ,

$$(9.9) \quad \sum_{c|d} cb_{c/2} = \sum_{2f|d} 2fb_f = 2B_{d/2},$$

since the terms with c odd vanish, and the even divisors of d are exactly $c = 2f$ with f a divisor of $d/2$. Argue by induction on the divisor d of e , the case $d = 1$ of (9.8) being $s_1 = -2B_{1/2} = 0$. Assuming $s_c = -b_{c/2}$ for every proper divisor c of d , (9.8) and (9.9) give

$$ds_d = -2B_{d/2} + \sum_{c|d, c \neq d} cb_{c/2} = -2B_{d/2} + (2B_{d/2} - db_{d/2}) = -db_{d/2},$$

hence $s_d = -b_{d/2}$; at $d = e$ this is (9.6). $\square$

Proposition 9.5 is the law behind Theorem 1.3 on T^2 . The counterpart in higher genus:

Corollary 9.10. *Let Σ be a closed orientable surface of genus ≥ 2 , β_0 an indivisible class, $e \geq 1$, and g an $e\beta_0$ -taut and $e\beta_0$ -regular Riemann-Finsler metric on Σ . Then*

$$\sum_{c|d} cs_c = 1 - 2B_{d/2} \quad \text{for every divisor } d \text{ of } e,$$

so that $s_1 = 1$ and $s_d = -b_{d/2}$ for every divisor $d > 1$ of e .

Proof. Tautness and regularity of the divisor classes are automatic, and $F(g, d\beta_0) = 1/d$ for every $d|e$ by Theorem 6.12. The class β_0 satisfies the cancellation property, $\pi_1(\Sigma)$ being torsion-free hyperbolic (Lemma 1.8), and Lemma 9.4 applies. So the derivation of (9.8) in the proof of Proposition 9.5 goes through with $dF(g, d\beta_0) = 1$ in place of 0, giving the displayed system. The system determines s_d , $d|e$, recursively from the b_c , and $s_1 = 1$, $s_d = -b_{d/2}$ for $d > 1$ solves it, by (9.9). $\square$

Closeness to flat, Clairaut or negative curvature metrics enters Theorem 1.3 only through tautness. Proposition 9.5 is not accessible to S^1 -equivariant Morse homology techniques, as these do not suitably track isotropy information. Nevertheless, there is a family resemblance with the resonance identities of closed geodesic and closed characteristic theory, where finite collections of periodic objects satisfy identities involving average Euler characteristics and mean indices; see, for example [55, 36] and the later noncontractible resonance identities [40]. The constraints of the present paper are of a different character: the resonance identities are real valued relations driven by the asymptotics of the index, while ours are exact integer identities among the signed counts of prime geodesic strings, driven by the value of a rational count.

Proof of Theorem 1.3. Tautness and regularity of the divisor classes of $e\beta_0$ are automatic, as noted after Proposition 9.5. On T^2 , Proposition 9.5 gives $s_e = -b_{e/2}$. For e odd this reads $s_e = 0$; each prime geodesic string contributes ± 1 , so their number is even. For $e = 2n$ it reads $s_{2n} = -b_n$, so $\#P_{2n} \geq |s_{2n}| = b_n$. In genus ≥ 2 , Corollary 9.10 gives $s_1 = 1$ and $s_d = -b_{d/2}$ for $1 < d|e$. So for e odd, $s_e = 0$ when $e > 1$, giving the pairing, while $s_1 = 1$ makes $\#P_1$ odd; and for $e = 2n$, $\#P_{2n} \geq |s_{2n}| = b_n$ as before. $\square$

10. TAUTNESS VIA GROMOV HYPERBOLICITY

In this section we prove Proposition 2.7. We first recall the needed notions, following [20]. Let Y be a metric space, and $I \subset \mathbb{R}$ an interval. A map $q : I \rightarrow Y$ is a (λ, c) -**quasigeodesic** if for all $s, t \in I$:

$$\lambda^{-1}|s - t| - c \leq d(q(s), q(t)) \leq \lambda|s - t| + c.$$

It is an r -**local** (λ, c) -**quasigeodesic** if its restriction to every subinterval of length r is a (λ, c) -quasigeodesic. We use the local-to-global theorem for quasigeodesics [26, Chapitre 3, Théorème 1.4], see also [20, III.H.1.13] for local geodesics: for a δ -hyperbolic geodesic space Y and constants (λ_0, c_0) , there are $r = r(\delta, \lambda_0, c_0)$ and $(\lambda, c) = (\lambda, c)(\delta, \lambda_0, c_0)$, such that every r -local (λ_0, c_0) -quasigeodesic of Y is a (λ, c) -quasigeodesic.

We also use the Gronwall inequality, in the following standard form, see [31] and [32, Chapter III]. Let F, G be time dependent vector fields on an open subset of $\mathbb{R}^m$, with $|F - G| \leq \epsilon'$, and with F Λ -Lipschitz in the space variable. Suppose y, z solve $\dot{y} = F(t, y)$, $\dot{z} = G(t, z)$, with $y(0) = z(0)$. Then, while both solutions are defined,

$$(10.1) \quad |y(t) - z(t)| \leq \frac{\epsilon'}{\Lambda} (e^{\Lambda|t|} - 1).$$

Indeed $\frac{d}{dt}|y - z| \leq |F(t, y) - F(t, z)| + |F(t, z) - G(t, z)| \leq \Lambda|y - z| + \epsilon'$, and (10.1) solves this differential inequality.

Proof of Proposition 2.7. Let $K(g_-) \leq -\kappa^2 < 0$. The universal cover $\tilde{X}$, with the lift of g_- , is a $\text{CAT}(-\kappa^2)$ space, so a $\delta(\kappa)$ -hyperbolic geodesic space [20, III.H]. Fix $r = r(\delta, 2, 1)$ and (λ, c) as in the local-to-global theorem, so both depend only on g_- .

Let o be a g' -geodesic of g' -unit speed, and $\tilde{o}$ a lift to $\tilde{X}$. As $\|g' - g_-\|_{C^1} < \epsilon$, its g_- -speed lies in $[1 - C\epsilon, 1 + C\epsilon]$, for a constant $C = C(g_-)$, as are all constants C in this proof. Christoffel symbols involve one derivative of the metric, so o has g_- -geodesic curvature at most $C\epsilon$. Compare o , over a parameter arc of length r , with the g_- -geodesic sharing its initial position and velocity. Written as first order systems on TX , in charts controlled by g_- , the two solve equations as in (10.1) with $\epsilon' = C\epsilon$ and $\Lambda = \Lambda(g_-)$; so their C^0 -distance over the arc is at most $C(r)\epsilon$. A g_- -geodesic of $\tilde{X}$ is minimizing, and covers distance $|s - t|$ times its speed. So on the arc, for ϵ small,

$$\frac{1}{2}|s - t| - 1 \leq (1 - C\epsilon)|s - t| - 2C(r)\epsilon \leq d(\tilde{o}(s), \tilde{o}(t)) \leq (1 + C\epsilon)|s - t|.$$

This is where the speed bound enters, converting parameter differences to g_- -distances. So $\tilde{o}$ is an r -local $(2, 1)$ -quasigeodesic, and by the local-to-global theorem a (λ, c) -quasigeodesic.

Now let o be a class β closed g' -geodesic, and $\tilde{o}$ a lift invariant under a deck transformation γ representing β . The translation length of γ is L_β — the g_- -length of the closed g_- -geodesic in class β . Fix a on the axis of γ and set $D = d(\tilde{o}(0), a)$. Over n periods the quasigeodesic inequality gives

$$n \ell_{g'}(o) \leq \lambda d(\tilde{o}(0), \gamma^n \tilde{o}(0)) + \lambda c \leq \lambda n L_\beta + \lambda(2D + c),$$

as o has g' -unit speed, and $n \rightarrow \infty$ yields $\ell_{g'}(o) \leq \lambda L_\beta$: the constant D drops out. So in every class the geodesics of every metric in the ball have g' -lengths at most λL_β . The ball is convex, so for g'_0, g'_1 in it the linear path stays in it, the length spread hypothesis of Theorem 2.5 holds in every class, and the path is a taut deformation. Theorem 2.8(3) gives the last clause. $\square$

11. TAUTNESS NEAR CLAIRAUT METRICS VIA KAM CONFINEMENT

A Riemannian metric on T^2 invariant under a free circle action is a **Clairaut metric**: in coordinates with x the orbit parameter and y the arc length along the trajectories orthogonal to the orbits, it reads

$$(11.1) \quad g_A = dy^2 + A(y)^2 dx^2 \quad \text{on } T^2 = \mathbb{R}^2/\Lambda, \quad \Lambda = \mathbb{Z}(1, 0) \oplus \mathbb{Z}(h, L),$$

with $A : \mathbb{R}/L\mathbb{Z} \rightarrow (0, \infty)$ smooth, L the length of the quotient circle, and $h \in \mathbb{R}$ the holonomy of the orthogonal trajectories around it. The flat metrics are the cases of constant A . We show that every Riemann-Finsler metric on T^2 sufficiently close to g_A is taut, with a length bound in each class, and is taut homotopic to g_A , hence to a flat metric. The deck vector of a class β is written $(m, n) \in \mathbb{Z}^2 - \{0\}$: the lift of a loop of class β is displaced by $m(1, 0) + n(h, L)$. We write $A_{\max}, A_{\min}$ for the extrema of A .

Proposition 11.2 (Near-Clairaut rigidity). *There is an absolute $k_0 \in \mathbb{N}$, and for every Clairaut metric g_A on T^2 , (11.1), a $\delta_A > 0$, with the following property. Every Riemann-Finsler metric g' on T^2 with $\|g' - g_A\|_{C^{k_0}} < \delta_A$ is taut, with every closed g' -geodesic o of class (m, n) satisfying*

$$(11.3) \quad \ell_{g'}(o) \leq C_A(|m| + |n|), \quad C_A := \max\{8(1 + |h|)A_{\max}^2/A_{\min}, 2L\}.$$

Any two metrics in the δ_A -ball are joined by a taut deformation, and every metric in the ball is taut homotopic to a flat metric.

Note that δ_A does not depend on β . For Finsler g' , $\|g' - g_A\|_{C^{k_0}}$ denotes the C^{k_0} -distance of the squared norms $|\cdot|_{g'}^2$ and $|\cdot|_{g_A}^2$ on $\{\frac{1}{2} \leq |\cdot|_{g_A} \leq 2\} \subset TT^2$. We set $\aleph := k_0$ and $h(g_A) := \delta_A$: these are the constants $\aleph, h$ of the Introduction, and with them the last two sentences of the proposition give Theorem 1.6.

The proof is effective: the order k_0 is determined by the smoothness threshold of the invariant curve theorem, and the threshold $\delta_0(A, r, \eta)$ of Theorem 11.4 is explicitly estimable from the quantitative finite smoothness forms of that theorem [35, 52]. So δ_A is in principle computable. We do not pursue this here: thresholds extracted by hand from KAM iterations are far smaller than the true breakdown of invariant circles, and a realistic δ_A would be a computer-assisted computation.

The classical theory of minimal geodesics on the torus (Morse [48], Hedlund [34], Bangert [8], Mather [44]) controls, for every metric, the *minimizers* in each class. Proposition 11.2 is orthogonal to it: the bound is on *all* class β closed geodesics, not just the minimizers, and no minimizer-theoretic argument reaches it. The mechanism is confinement by invariant tori of the geodesic flow which are graphs over T^2 and hence, in pairs, separate the unit tangent bundle. Intrinsically this graph property is the second Birkhoff theorem, established for Lagrangian invariant tori by Bialy–Polterovich and Polterovich [18, 51, 15, 14] and, without a dynamical assumption, by Arnaud [6]; for the merely continuous invariant tori of T^2 geodesic flows it is due to Bialy [16] (see also [17]). In the perturbative regime below it comes for free from KAM. The smallness of δ_A is necessary: the metric of Example 1 is not β -taut for any nontrivial β .

We expect taut metrics on T^2 whose geodesic flow carries no incompressible invariant torus at all. Exhibiting one is a converse KAM problem, not taken up here. In higher genus the negatively curved metrics are such examples: their geodesic flow is Anosov, so it has no invariant torus, and they are taut. For the metrics of Proposition 2.7 the quasigeodesic argument of Section 10 leaves room only for incompressible invariant tori all of whose geodesics are bi-asymptotic, in the universal cover, to one axis of the deck group; we do not know whether these occur.

11.1. Setup. Here and below C denotes a positive constant depending only on A , whose value may change between occurrences. The unit-speed geodesic flow of g' is the Hamiltonian flow of $H'(x, y, p) = \frac{1}{2}F'^*(x, y, p)^2$ on $\Sigma' = \{H' = \frac{1}{2}\} \subset T^*T^2$, where F'^* is the dual norm; a periodic orbit of period T projects to a closed geodesic of length T . The dual Hamiltonian of g_A is

$$H_A(x, y, p_x, p_y) = \frac{1}{2}(p_x^2/A(y)^2 + p_y^2),$$

with unit level Σ_A . The fiberwise Legendre dual costs at most one derivative, so, with $\delta := \|g' - g_A\|_{C^{k_0}}$,

$$\|H' - H_A\|_{C^{k_0-1}(\mathcal{A})} \leq C\delta \quad \text{on} \quad \mathcal{A} = \{\tfrac{1}{4} \leq H_A \leq 1\}.$$

For δ small, $D_p^2 H'$ is close to $D_p^2 H_A$, so each fiber $\{p : H'(x, y, p) = \frac{1}{2}\}$ of $\Sigma' \rightarrow T^2$ is a strictly convex curve enclosing the origin, C^1 -close to the ellipse $\{H_A(y, p) = \frac{1}{2}\}$.

The Clairaut flow. As H_A is x -independent, p_x is conserved. On the band

$$B := \{|p_x| \leq \frac{3}{4}A_{\min}\} \cap \Sigma_A$$

we have $p_y^2 = 1 - p_x^2/A^2 \geq 7/16$, so B has two sheets $B_{\pm} = B \cap \{\pm p_y > 0\}$, each parametrized by $(x, y, p_x) \in T^2 \times [-\frac{3}{4}A_{\min}, \frac{3}{4}A_{\min}]$ and foliated by the invariant tori $T_{\pm, p} := \{p_x = p\} \cap B_{\pm}$, graphs over T^2 . On $B_{\pm}$ the velocity $\dot{y} = p_y$ has a fixed sign and $|\dot{y}| \geq \sqrt{7}/4$, so the flow returns to the section $\{y = 0\}$ in bounded time. Over one y -period, x advances by

$$\omega(p) := \int_0^L \frac{p dy}{A(y)\sqrt{A(y)^2 - p^2}},$$

the same on both sheets, and the point $(x, \pm L)$ is identified with $(x \mp h, 0)$. So the return maps are

$$P_{\pm}(x, p) = (x + \omega(p) \mp h, p),$$

integrable twist maps of the annulus $\mathbb{R}/\mathbb{Z} \times [-\frac{3}{4}A_{\min}, \frac{3}{4}A_{\min}]$:

$$\omega'(p) = \int_0^L \frac{A(y) dy}{(A(y)^2 - p^2)^{3/2}} \geq \int_0^L A(y)^{-2} dy > 0.$$

Along the H' -flow, $\dot{p}_x = -\partial_x(H' - H_A) = O(\delta)$, so over the bounded return time p_x moves by $O(\delta)$, and $\dot{y} = \partial_{p_y} H' = p_y + O(\delta)$ keeps its sign. So on each sheet the H' -flow on Σ' has a return map $P'_{\pm}$ to $\{y = 0\}$, defined on the annulus $|p_x| \leq \frac{3}{4}A_{\min} - C\delta$, exact area-preserving as a Poincaré map of a Hamiltonian flow on an energy level, and C^{k_0-2} -close to $P_{\pm}$: one derivative is lost in the Legendre transform and one in the return map.

11.2. The KAM input. For a real number $r > 0$, call $p \in [-\frac{5}{8}A_{\min}, \frac{5}{8}A_{\min}]$ *good* if $\omega(p) - h$ and $\omega(p) + h$ are both Diophantine of exponent 3 with constant r : $|\omega(p) \mp h - a/b| \geq rb^{-3}$ for all integers a and $b \geq 1$. These are the rotation numbers of P_+ and P_- on the circle $\{p_x = p\}$. As ω is a diffeomorphism onto its image, and the Diophantine numbers of constant r have measure tending to full as $r \rightarrow 0$, for r small there is a good $p_0 \in [\frac{1}{2}A_{\min}, \frac{5}{8}A_{\min}]$; as ω is odd and Diophantine numbers come in pairs $\pm$, $-p_0$ is good as well.

Theorem 11.4 (Invariant tori of the band). *There is an absolute k_0 such that for every Clairaut metric g_A and every $r, \eta > 0$ there is $\delta_0 = \delta_0(A, r, \eta) > 0$ with the following property. If $\|g' - g_A\|_{C^{k_0}} \leq \delta_0$, then for every good p_0 and each sign, the H' -flow on Σ' has an invariant torus $T'_{\pm, p_0}$, a graph over T^2 , C^1 -close to $T_{\pm, p_0}$, on which $|p_x - p_0| \leq 2\eta$.*

This is Moser's invariant curve theorem [49], in Herman's finite smoothness form [35], applied to the exact area-preserving map $P'_{\pm}$, which is C^{k_0-2} -close to the integrable twist map $P_{\pm}$, with twist bounded below by $\int_0^L A^{-2} dy$ and Diophantine rotation number $\omega(p_0) \mp h$ of exponent 3. It gives an invariant curve within η of the circle $\{p_x = p_0\}$ in C^1 ; its saturation under the H' -flow is the torus $T'_{\pm, p_0}$, a graph by C^1 -closeness to the graph $T_{\pm, p_0}$, and p_x drifts by $O(\delta) \leq \eta$ along the flow between returns. This route determines the absolute constant k_0 : two derivatives lost as above, and the remainder the smoothness threshold of the curve theorem for exponent 3. Both inputs are explicit, so k_0 is effective; we do not optimize its value. For the KAM lineage of the statement—not needed for the proof—see Arnold [7] for the analytic case, Pöschel [52, 53] for quantitative versions in finite smoothness, and Broer–Huitema [21] for the isoenergetic form; the twist of the return map is the isoenergetic nondegeneracy of H_A .

11.3. Proof of Proposition 11.2.

Proof. Fix r small, a good $p_0 \in [\frac{1}{2}A_{\min}, \frac{5}{8}A_{\min}]$, and η small; let $\delta_1 \leq \delta_0(A, r, \eta)$ be a threshold, reduced if necessary for the $O(\delta)$ estimates below, and let $\delta \leq \delta_1$. Theorem 11.4 gives the four invariant tori

$$T'_{+,p_0}, \quad T'_{-,p_0}, \quad T'_{+,-p_0}, \quad T'_{-,-p_0} \subset \Sigma',$$

graphs over T^2 , on which $|p_x \mp p_0| \leq 2\eta$.

Confinement. The fiber of $\Sigma' \rightarrow T^2$ over a point is a circle of directions, and the four tori mark four points on it, in the cyclic order of the directions they approximate: $T'_{+,p_0}, T'_{-,p_0}, T'_{-,-p_0}, T'_{+,-p_0}$. The complement of the tori in Σ' therefore has four components, each an interval bundle over T^2 : M_+ , between T'_{+,p_0} and T'_{-,p_0} , containing the direction ∂_x ; M_- , between $T'_{-,-p_0}$ and $T'_{+,-p_0}$, containing $-\partial_x$; L_+ , between $T'_{+,-p_0}$ and T'_{+,p_0} , containing ∂_y ; and L_- , between T'_{-,p_0} and $T'_{-,-p_0}$, containing $-\partial_y$. The fibers of Σ' are strictly convex curves, so a linear function of p is monotone on the two arcs joining its maximum point to its minimum point; on an arc containing its maximum point it is at least its values at the endpoints, and on an arc containing its minimum point at most. The maximum of p_x lies in M_+ , its minimum in M_- , and those of p_y in $L_{\pm}$. On M_+ , $p_x \geq p_0 - 2\eta \geq \frac{1}{2}A_{\min} - 2\eta$, hence

$$\dot{x} = \partial_{p_x} H' = p_x/A^2 + O(\delta) \geq A_{\min}/(8A_{\max}^2)$$

for η and δ small; on M_- likewise $\dot{x} \leq -A_{\min}/(8A_{\max}^2)$. On L_+ , $-p_0 - 2\eta \leq p_x \leq p_0 + 2\eta$, so $|p_x| \leq \frac{5}{8}A_{\min} + 2\eta$ and

$$\dot{y} = p_y + O(\delta) \geq \sqrt{1 - (5/8 + 2\eta/A_{\min})^2} - O(\delta) \geq \frac{1}{2}$$

for η and δ small; on L_- likewise $\dot{y} \leq -\frac{1}{2}$. On the four tori $|p_x| \leq \frac{5}{8}A_{\min} + 2\eta$ as well, so $|\dot{y}| \geq \frac{1}{2}$ there. An orbit of the H' -flow never crosses an invariant torus, so it stays in one region, or on one torus. This is where two degrees of freedom enter: the tori have codimension one in Σ' , and a pair of graph-tori separates it.

Length bound. Let o be a closed g' -geodesic of class (m, n) and period $T = \ell_{g'}(o)$, with lift $(\tilde{x}, \tilde{y})$, so that $\tilde{x}(T) - \tilde{x}(0) = m + nh$ and $\tilde{y}(T) - \tilde{y}(0) = nL$. If o lies in $M_{\pm}$ then $|m| + |h||n| \geq |m + nh| = |\int_0^T \dot{x} dt| \geq TA_{\min}/(8A_{\max}^2)$. If o lies in $L_{\pm}$ or on a torus then $|n|L \geq T/2$. This is (11.3), with $\delta_A := \delta_1$.

Tautness and the deformation. By (11.3), $S(g', \beta)$ consists of geodesics of length at most $C_A(|m| + |n|)$, so it is compact by Lemma 3.2: g' is taut. For g', g'' in the δ_A -ball, interpolate the squared norms, $F_t^2 = (1-t)F'^2 + tF''^2$: strong convexity and 2-homogeneity are preserved, and by convexity of the ball each g_t satisfies (11.3). Hence for each β the length spread satisfies

$$\sup_{t \in [0,1]} \left(\sup_{S(g_t, \beta)} \ell_{g_t} - \inf_{S(g_t, \beta)} \ell_{g_t} \right) \leq C_A(|m| + |n|) < \infty,$$

and Theorem 2.5 shows $\{g_t\}$ is β -taut for every β , i.e. a taut deformation.

The flat metric. Finally, g_A is taut homotopic to a flat metric, through the Clairaut metrics g_{A_t} , $A_t := ((1-t)A^2 + t)^{1/2}$, $t \in [0, 1]$, the path of the proof of Proposition 1.5; concatenated with the deformation inside the ball, this gives the remaining clause. For a fiber class $(m, 0)$ that proof applies verbatim, its constants being uniform in t . Let (m, n) be a class with $n \neq 0$. A closed g_{A_t} -geodesic with y -winding has $|p_x| < \min A_t$: for $|p_x| > \min A_t$ the geodesic is confined between two consecutive solutions of $A_t(y) = |p_x|$, and for $|p_x| = \min A_t$ the y -period

$$\tau_t(p) := \int_0^L \frac{dy}{\sqrt{1 - p^2/A_t(y)^2}}$$

is infinite. So the geodesic lies on a rotational torus $\{p_x = p, \pm p_y > 0\}$ of g_{A_t} , with $|n|\omega_t(p) = m + nh$, where ω_t is ω with A_t in place of A : the lift advances by $|n|\omega_t(p)$ in x over its $|n|$ periods in y , and by the deck vector $m(1, 0) + n(h, L)$. As ω_t is an increasing bijection of $(-\min A_t, \min A_t)$ onto $\mathbb{R}$, this

determines $p = p_t$, continuous in t . The flow lines of that torus are the x -translates of one another, so the class (m, n) geodesic strings of g_{A_t} form a circle, each of length $|n| \tau_t(p_t)$, continuous in t . The pairs (o, t) thus form a compact set, and the path is a taut deformation. $\square$

Remark 11.5. Dimension two is essential and enters only in the confinement step: on T^n , $n \geq 3$, KAM tori have codimension $n - 1 \geq 2$ in the energy level and separate nothing; the analogue of Proposition 11.2 is open there and adjacent to Arnold diffusion. Nothing soft can replace the confinement: pointwise bounds $|K| \leq \delta$ do not exclude long class β geodesics—the mushroom of Example 1, scaled by a large constant, has $|K| \leq \delta$ and is not β -taut for any β —so the dynamical confinement does real work. For flat g_A every invariant torus of Σ_A is rotational, every direction is confined, and the argument sharpens the bound to $(1 + \varepsilon)L_\beta$ on a smaller ball, with L_β the minimal g_A -length in the class; we do not need this.

12. EXAMPLES

12.1. Clairaut metrics on T^n .

Proof of Proposition 1.5. The momenta $p = a(y)\dot{x}$ are conserved, and for a unit speed geodesic

$$\dot{y}^2 + V_p(y) = 1,$$

where

$$V_p(y) := \langle p, a(y)^{-1}p \rangle.$$

A class β geodesic has zero y -winding: the lift of its S^1 -coordinate satisfies $\int_0^\ell \dot{y} dt = 0$, for ℓ the length. So $\dot{y}$ vanishes somewhere, and there $V_p = 1$, by unit speed. Hence $|p| \geq c > 0$, with c depending only on a .

Write $\beta = (m, 0)$ as in the statement of the proposition: the lift of the x -coordinate to $\mathbb{R}^{n-1}$ advances by m over the period, that is $\int_0^\ell \dot{x} dt = m$. Now $\dot{x} = a(y)^{-1}p$, and the eigenvalues of a^{-1} are bounded below by $\lambda > 0$; so over the full length ℓ ,

$$|m| = \left| \int_0^\ell \dot{x} dt \right| \geq \lambda \ell |p| \geq \lambda c \ell,$$

so no cancellation in the advance is possible, and the length is bounded in class β . Tautness follows by Lemma 3.2. For the deformation, interpolate $a_t(y) = (1 - t)a(y) + t\bar{a}$, with $\bar{a}$ the constant matrix splitting $T^n = T^{n-1} \times S^1$ as a Riemannian product. Now, each a_t is Clairaut, the constants above are uniform in t , so the space of pairs (o, t) , with o a class β g_t -geodesic, is compact — a β -taut deformation. For the product flat metric, Corollary 7.5 with $Y = S^1$ gives $F = 0$, as $\chi(S^1) = 0$; so $F(g, \beta) = 0$ by Theorem 2.8(3). $\square$

12.2. A period doubling far from flat. We prove Theorem 1.13. Throughout, $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ with coordinates (x, y) , and the metrics have the form $g = dy^2 + A(x, y)^2 dx^2$, for which we use the following facts. The Gauss curvature is $K = -\partial_y^2 A/A$. A circle $\{y = c\}$ is a g -geodesic if and only if $\partial_y A(x, c) \equiv 0$, its geodesic curvature being $\pm \partial_y A/A$, and then its length is $\int_0^1 A(x, c) dx$. When moreover $A(x, c) \equiv 1$, the circle is parametrized by x at unit speed, and its linearized return map $\mathcal{R}$ is the monodromy over $x \in [0, 1]$ of the Hill equation

$$J'' + K(x)J = 0, \quad K(x) = -\partial_y^2 A(x, c),$$

for the normal Jacobi fields $J\partial_y$. For any closed geodesic, parametrized by arc length s , $\mathcal{R}$ is likewise the monodromy of $J'' + K(s)J = 0$ over one period.

The Clairaut metric. The profile $A(y) = 1 - \frac{1}{2} \sin^2 \pi y = \frac{3}{4} + \frac{1}{4} \cos 2\pi y$ has critical points $y = 0$, where $A = 1$, and $y = 1/2$, where $A = 1/2$, and no others. So the class β_0 circles that are g_A -geodesics are $o := \{y = 0\}$, of length 1, and $o' := \{y = 1/2\}$, of length $1/2$. The curvature is

$$K = -\frac{A''}{A} = \frac{4\pi^2 \cos 2\pi y}{3 + \cos 2\pi y},$$

equal to π^2 along o and to $-2\pi^2$ along o' . Along o the Jacobi equation $J'' + \pi^2 J = 0$ has monodromy $-I$ over $x \in [0, 1]$: o is nondegenerate, $\det(I - \mathcal{R}_o) = 4$, $i(o) = -1$, and the double cover o^2 is degenerate. The circle o' is the minimizer of its class, as $A \geq 1/2$ with equality only at $y = 1/2$, and $K < 0$ along it: it is positive hyperbolic, with $i(o') = +1$.

Proof of Theorem 1.13. (1). g_A is a Clairaut metric, so Proposition 11.2 gives $\delta := \delta_A$ such that every g with $\|g - g_A\|_{C^\infty} < \delta$ is taut and taut homotopic to g_A , hence to a flat metric. Since $A_\epsilon - A = -\frac{\epsilon}{2} \cos(2\pi x) \sin^2 \pi y$, we have $\|g_\epsilon - g_A\|_{C^\infty} \leq C\epsilon$ with $C = C(\mathbb{N})$, and g_ϵ lies in the ball for $\epsilon < \epsilon_0 \leq \delta/C$.

(2). For every ϵ ,

$$\partial_y A_\epsilon = -\frac{\pi}{2}(1 + \epsilon \cos 2\pi x) \sin 2\pi y$$

vanishes on $y = 0$ and on $y = 1/2$, so o and o' are g_ϵ -geodesics, prime as β_0 is indivisible, of lengths 1 and $\int_0^1 A_\epsilon(x, 1/2) dx = 1/2$. Along o , $A_\epsilon(x, 0) \equiv 1$ and

$$K_\epsilon(x) = -\partial_y^2 A_\epsilon(x, 0) = \pi^2(1 + \epsilon \cos 2\pi x).$$

With $z = \pi x$ the Jacobi equation $J'' + K_\epsilon J = 0$ is Mathieu's equation

$$\frac{d^2 J}{dz^2} + (a - 2q \cos 2z)J = 0, \quad a = 1, \quad q = -\epsilon/2,$$

and $\mathcal{R}_o$ is its monodromy over $z \in [0, \pi]$. The first instability interval of Mathieu's equation lies between its two characteristic values with π -antiperiodic solutions, which are $1 \pm |q| + O(q^2)$ [2, Section 20.2]. So there is $q_1 > 0$ such that $a = 1$ lies inside the interval for $0 < |q| < q_1$, and there the monodromy has real eigenvalues $-\mu^{\pm 1}$, $\mu > 1$ [43, Chapter II, Theorem 2.1]: o is inverse-hyperbolic for $0 < \epsilon < 2q_1$. Numerically $\text{tr } \mathcal{R}_o$ decreases from -2 at $\epsilon = 0$ to $-2.6 \dots$ as $\epsilon \rightarrow 1$, with $\text{tr } \mathcal{R}_o + 2 \approx -0.617\epsilon^2$ for small ϵ . Along o' , $A_\epsilon(x, 1/2) = \frac{1}{2}(1 - \epsilon \cos 2\pi x)$ and

$$K = -\frac{\partial_y^2 A_\epsilon}{A_\epsilon} = -2\pi^2 \frac{1 + \epsilon \cos 2\pi x}{1 - \epsilon \cos 2\pi x} < 0,$$

so o' is positive hyperbolic: negative curvature along a closed geodesic makes both diagonal entries of its Hill monodromy, in arc length, exceed 1, as in the proof of Theorem 12.4 below. It is the minimizer of its class: every class β_0 loop has length at least $\int_0^1 \min_y A_\epsilon(x, y) dx = \int_0^1 A_\epsilon(x, 1/2) dx = 1/2$, with equality only for o' . Taking $\epsilon_0 \leq 2q_1$ gives (2).

(3). Fix $0 < \epsilon < \epsilon_0$. The string o is nondegenerate and inverse-hyperbolic for g_ϵ , so every metric C^2 -close to g_ϵ has a closed geodesic near o , of class β_0 and with $\text{tr } \mathcal{R} < -2$, by the implicit function theorem applied to a local return map, as in the proof of Theorem 12.4; it is prime, as β_0 is indivisible. Let U be the set of metrics in the C^∞ δ -ball of g_A that are C^2 -close to g_ϵ in this sense: a C^∞ -neighborhood of g_ϵ . Its members are taut by (1). For a $2\beta_0$ -regular $g \in U$, which is β_0 -regular automatically, $b_1 \geq 1$ in the notation of Section 9, and Proposition 9.5 gives $s_2 = -b_1 \leq -1$: g has a prime geodesic string in class $2\beta_0$. The $2\beta_0$ -regular metrics are C^∞ -generic in U by the bumpy metric theorem, cited before Proposition 9.5. $\square$

Remark 12.1. The maximal curvature π^2 of g_A is an isometry invariant, and every metric of the ball in (1), shrunk if necessary, has maximal curvature above $\pi^2/2$: none is C^2 -close to a flat metric, in any coordinates. The confinement of Proposition 11.2 is around g_A , not around a flat metric: the mechanism needs an integrable metric, not a flat one. The classes $e\beta_0$, $e \geq 3$, carry prime strings

by classical twist dynamics, Remark 12.3; the class $2\beta_0$ is the one for which the identity $s_2 = -b_1$ is needed.

Census. The class β_0 geodesics of g_A are o , o' , and the geodesics oscillating about o . For the latter the Clairaut momentum $p = A^2 dx/dt$, with t the arc length, lies in $(1/2, 1)$; y oscillates between the levels $A(y) = p$, and the x -advance over one oscillation is

$$\Delta x(p) = 4 \int_0^{y_p} \frac{p dy}{A(y) \sqrt{A(y)^2 - p^2}}, \quad A(y_p) = p, \quad 0 < y_p < \frac{1}{2}.$$

An oscillating geodesic closes up in class $e\beta_0$ after m oscillations exactly when $m \Delta x(p) = e$, and then comes in a circle of geodesic strings, by the x -symmetry of g_A . Numerically Δx decreases from $+\infty$ as $p \rightarrow 1/2$, the separatrix of o' , to 2 as $p \rightarrow 1$, the linearized value $2\pi/\sqrt{K} = 2$ along o . Granting this: the class β_0 strings of g_A are exactly o and o' , so that $F(g_A, \beta_0) = 1 - 1 = 0$; the class $2\beta_0$ strings are the two double covers; and each class $e\beta_0$, $e \geq 3$, contains circles of prime strings, one for each $m < e/2$ coprime to e . By the length bound (11.3) and the nondegeneracy of o and o' , every metric in a smaller C^∞ -ball of g_A has exactly two class β_0 strings, the continuations of o and o' : a third, along a sequence of metrics tending to g_A , would converge to o or o' , against the uniqueness of their continuations. For ϵ small enough that g_ϵ lies in this ball, shrink U into it: its members have exactly two class β_0 strings, one of each hyperbolic type, with $F(g, \beta_0) = 0$ realized with $b_1 = 1$, and its $2\beta_0$ -regular members have $s_2 = -1$: an odd number of prime class $2\beta_0$ strings, all near o^2 , by the same compactness.

Remark 12.2. At $\epsilon = 5/\pi^2$, the metric of Figure 1, beyond the range of the proof, numerics give the same picture. The Hill discriminant is $\text{tr } \mathcal{R}_o = -2.158 \dots$. A computation of the return map to $\{x = 0\}$ finds exactly one prime geodesic string γ of class $2\beta_0$, elliptic ($\text{tr } \mathcal{R}_\gamma = -0.492 \dots$), of length $1.969 \dots$, born of o at the period doubling, and no class β_0 or $2\beta_0$ geodesic strings other than o , o' , their double covers and γ among the geodesics winding monotonically in x . Tautness, and the census beyond the monotone window, were checked numerically only. For small ϵ , granting the census, the count of prime class $2\beta_0$ strings is odd for the $2\beta_0$ -regular members of U , and equal to one for a generic period doubling, which produces a single period two orbit.

Remark 12.3. The classes $e\beta_0$, $e \geq 3$, need no arithmetic. The librational tori of g_A about o have rotation number $1/\Delta x(p)$ per revolution; Δx is analytic and nonconstant on $(1/2, 1)$, so the twist $\Delta x'(p)$ vanishes only on a discrete set, and for small ϵ the tori with Diophantine rotation number and nonzero twist survive. The Poincaré–Birkhoff theorem [19], on the annulus between two surviving tori with rotation numbers on either side of $1/e$, gives at least two closed geodesics of rotation number $1/e$, one oscillation in e revolutions: prime strings of class $e\beta_0$, in pairs, as part (1) of Theorem 1.3 requires for e odd. The class $2\beta_0$ is the one this does not reach: $1/2$ is the rotation number of o itself, not of an annulus around it. There the identity $s_2 = -b_1$ forces a prime string.

12.3. A prime geodesic string in a divisible class, near negative curvature. Let (Σ, g_-) be a closed orientable Riemannian surface of negative curvature, o_- a simple closed g_- -geodesic of length L , and β_0 its free homotopy class. The class β_0 is indivisible: the geodesic string of a class $k\alpha$, $k \geq 2$, is the k -fold cover of the one of class α , so its underlying loop is not embedded. In particular every geodesic string in class β_0 is prime.

For g_- itself, Theorem 1.3 forces nothing in the divisible classes: every closed g_- -geodesic is positive hyperbolic, as negative curvature makes both diagonal entries of the Hill monodromy exceed 1 (see the proof below), so $b_d = 0$ for every d , in the notation of Section 9. This changes immediately inside the C^1 -ball of Proposition 2.7, by a perturbation of definite C^2 -size; compare the C^∞ -small one of Theorem 1.13.

Theorem 12.4. *Let $\epsilon = \epsilon(g_-)$ be as in Proposition 2.7. For every $0 < \epsilon' \leq \epsilon$ there is a $2\beta_0$ -regular Riemannian metric g with $\|g - g_-\|_{C^1} < \epsilon'$, having an inverse-hyperbolic prime geodesic string in class*

β_0 . Consequently, by Theorem 1.3 (2), the divisible class $2\beta_0$ contains a prime g -geodesic string; and as $s_1 = 1$ (Corollary 9.10), the class β_0 contains at least three.

Proof. As Σ is orientable, o_- is two-sided, and the normal exponential map embeds $(\mathbb{R}/L\mathbb{Z}) \times (-w_0, w_0)$ for some $w_0 > 0$. This gives Fermi coordinates (x, y) in which

$$g_- = dy^2 + A_-(x, y)^2 dx^2, \quad A_-(x, 0) \equiv 1, \quad \partial_y A_-(x, 0) \equiv 0,$$

and the Gauss curvature of any metric of this form is $K = -\partial_y^2 A/A$; for g_- hyperbolic, $A_- = \cosh y$.

Fix $\chi \in C_c^\infty((-1, 1))$ with $\chi \equiv 1$ on $[-1/2, 1/2]$, and $q \in C^\infty(\mathbb{R}/L\mathbb{Z})$ to be chosen, independently of w . For $0 < w < w_0$ set

$$g = dy^2 + A^2 dx^2, \quad A = A_- + \frac{1}{2} q(x) y^2 \chi(y/w),$$

extended by g_- outside the collar. Still $A(x, 0) \equiv 1$ and $\partial_y A(x, 0) \equiv 0$. So the core curve $\{y = 0\}$ is a unit speed g -geodesic of length L , prime of class β_0 ; write o for its string. Along it the Gauss curvature is $K(x) = K_-(x) - q(x)$, with $K_-(x)$ the g_- -curvature. The perturbation is supported in $\{|y| \leq w\}$, so

$$\|g - g_-\|_{C^0} \leq Cw^2, \quad \|g - g_-\|_{C^1} \leq Cw,$$

with $C = C(q, \chi)$ independent of w . So for w small, g is in the C^1 ϵ' -ball. The C^2 -distance does not decay: the curvature along the core moves by q .

As in Section 12.2, $\mathcal{R}_o$ is the monodromy over the period L of the Hill equation $J'' + K(x)J = 0$. Set $a := 2\pi/L$, and consider first the discontinuous choice

$$q = K_- - a^2 \text{ on } [0, L/2], \quad q = 0 \text{ on } [L/2, L],$$

so that $K = a^2$ on $[0, L/2]$, and $K = K_- < 0$ on $[L/2, L]$. Since $a \cdot L/2 = \pi$, the fundamental matrix of $J'' + a^2 J = 0$ over $[0, L/2]$ is exactly $-I$. Let H be the fundamental matrix of $J'' = (-K_-)J$ over $[L/2, L]$. Both diagonal entries of H exceed 1: the solution with $J = 1$, $J' = 0$ at $L/2$ is convex while positive, hence increasing, so $J(L) > 1$; the solution with $J = 0$, $J' = 1$ at $L/2$ has increasing derivative, so $J'(L) > 1$. Hence

$$\text{tr } \mathcal{R}_o = \text{tr } (H \cdot (-I)) = -\text{tr } H < -2;$$

for g_- hyperbolic, explicitly $\text{tr } \mathcal{R}_o = -2 \cosh(L/2)$. The monodromy depends continuously on K in the L^1 -norm: this is the Gronwall argument of (10.1), with the C^0 -distance of the vector fields replaced by the L^1 -distance in time of the coefficients. So a smooth q , equal to the above off a set of small measure and bounded independently of it, keeps $\text{tr } \mathcal{R}_o < -2$: the string o is inverse-hyperbolic, in particular nondegenerate.

The map $\mathcal{R}$ of a closed geodesic varies continuously with the metric in the C^2 -topology, and a non-degenerate closed orbit persists under such perturbations, by the implicit function theorem applied to a local return map; see e.g. [25, Section 5.2]. So there is a C^2 -neighborhood U of g , which we take inside the C^1 ϵ' -ball of g_- , so that every $g' \in U$ has a closed geodesic string near o , of class β_0 , with $\text{tr } \mathcal{R} < -2$. This string is prime, as β_0 is indivisible. By the Riemannian bumpy metric theorem, cited before Proposition 9.5, U contains a $2\beta_0$ -regular Riemannian metric; replace g by it. So g is $2\beta_0$ -regular, lies in the ϵ' -ball, and $b_1 \geq 1$ for g .

Now g is $2\beta_0$ -taut, by Proposition 2.7, and $2\beta_0$ -regular, so Theorem 1.3 applies with $e = 2$. Part (2), with $n = 1$, gives $\#P_2 \geq b_1 \geq 1$: a prime geodesic string in class $2\beta_0$. Moreover $s_1 = 1$, by Corollary 9.10, while the inverse-hyperbolic string contributes $i = -1$ to s_1 , by Lemma 9.4. So P_1 contains at least two strings of index $+1$ besides it: $\#P_1 \geq 3$. $\square$

Corollary 12.5. *For every $0 < \epsilon' \leq \epsilon$ there is a nonempty C^2 -open set U of Riemannian metrics, contained in the C^1 ϵ' -ball of g_- , so that every $2\beta_0$ -regular $g' \in U$ has a prime geodesic string in the divisible class $2\beta_0$. The $2\beta_0$ -regular metrics are C^∞ -generic in U .*

Proof. Take U as in the proof of Theorem 12.4, restricted to Riemannian metrics: every $g' \in U$ has an inverse-hyperbolic prime geodesic string in class β_0 , and Part (2) of Theorem 1.3 applies to its $2\beta_0$ -regular members, taut by Proposition 2.7. Genericity is the bumpy metric theorem, cited before Proposition 9.5. $\square$

Existence is thus forced on an open set of metrics, modulo a generic regularity condition, while for g_- itself every geodesic string in a divisible class is an iterate. This open forcing is what the invariance of F adds: a bifurcation analysis gives branches at special parameters, and min-max methods do not track isotropy.

Iterating the law through the powers of 2 gives prime geodesic strings in arbitrarily divisible classes. The iteration step is the following localization of the Hill engineering above.

Lemma 12.6 (Flip lemma). *Let g be a Riemannian metric on a closed orientable surface Σ , δ a prime closed g -geodesic, and G a finite set of closed g -geodesics, none with the image of δ . For every $\rho > 0$ there is a Riemannian metric g' with $\|g' - g\|_{C^1} < \rho$, agreeing with g outside an arbitrarily small neighborhood of a sub-arc of δ , disjoint from the members of G , so that: δ is a closed g' -geodesic, unchanged as a parametrized curve, now of inverse-hyperbolic type with $\text{tr } \mathcal{R}_\delta < -2$; and every member of G is a closed g' -geodesic with unchanged linearized return map.*

Proof. All self-intersections of δ are transverse, so finitely many: a tangential return with equal velocities contradicts that δ is prime, and one with opposite velocities forces a vanishing velocity at the symmetric midpoint. Likewise δ meets the members of G in finitely many points, as a tangency would force the geodesics to coincide up to orientation. So δ has an embedded sub-arc avoiding all these points, with a Fermi strip around it meeting δ only along the sub-arc and meeting no member of G . In the strip $g = dy^2 + A^2 dx^2$ with $A(x, 0) \equiv 1$ and $\partial_y A(x, 0) \equiv 0$, as in the proof of Theorem 12.4, and we perturb by

$$A \mapsto A + \frac{1}{2} q(x) y^2 \chi(y/w),$$

with q supported in a window W interior to the sub-arc. For any such q : the curve δ is unchanged, and is still a closed g' -geodesic of the same length; the members of G do not meet the support of the perturbation, so they and their linearized return maps are unchanged; and $\|g' - g\|_{C^1} \leq C(q)w$.

On an orientable surface the normal bundle of δ is trivial, so $\mathcal{R}_\delta$ is the monodromy of the scalar Hill equation $J'' + K(t)J = 0$ over the full period, and $\mathcal{R}_\delta = M \cdot E$, with E the fundamental matrix over W and M , over the rest of the period, unchanged by the perturbation. First we arrange $M_{12} \neq 0$, shrinking W if necessary: varying the endpoint t_1 of W , $\partial M_{12}/\partial t_1 = -M_{11}$, while $M_{12} = 0$ forces $M_{11}M_{22} = \det M = 1$, so the zeros of M_{12} in t_1 are isolated. Now take $q = K - c^2$ on W , of length ℓ' , so that momentarily ignoring smoothness at the ends of W ,

$$\text{tr } \mathcal{R}_\delta = (M_{11} + M_{22}) \cos(c\ell') + \left(\frac{M_{21}}{c} - cM_{12} \right) \sin(c\ell').$$

Along $c\ell' = \pi/2 + m\pi$, with the parity of m fixed by the sign of M_{12} , this tends to $-\infty$ as $m \rightarrow \infty$. Fix such a c with trace < -3 . Smoothing q at the ends of W , over small intervals on which the coefficient stays bounded by $\max(|K|, c^2)$, perturbs the monodromy continuously for this fixed c ; then take w small. So $\text{tr } \mathcal{R}_\delta < -2$: δ is inverse-hyperbolic for g' . $\square$

The localization also removes the simplicity assumption on o_- : in the rest of this subsection β_0 is any indivisible class, and $\epsilon = \epsilon(g_-)$ is as before.

Theorem 12.7. *For every $0 < \epsilon' \leq \epsilon$ and every $k \geq 0$ there is a Riemannian metric g with $\|g - g_-\|_{C^1} < \epsilon'$ having an inverse-hyperbolic prime geodesic string in the class $2^j \beta_0$ for every $0 \leq j \leq k$. In particular each divisible class $2^j \beta_0$, $1 \leq j \leq k$, contains a prime geodesic string.*

Proof. Induction on k , with $\|g_k - g_-\|_{C^1} < \epsilon'(1 - 2^{-k-1})$. For $k = 0$: the closed g_- -geodesic of class β_0 is prime, as β_0 is indivisible, and Lemma 12.6 applied to it, with $G = \emptyset$ and $\rho = \epsilon'/2$, gives g_0 . (When β_0 is the class of a simple closed geodesic, Theorem 12.4, with $\epsilon'/2$ in place of ϵ' , does this as well.)

Let g_k have inverse-hyperbolic prime geodesic strings $\gamma_0, \dots, \gamma_k$, with γ_j of class $2^j\beta_0$ and $\text{tr } \mathcal{R}_{\gamma_j} < -2$, as Lemma 12.6 provides. Having $\text{tr } \mathcal{R} < -2$, they persist, with their classes and with $\text{tr } \mathcal{R} < -2$, under perturbations small in C^2 . A string of class $2^j\beta_0$ with $\text{tr } \mathcal{R} < -2$ is prime: a non-prime one is o^m with $m \mid 2^j$, by Lemma 1.8, so m is even, and $\text{tr } \mathcal{R}_o^m \geq -2$ for even m and any $\mathcal{R}_o \in Sp(2, \mathbb{R})$. By the Riemannian bumpy metric theorem we may then choose a C^∞ -small such perturbation g' which is moreover $2^{k+1}\beta_0$ -regular, with $\|g' - g_k\|_{C^1} < \epsilon'2^{-k-3}$, so that g' lies in the C^1 ϵ -ball of g_- and is taut by Proposition 2.7. Let γ'_j be the continuation of γ_j . For g' : $b_{2^k} \geq 1$, and by Corollary 9.10, $s_{2^{k+1}} = -b_{2^k} \leq -1$. So g' has a prime geodesic string δ of class $2^{k+1}\beta_0$. The strings $\gamma'_0, \dots, \gamma'_k, \delta$ are pairwise distinct geodesics: two prime strings with the same image have equal or inverse classes, while the classes $2^j\beta_0$ are pairwise distinct and pairwise non-inverse, as in the proof of Lemma 1.8. Apply Lemma 12.6 to δ , with $G = \{\gamma'_0, \dots, \gamma'_k\}$ and $\rho < \epsilon'2^{-k-3}$. The result is g_{k+1} , with $\|g_{k+1} - g_-\|_{C^1} < \epsilon'(1 - 2^{-k-2})$. $\square$

Remark 12.8. As in Corollary 12.5, the conclusion is C^2 -open: the strings $\gamma_0, \dots, \gamma_k$ of the proof have $\text{tr } \mathcal{R} < -2$, so every Riemannian metric C^2 -close to g , inside the C^1 ϵ -ball of g_- and so taut, has geodesic strings with $\text{tr } \mathcal{R} < -2$ in all the classes $2^j\beta_0$, $j \leq k$, and its $2^{k+1}\beta_0$ -regular members have a prime geodesic string in class $2^{k+1}\beta_0$, as in the proof above; the strings with $\text{tr } \mathcal{R} < -2$ are prime, as shown there.

Proof of Corollary 1.4. Given k and a C^1 -neighborhood of g_- , take $0 < \epsilon' \leq \epsilon$ with the C^1 ϵ' -ball of g_- inside it, and g as in Theorem 12.7. By Remark 12.8, a C^2 -neighborhood of g , which we take inside the ϵ' -ball, consists of metrics with inverse-hyperbolic prime geodesic strings in all the classes $2^j\beta_0$, $1 \leq j \leq k$. $\square$

The odd divisible classes are beyond this method: there the law forces pairing, not existence.

APPENDIX A. FULLER INDEX AND CONTINUATION

In this appendix we recall the part of Fuller's periodic orbit index used in the paper, and fix the notation for orbit strings and for continuation under small perturbations of vector fields. Some of this material overlaps with [57]; we nevertheless give a brief self-contained account, especially since the noncompact setting needed here is not treated there, and the details serve to fix notation used throughout the paper.

Let X be a smooth manifold, and fix a complete auxiliary Riemannian metric h on X . Let

$$\mathcal{V}(X)$$

denote the space of smooth vector fields on X , with the C^0 -topology on compact subsets.

Let

$$LX = C^\infty(S^1, X)$$

denote the smooth free loop space, with the C^1 topology. For $V \in \mathcal{V}(X)$, define

$$S(V) = \{o \in LX \mid o \text{ is nonconstant, } \dot{o}(t) = TV(o(t)) \text{ for some } T > 0\}.$$

The number T is the period of o , written

$$T = T(o);$$

it is uniquely determined.

The circle $S^1 = \mathbb{R}/\mathbb{Z}$ acts on $S(V)$ by reparametrization:

$$(s \cdot o)(t) = o(t + s).$$

We define the space of orbit strings by

$$\mathcal{O}(V) = S(V)/S^1.$$

If β is a free homotopy class in X , we write

$$S(V, \beta) \subset S(V), \quad \mathcal{O}(V, \beta) \subset \mathcal{O}(V)$$

for the corresponding subspaces. When no confusion is possible, we use the same letter o for a parametrized orbit and for its associated orbit string.

Let

$$q : LX \rightarrow LX/S^1$$

be the quotient map. If

$$U \subset LX/S^1,$$

we write

$$\widehat{U} = q^{-1}(U) \subset LX.$$

Definition A.1. We say that U is *h-controlled* if there are constants

$$0 < \ell_- < \ell_+ < \infty$$

and a compact set $K \subset X$ such that every loop $o \in \widehat{U}$ satisfies

$$\text{image}(o) \subset K, \quad \ell_- \leq \ell_h(o) \leq \ell_+.$$

The lower bound excludes collapse to constant loops; in the applications it is usually supplied by a nontrivial projected free homotopy class.

For an orbit string $o \in \mathcal{O}(V)$, its multiplicity

$$\text{mult}(o) \in \mathbb{N}$$

is the unique positive integer such that o is the $\text{mult}(o)$ -fold cover of a prime orbit string. Equivalently, if $T(o)$ is the period of o and $T_{\text{prim}}(o)$ is the period of the underlying prime orbit string, then

$$\text{mult}(o) = \frac{T(o)}{T_{\text{prim}}(o)}.$$

A.1. Fuller index. Suppose

$$N \subset \mathcal{O}(V)$$

is finite and consists of nondegenerate orbit strings. Fuller associates to (N, V) the rational number

$$i(N, V) = \sum_{o \in N} \frac{i(o)}{\text{mult}(o)}.$$

Here, for a nondegenerate orbit string o of the flow of V , on an ambient manifold of dimension $2m+1$ (all ambient manifolds in this paper are cosphere bundles, of odd dimension), and with $\mathcal{R}_o$ the linearized time- $T(o)$ return map, computed on a local hypersurface transverse to the image of o — the $\text{mult}(o)$ -th iterate of the first return map — we set the **local Fuller index**

$$(A.2) \quad i(o) := (-1)^m \text{sign det}(I - \mathcal{R}_o).$$

Up to the dimensional sign $(-1)^m$ this is the fixed point index of the time- $T(o)$ return map; the sign is chosen so that minimizing closed geodesics contribute positively, see below.

By a compact open subset N of an orbit-string space, we mean a subset which is compact and open in the relative topology. In this case Fuller again associates to this a rational index $i(N, V)$. The latter can be defined perturbatively, and using the formula as above.

In the special case of a nondegenerate closed geodesic orbit on an orientable manifold X , with ambient manifold S_g^*X , so that $m = \dim X - 1$, the Morse index theorem gives the classical relation $\text{sign det}(I - \mathcal{R}_o) = (-1)^{\text{morse}(o)+m}$,⁶ see for example [41, Chapter 12], so that by (A.2):

$$i(o) = (-1)^{\text{morse}(o)}.$$

Consequently, if N consists of finitely many nondegenerate closed orbits of the geodesic Reeb flow R^{λ_g} on C , we get:

$$(A.3) \quad i(N, R^{\lambda_g}) = \sum_{o \in N} \frac{(-1)^{\text{morse}(o)}}{\text{mult}(o)}.$$

Remark A.4. The relation above is where orientability of X is used. It fails for one-sided closed geodesics, those with nonorientable normal bundle. On the flat Klein bottle $\mathbb{R}^2 / \langle (x, y) \mapsto (x + 1, -y), (x, y) \mapsto (x, y + 1) \rangle$ the one-sided circle $\{y = 0\}$ is a closed geodesic. Its closed normal variation fields W , in a parallel unit normal, are antiperiodic, $W(1) = -W(0)$. The index form is $\int W'^2$, positive on nonzero such W : the Morse index is 0 and the geodesic is nondegenerate. But $\mathcal{R}_o = -\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\det(I - \mathcal{R}_o) = 4$, so $i(o) = -1$. We do not pursue the nonorientable case.

A.2. Continuation. Let

$$\{V_t\}_{t \in [0,1]} \subset \mathcal{V}(X)$$

be a continuous family. Define

$$S(\{V_t\}) = \{(o, t) \in LX \times [0, 1] \mid o \in S(V_t)\},$$

and

$$(A.5) \quad \mathcal{O}(\{V_t\}) = S(\{V_t\})/S^1,$$

where S^1 acts on the loop coordinate by reparametrization as before; for a free homotopy class β in X , $S(\{V_t\}, \beta)$ and $\mathcal{O}(\{V_t\}, \beta)$ denote the subspaces of pairs with o of class β .

Let

$$\mathcal{N} \subset \mathcal{O}(\{V_t\})$$

be compact and open. Suppose that its endpoint slices are

$$N_0 = \mathcal{N} \cap \mathcal{O}(V_0), \quad N_1 = \mathcal{N} \cap \mathcal{O}(V_1).$$

Then

$$(A.6) \quad i(N_0, V_0) = i(N_1, V_1).$$

We refer to this as the **basic invariance** of the Fuller index.

For applications, we need a local version of this statement under small perturbations. This is a variation of [29, Lemma 4.1], with the added control needed for noncompact X .

Lemma A.7 (Isolation lemma). *Let h be a complete auxiliary Riemannian metric on X . Let $V_0 \in \mathcal{V}(X)$ be a nonsingular vector field, and let*

$$N_0 \subset \mathcal{O}(V_0)$$

be compact and open. Let

$$U \subset LX/S^1$$

be an h -controlled open neighborhood of N_0 such that

$$\overline{U} \cap \mathcal{O}(V_0) = N_0.$$

Let W be an open neighborhood of N_0 satisfying

$$\overline{W} \subset U.$$

⁶ $\text{morse}(o)$ refers to the Morse index of a representative of the class o .

Then there exists $\epsilon > 0$ with the following property. If V is a smooth vector field satisfying

$$\sup_{x \in K} |V(x) - V_0(x)|_h < \epsilon,$$

where K is the compact set from the h -control of U , then

$$\mathcal{O}(V) \cap U \subset W.$$

In particular,

$$\mathcal{O}(V) \cap U$$

is compact and open in $\mathcal{O}(V)$.

Proof. Let K , ℓ_- , and ℓ_+ be the compact set and length bounds from the h -control of U . Since V_0 is nonsingular and K is compact, there are constants

$$0 < B < A < \infty$$

such that

$$B \leq |V_0(x)|_h \leq A$$

for all $x \in K$. We choose $\epsilon < B/2$. Then every vector field V satisfying

$$\sup_K |V - V_0|_h < \epsilon$$

obeys

$$|V(x)|_h \geq B/2, \quad |V(x)|_h \leq A + \epsilon$$

for all $x \in K$.

Suppose the conclusion failed for the chosen W . Then there would be vector fields

$$V_j \rightarrow V_0$$

uniformly on K , and orbit strings

$$o_j \in (\mathcal{O}(V_j) \cap U) - W.$$

Choose parametrized representatives, still denoted o_j , satisfying

$$(A.8) \quad \dot{o}_j(s) = T_j V_j(o_j(s)).$$

Since $o_j \in \widehat{U}$, controlledness gives

$$\text{image}(o_j) \subset K, \quad \ell_- \leq \ell_h(o_j) \leq \ell_+.$$

For j sufficiently large,

$$B/2 \leq |V_j|_h \leq A + \epsilon$$

on K . Hence

$$\frac{\ell_-}{A + \epsilon} \leq T_j \leq \frac{2\ell_+}{B}.$$

The velocities are also uniformly bounded:

$$|\dot{o}_j(s)|_h = T_j |V_j(o_j(s))|_h \leq \frac{2\ell_+(A + \epsilon)}{B}.$$

Thus the o_j are equicontinuous and have images in the fixed compact set K . By the Arzelà–Ascoli theorem, after passing to a subsequence,

$$o_j \rightarrow o_\infty$$

uniformly. After passing to a further subsequence, $T_j \rightarrow T_\infty$, with $T_\infty > 0$.

Hence o_j is a Cauchy sequence in the C^1 space of maps, by the defining equation (A.8). Then C^1 completeness gives that

$$\dot{o}_\infty = T_\infty V_0(o_\infty).$$

Thus o_∞ is a closed V_0 -orbit string. Moreover

$$o_\infty \in \overline{U} - W.$$

This contradicts

$$\overline{U} \cap \mathcal{O}(V_0) = N_0 \subset W.$$

Therefore

$$\mathcal{O}(V) \cap U \subset W$$

for all V sufficiently close to V_0 on K .

It remains to prove compactness. Let o_j be any sequence in $\mathcal{O}(V) \cap U$. The same estimates as above give uniformly bounded periods and velocities, with all images contained in K . Arzelà–Ascoli therefore gives a convergent subsequence. Its limit is again a closed V -orbit. Since the inclusion just proved gives

$$\mathcal{O}(V) \cap U = \mathcal{O}(V) \cap \overline{W}$$

and $\overline{W} \subset U$, the limit lies in $\mathcal{O}(V) \cap U$. Thus $\mathcal{O}(V) \cap U$ is compact. It is open because U is open. $\square$

We call such an ϵ a **Fuller constant** for the data

$$(V_0, N_0, U, W).$$

It also depends on the auxiliary control data K, ℓ_-, ℓ_+ , which will always be understood from the chosen U .

Lemma A.9 (Fuller continuation). *Let V_0, N_0, U, W , and ϵ be as in Lemma A.7. Let*

$$\{V_t\}_{t \in [0,1]}, \quad V_{t=0} = V_0$$

be a continuous family such that for every $t \in [0, 1]$,

$$\sup_{x \in K} |V_t(x) - V_0(x)|_h < \epsilon,$$

where K is the compact set from the h -control of U . Define

$$\mathcal{N} = (U \times [0, 1]) \cap \mathcal{O}(\{V_t\}).$$

Then $\mathcal{N}$ is compact and open in the orbit cobordism $\mathcal{O}(\{V_t\})$. Its endpoint slices are

$$N_0 = U \cap \mathcal{O}(V_0), \quad N_1 = U \cap \mathcal{O}(V_1).$$

Consequently,

$$i(N_0, V_0) = i(N_1, V_1).$$

We call N_1 the Fuller continuation of N_0 .

Proof. Since U is open,

$$\mathcal{N} = (U \times [0, 1]) \cap \mathcal{O}(\{V_t\})$$

is open in $\mathcal{O}(\{V_t\})$.

By Lemma A.7, applied to each V_t , we have

$$\mathcal{O}(V_t) \cap U \subset W$$

for every $t \in [0, 1]$. Since

$$\overline{W} \subset U,$$

this implies

$$\mathcal{N} = (\overline{W} \times [0, 1]) \cap \mathcal{O}(\{V_t\}).$$

The compactness then works as in Lemma A.7. The endpoint slices are

$$\mathcal{N} \cap \mathcal{O}(V_0) = U \cap \mathcal{O}(V_0) = N_0$$

and

$$\mathcal{N} \cap \mathcal{O}(V_1) = U \cap \mathcal{O}(V_1) = N_1.$$

Fuller's basic invariance applied to the compact open cobordism $\mathcal{N}$ gives

$$i(N_0, V_0) = i(N_1, V_1).$$

□

Example 7 (Geodesic flows and the Sasaki metric). Let (Y, g_Y) be a complete Riemannian manifold, and let

$$C = S^*Y$$

be its unit cotangent bundle, with projection $\text{pr} : C \rightarrow Y$ and Liouville contact form λ_{g_Y} . The Reeb vector field $R^{\lambda_{g_Y}}$ is the geodesic Reeb field.

Let g_S be the Sasaki metric on S^*Y . It is defined using the Levi-Civita connection of g_Y . Thus

$$TC = T^{\text{vert}}C \oplus T^{\text{hor}}C$$

is an orthogonal splitting, where

$$T^{\text{vert}}C = \ker(\text{pr}_*)$$

and $T^{\text{hor}}C$ is the Levi-Civita horizontal subbundle. If $\xi, \eta \in T(S^*Y)$ are horizontal, then

$$g_S(\xi, \eta) = g_Y(\text{pr}_* \xi, \text{pr}_* \eta),$$

and the vertical part is measured using the metric induced by g_Y on the cotangent fibers.

The basic property needed in the paper is the following. If o is a nonconstant constant-speed g_Y -geodesic and $\tilde{o}$ is its unit cotangent lift, then

$$(A.10) \quad \ell_{g_S}(\tilde{o}) = \ell_{g_Y}(o).$$

Indeed, the covector component of $\tilde{o}$ is parallel along o , so the vertical part of $\dot{\tilde{o}}$ vanishes and the Sasaki speed equals the base speed.

If $\beta \in \pi_1^{\text{inc}}(Y)$ and $U \subset S(g_Y, \beta)$ is g_Y -length bounded, then the set of canonical lifts

$$\tilde{U} \subset S(R^{\lambda_{g_Y}})$$

is g_S -length bounded by (A.10).

APPENDIX B. A LOCAL FIBERED CALCULATION OF F

We prove in this section the technical lemma computing the local index, used in Theorem 7.1.

Let

$$P : S^*X \rightarrow \mathbb{R}$$

be the verticality function as in the proof of Theorem 7.1,

$$P(\xi) = |\pi^{\text{vert}} v_\xi|_g^2.$$

Here v_ξ is the g -dual unit vector to ξ , and

$$\pi^{\text{vert}} : TX \rightarrow T^{\text{vert}}X$$

is the g -orthogonal projection. The notation β and β_Z is likewise as in Theorem 7.1.

Lemma B.1 (Local product calculation). *Let*

$$Z \hookrightarrow (X, g) \xrightarrow{p} (Y, g_Y)$$

be a β -periodic fibration as in Theorem 7.1, with fiber metrics g_y on $Z_y = p^{-1}(y)$. Let

$$\tilde{f} = f \circ p \circ \text{pr}$$

for a Morse function $f : Y \rightarrow \mathbb{R}$, and define the vector field on S^*X

$$V_\varepsilon = R^{\lambda_g} - \varepsilon \nabla_{g_S}(P + \tilde{f}).$$

Fix

$$y \in \text{Crit}(f),$$

and let

$$N_{\varepsilon,y} \subset N_\varepsilon = U \cap \mathcal{O}(V_\varepsilon)$$

be the compact open set of elements of N_ε lying over the fiber Z_y , as in the proof of Theorem 7.1: the closed V_ε -orbit strings over Z_y of projected class β . Then

$$i(N_{\varepsilon,y}, V_\varepsilon) = (-1)^{\text{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha),$$

where $S_y \subset \pi_1^{\text{inc}}(Z_y)$ is the finite set of fiber classes mapping to β , as in Theorem 7.1.

Proof. Let

$$T^{\text{vert}}X = \ker(dp), \quad H = (T^{\text{vert}}X)^\perp_g.$$

Since the submersion is parallel, the two distributions

$$T^{\text{vert}}X \quad \text{and} \quad H$$

are preserved by the Levi-Civita connection. Since $T^{\text{vert}}X$ and $H = (T^{\text{vert}}X)^\perp_g$ are parallel complementary distributions, the local de Rham decomposition theorem, with completeness of g and simple connectivity of D_y as in the proof of Theorem 7.1, gives a product decomposition

$$p^{-1}(D_y) \cong Z_y \times D_y, \quad g = g_y \oplus g_Y|_{D_y},$$

for a normal ball neighborhood $D_y \subset Y$ centered at y . See Kobayashi–Nomizu [39, Chapter IV].

For this local calculation we suppress the fixed- C identification and write S^*X for the g -unit cotangent bundle. Let

$$C^{\text{vert}} = \{P = 1\} \subset S^*X$$

be the vertical locus. Over Z_y , this locus is naturally identified with

$$S^*Z_y \subset S^*X.$$

Near

$$S^*Z_y \subset C^{\text{vert}},$$

the product decomposition gives the normal-bundle model

$$(B.2) \quad S^*Z_y \times D_y \times B_y^*,$$

where

$$B_y^* \subset T_y^*Y$$

is a small ball in the horizontal cotangent direction. In this model, the vertical locus is

$$S^*Z_y \times D_y \times \{0\}.$$

A covector near the vertical locus decomposes as

$$\xi = \xi_{\text{vert}} + \eta, \quad \eta \in B_y^*.$$

The unit condition is

$$|\xi_{\text{vert}}|_{g_y^*}^2 + |\eta|_{g_Y^*}^2 = 1.$$

Therefore

$$P(\xi) = |\xi_{\text{vert}}|_{g_y^*}^2 = 1 - |\eta|_{g_Y^*}^2.$$

Thus P has a Morse–Bott maximum along C^{vert} .

In the local model (B.2),

$$N_{\varepsilon,y} = N_y^{\text{fib}} \times \{y\} \times \{0\},$$

where $N_y^{\text{fib}} = \bigsqcup_{\alpha \in S_y} \tilde{\mathcal{O}}(g_y, \alpha)$ is the closed orbit set of the fiber geodesic Reeb flow on S^*Z_y in the classes of S_y .

The Fuller index is local, so we may work in the product model $p^{-1}(D_y) \cong Z_y \times D_y$ with the metric $g_y \oplus g_Y$. Replace g_y there by a C^∞ -small perturbation g'_y whose class α geodesics of bounded length, $\alpha \in S_y$, are nondegenerate, hence finitely many (the bumpy metric theorem, localized as in the proof of Theorem 1.2 (1)); the metric $g'_y \oplus g_Y$ is still a parallel product, so the description of $N_{\varepsilon,y}$ above persists. Along the linear path from g_y to g'_y , Lemma A.9 applies to the vector fields V_ε of the model and to the fiber Reeb flow alike, so neither side of the claimed identity changes, the right side being computed on the continued compact open set, which at g_y is N_y^{fib} . So we may assume the fiber strings nondegenerate.

We now describe the local return map at an element of $N_{\varepsilon,y}$. Along the vertical locus over Z_y , the fiber component of V_ε is the geodesic Reeb field $R^{\lambda_{g_y}}$ on S^*Z_y . The base component is

$$-\varepsilon \nabla_{g_Y} f$$

on D_y . The normal covector component is the outward radial component coming from

$$-\varepsilon \nabla_{g_S} P$$

on B_y^* .

Let

$$o \in N_y^{\text{fib}},$$

and choose a product transverse section

$$\Sigma = \Sigma^{\text{fib}} \times D_y \times B_y^*,$$

where

$$\Sigma^{\text{fib}} \subset S^*Z_y$$

is transverse to the fiber Reeb flow near o , and let $T(o)$ be the return time. In the coordinates (fiber, base, normal) of the product section, denote by σ the local return map of V_ε ; it is block triangular over the normal variable η . The normal component is a map κ of B_y^* alone, the time- $T(o)$ map of the radial field induced by $-\varepsilon \nabla_{g_S} P$, while the fiber and base components depend on η but restrict at $\eta = 0$ to, respectively, the local return map Φ_{fib} of the fiber geodesic flow generated by $R^{\lambda_{g_y}}$, and the time- $T(o)$ map ψ_y of $-\varepsilon \nabla_{g_Y} f$ near y .

The fixed point of σ corresponding to o is $(w, y, 0)$, where w is the point where o meets the fiber section. Consequently the linearized return map $\mathcal{R}_o = d\sigma$ at this fixed point is block triangular with diagonal blocks $d\Phi_{\text{fib}}$, $d\psi_y$ and $d\kappa$, so that

$$\det(I - \mathcal{R}_o) = \det(I - d\Phi_{\text{fib}}) \cdot \det(I - d\psi_y) \cdot \det(I - d\kappa);$$

compare also the multiplicativity of the fixed point index under products, Dold [27, Chapter VII, Section 5]. Here, for a map ϕ with an isolated fixed point x , we write $i_{\text{fp}}(\phi, x) \in \mathbb{Z}$ for its fixed point index: the local degree at x of the map $\text{id} - \phi$, computed in any chart; at a nondegenerate fixed point,

$$i_{\text{fp}}(\phi, x) = \text{sign} \det(I - d\phi_x).$$

Hence

$$i_{\text{fp}}(\sigma, (w, y, 0)) = i_{\text{fp}}(\Phi_{\text{fib}}, w) \cdot i_{\text{fp}}(\psi_y, y) \cdot i_{\text{fp}}(\kappa, 0).$$

In normal coordinates at y , the map $\text{id} - \psi_y$ equals $\varepsilon T(o) \nabla_{g_Y} f$ to first order, so has the same local degree as $\nabla_{g_Y} f$. Thus, because y is a Morse nondegenerate critical point,

$$i_{\text{fp}}(\psi_y, y) = (-1)^{\text{morse}(y)}.$$

Since the normal component of

$$-\varepsilon \nabla_{g_S} P$$

has outward radial linearization on a vector space of dimension $\dim Y$, the map $\text{id} - \kappa$ has local degree $(-1)^{\dim Y}$. Hence

$$i_{\text{fp}}(\kappa, 0) = (-1)^{\dim Y}.$$

Therefore

$$i_{\text{fp}}(\sigma, (w, y, 0)) = (-1)^{\dim Y + \text{morse}(y)} i_{\text{fp}}(\Phi_{\text{fib}}, w).$$

The multiplicity of o is unchanged by taking the product with the isolated base and normal fixed points. Finally we pass from fixed point indices to the Fuller weights of (A.2): the ambient manifolds are S^*X on the left and S^*Z_y on the right, of dimensions $2(\dim Z + \dim Y) - 1$ and $2\dim Z - 1$, so the two dimensional signs differ by exactly $(-1)^{\dim Y}$, cancelling the factor above; with $i(o; V)$ the local index (A.2) of o as an orbit string of V :

$$\frac{i(o; V_\varepsilon)}{\text{mult}(o)} = (-1)^{\text{morse}(y)} \frac{i(o; R^{\lambda_{g_y}})}{\text{mult}(o)}.$$

Summing over all fiber orbit strings in the local block gives

$$i(N_{\varepsilon, y}, V_\varepsilon) = (-1)^{\text{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha).$$

This proves the claim. $\square$

APPENDIX C. SKY CATASTROPHES

We now recall the topological formulation of sky catastrophes as studied in [57]. This is the formulation behind Definition 2.4 and Section 2.2.

Definition C.1. Let

$$\{V_t\}_{t \in [0,1]} \subset \mathcal{V}(X)$$

be a continuous family. We say that the family has a sky catastrophe in class β if there is an endpoint orbit string

$$y \in \mathcal{O}(V_0, \beta) \cup \mathcal{O}(V_1, \beta) \subset \mathcal{O}(\{V_t\}, \beta)$$

such that no compact open subset of $\mathcal{O}(\{V_t\}, \beta)$ contains y .

For suitably regular families, this agrees with the preliminary Definition 2.11.

Definition C.2. Let $\{V_t\}$ be a continuous family of Reeb vector fields on the fixed cosphere model C of Section 5, and $\beta \in \pi_1^{\text{inc}}(X)$. Denote by $\mathcal{O}_\beta(\{V_t\}) \subset \mathcal{O}(\{V_t\})$ the subspace of orbit strings whose projection to X has class β . We say that $\{V_t\}$ has a **sky catastrophe over the class β** if some endpoint orbit string in $\mathcal{O}_\beta(\{V_t\})$ is contained in no compact open subset of $\mathcal{O}_\beta(\{V_t\})$.

This projected form is the correct one for lifts: on a surface different class β strings can lift to different classes, shifted by the fiber class according to a turning number, while in dimension at least 3 the fiber of $C \rightarrow X$ is simply connected, and a sky catastrophe over β is one in the single lifted class.

For a family $\{g_t\}$ of complete metrics as in Definition 2.4, with β -taut endpoints, let $\{V_t\}$ be the induced family of Reeb vector fields on C . The lift homeomorphism of Section 5.1, in the family form of Section 5.2, identifies $\mathcal{O}_\beta(\{V_t\})$ with the downstairs cobordism $\mathcal{O}(\{g_t\}, \beta) = S(\{g_t\}, \beta)/S^1$, endpoint slices with endpoint slices, and compact open subsets with S^1 -saturated compact open subsets of $S(\{g_t\}, \beta)$; the saturation of a compact open subset is compact and open. The endpoint slices are compact, so the pointwise condition above and the condition of Definition 2.4 agree, by a finite subcover. So $\{g_t\}$ is a β -taut deformation exactly when $\{V_t\}$ has no sky catastrophe over the class β , as stated after Definition 2.4.

ACKNOWLEDGEMENTS

I am grateful to the IAS for resources provided during my stay as a member, where some of this paper was written. Thanks also to Mohammed Abouzaid, Kaoru Ono, Viktor Ginzburg, Yasha Eliashberg, Dennis Sullivan, Spencer Cattalani and Marco Mazzucchelli for interesting related discussions, and to Leonid Polterovich for comments on the history of the graph property of invariant tori, and related discussions. I have benefited from Anthropic's Claude: Theorem 1.13, Examples Section, KAM confinement analysis, Proposition 2.7, and surface group theory analysis are mostly due to Claude.

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